The working (6,7): a complete named derivation

79 graphical steps | Four explicit sphere moves | Portrait US Letter

VERY BRIEF SUMMARY. Remove the incident half twists, use a five-point disk, turn the remaining positive factors into FOUR exterior twists by PC lanterns, roses, daisies and Hurwitz transports, then cap their fourth power.

The previously missing geometric boundary closure is now supplied. The final equality is obtained from actual named substitutions and factor transports, rather than a whole-product action recognition. All four sphere moves retain their disk discrepancies, exact sphere certificates and separately checked genus-two lift.

What is still open. This detailed route has 79 retained graphical steps: three fewer than the previous82-step proof and twenty-two fewer than the101-step version and shorter than the earlier 177-step archive. It does not yet meet the goal of fewer than twenty meaningful operations. Reusable named patterns and fixed-factor drags are identified, with all adjacent checks retained in the certificates.

How to read it. Read each factorization downward. Curves are paired with the braid representing that exact support; blue marked-point dots continue at braid separators. The unnumbered support points use the same size and spacing in every factor. Only the changed interval is printed at a move, with the unchanged factors counted above and below. Every actual SPHERE MOVE has a yellow banner. Individual supports are not cumulative joint cut-system certificates.

Logical scope. Common frames and cyclic starts are explicit conjugations. The terminal four-exterior product is central, so reversing those conjugations recovers the source equality without assuming it was central. Marked points are not framed boundaries; signed framing balances are retained for PC substitutions. A single five-point cap has lift -I; the sixth- and fourth-power caps used here have +I.

Contents

Full diagrams, named sub-derivations and sphere discrepancies.

Complete named diagram sequence

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IX. Sphere discrepancies and certificates 78

79 graphical steps retained. Four sphere moves labeled; full discrepancies in Section IX.

What the steps will do

1. Reach the five-point chart. Keep the original F1 square intact. Carry the transported F7 through six neighbors, then apply the actual sphere lantern directly to the original separating square and two distinct half twists. No F1 root split, preconditioning or matched-half combine appears. Change the exterior point explicitly, retaining and separately capping the six framing corrections.

2. Use the existing roots. Keep the original123/124 squares intact through a powered cabled lantern with the shared12 absorbed; two cyclic starts replace the old six-operation prefix by three. Then borrow Q14 from the actual four-point daisy, transport it to match the original half twists, return the same Q14 and remove the last half pair. A cyclic start after the square and outer bypasses a standalone commutation; the short-side drag crosses that outer and later rejoins the same Q15 state. A partial PC daisy exposes the later squared root petals early; they travel through the same square drag and rejoin the old route literally. No auxiliary half twists are introduced.

3. Extract two exterior twists in the composed chart. Merge the positive T45 frame into the preceding exterior-lantern chart. Transport the actual loan, root carries and cabled lantern in that chart, checking every individual input and retaining all original source certificates. This bypasses one standalone global conjugation and rejoins the old route literally.

4. Return the loan as six actual pairs. The positive T45 chart is already in place. Commute the central exterior and transport the same outer upward by inverse Hurwitz moves. The actual six pairs match the expanded PC rose without a preliminary petal reorder or a temporary three-point root.

5. Carry the first outer upward. Choose the cyclic cut before gathering the first daisy, then carry the same outer UP through its petals. This changes the actual route and avoids dragging it through the second outer. The first PC daisy accepts its doubled petal directly. Its output keeps the spare positive twist visible, with the complete generic split and transport expansion retained in the certificate.

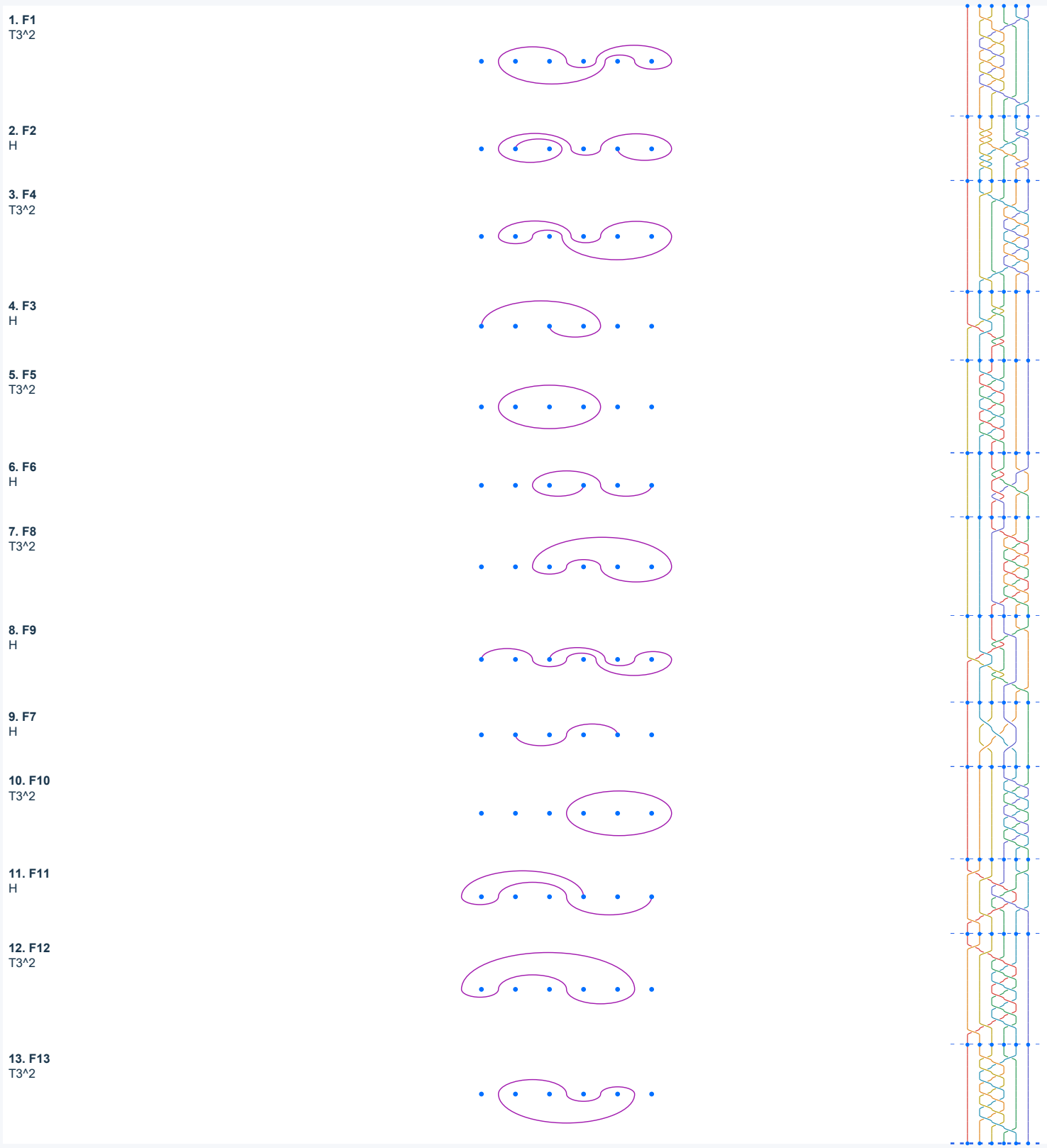
6. Close the second daisy in its cyclic order. Start at the new exterior, move the extra petal past only two central exteriors, and match the actual cyclic PC daisy. All individual inputs, seven internal interchanges, the internal cyclic conjugation and complete nonabelian kernels are retained. Four exterior twists remain visible, then combine and undergo the separate fourth-power sphere cap with +I lift.

I. The fixed-point boundary chart

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before The fixed-point boundary chart

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.



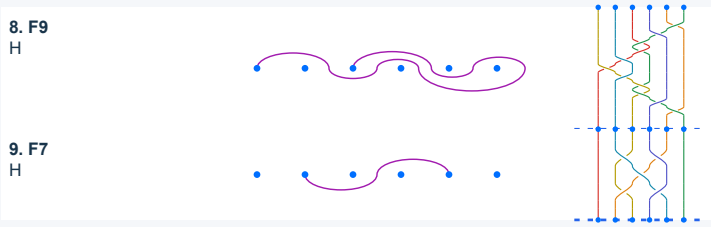
I - step 001

Change the working ordering by an inverse Hurwitz move

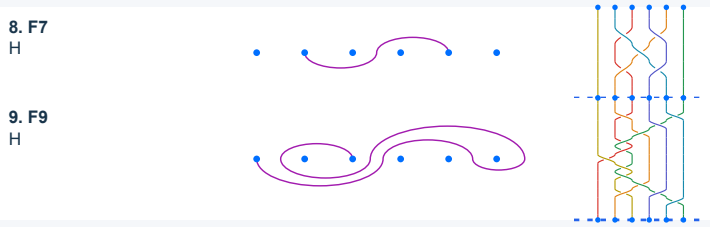
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



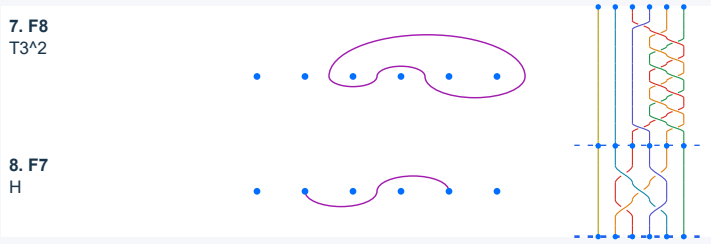
I - step 002

Change the working ordering by an inverse Hurwitz move

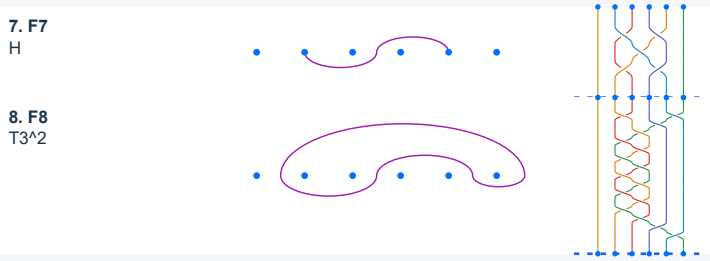
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



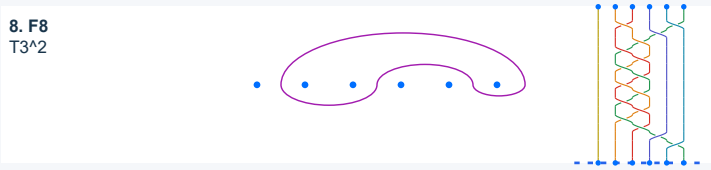
I - step 003

Split F8 into ordinary roots

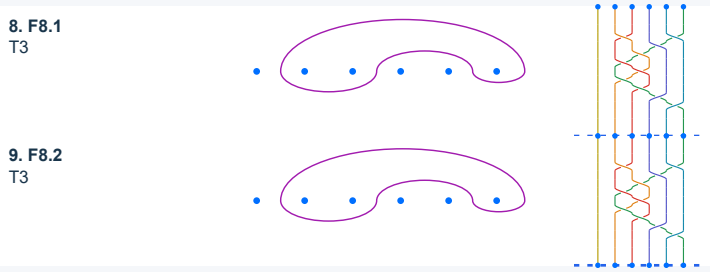
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



I - step 004

Split F10 into ordinary roots

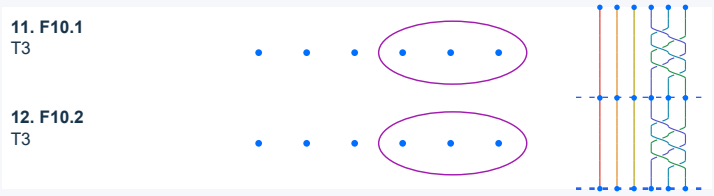
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



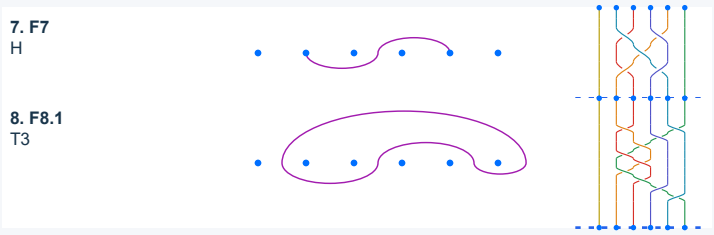
I - step 005

F8.1 up past F7

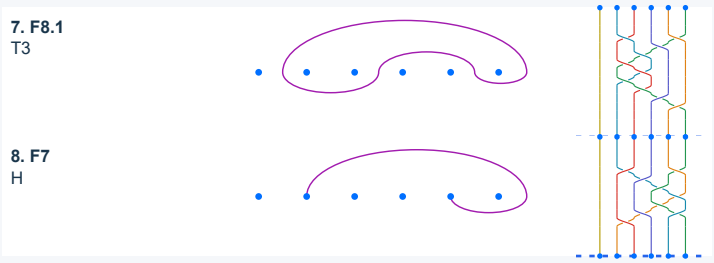
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



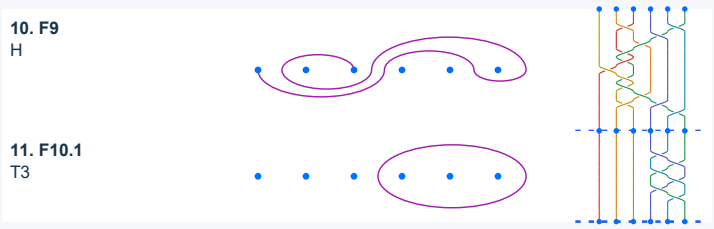
I - step 006

F10.1 up past F9

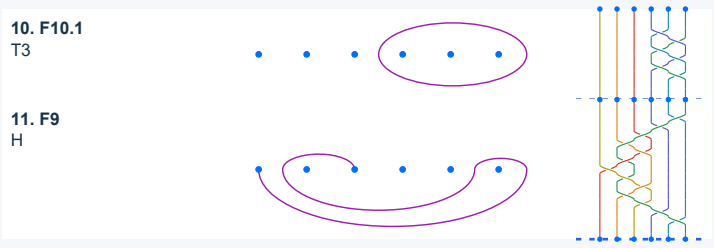
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



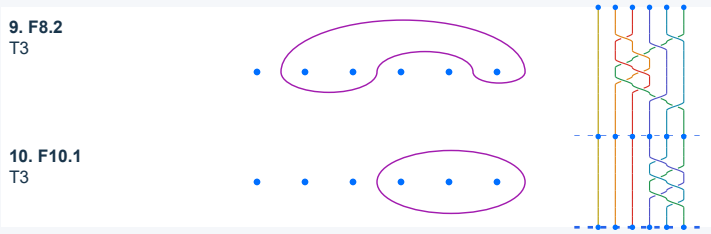
I - step 007

F10.1 up past F8.2

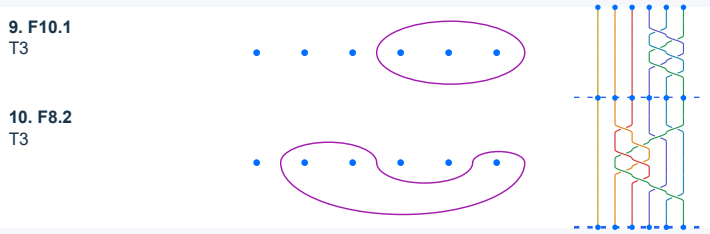
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



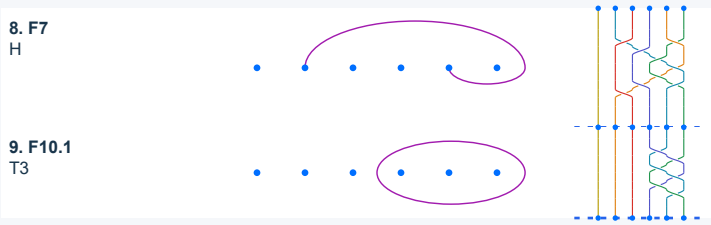
I - step 008

F10.1 up past F7

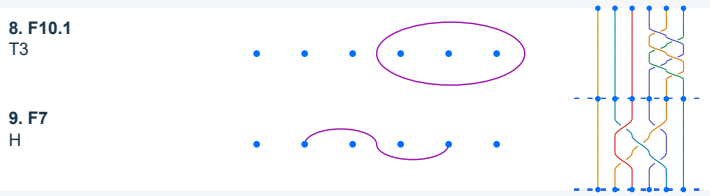
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



I - step 009

Use the exact disk presentation of the transported F7 arc

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



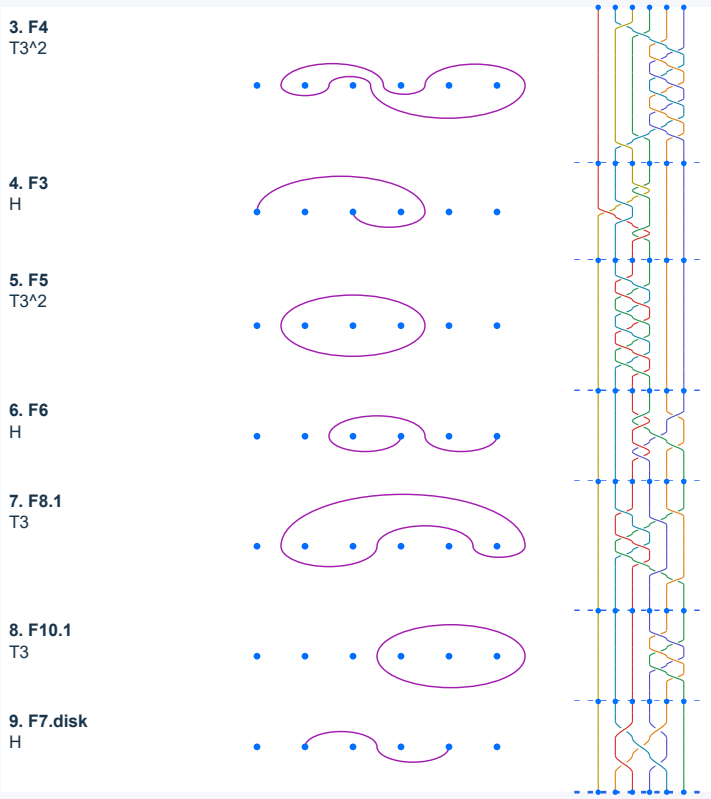
I - step 010

Carry F7 up through SIX factors; keep the original F1 square intact

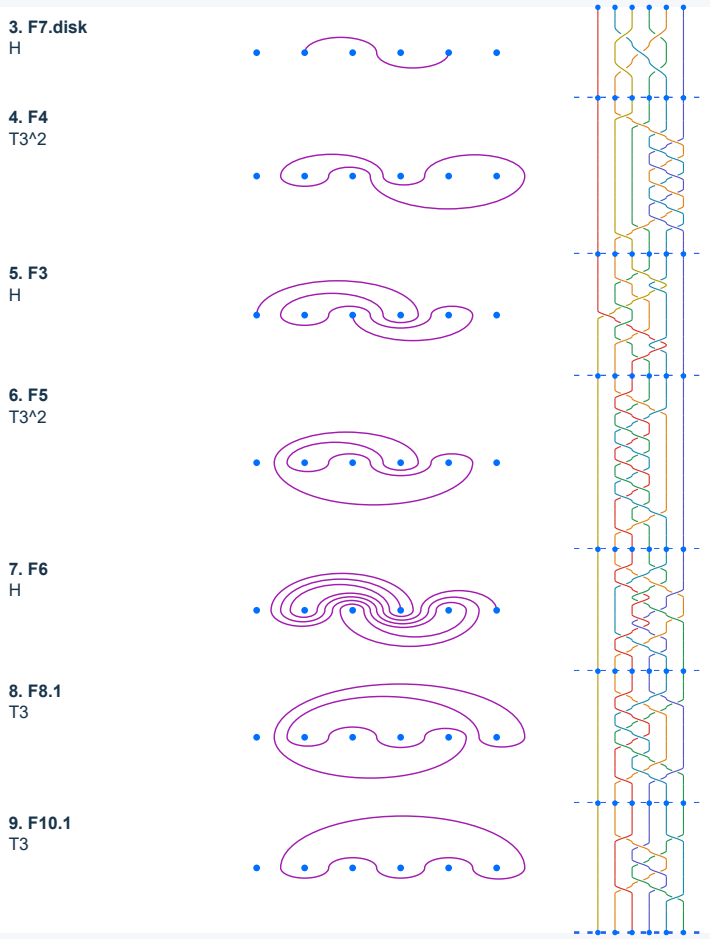
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



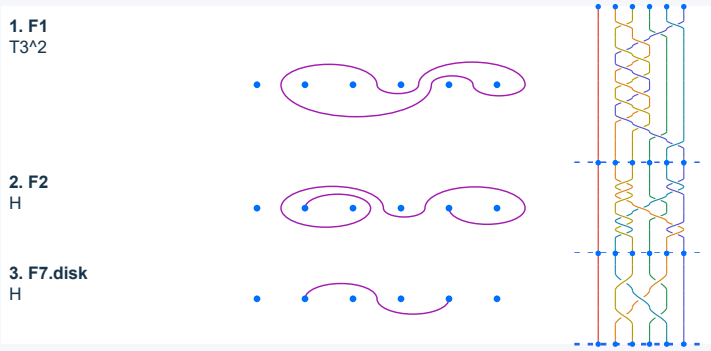
I - step 011

SPHERE LANTERN: keep F1 squared; absorb the two distinct actual half twists

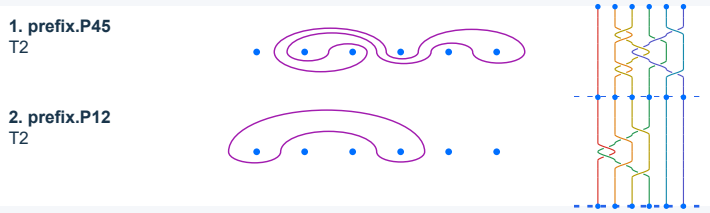
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. Direct sphere lantern: the original F1 square stays intact; the two half twists are DISTINCT. Their common chart, full-twist PC parent, framing balance and full sphere discrepancy are retained. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



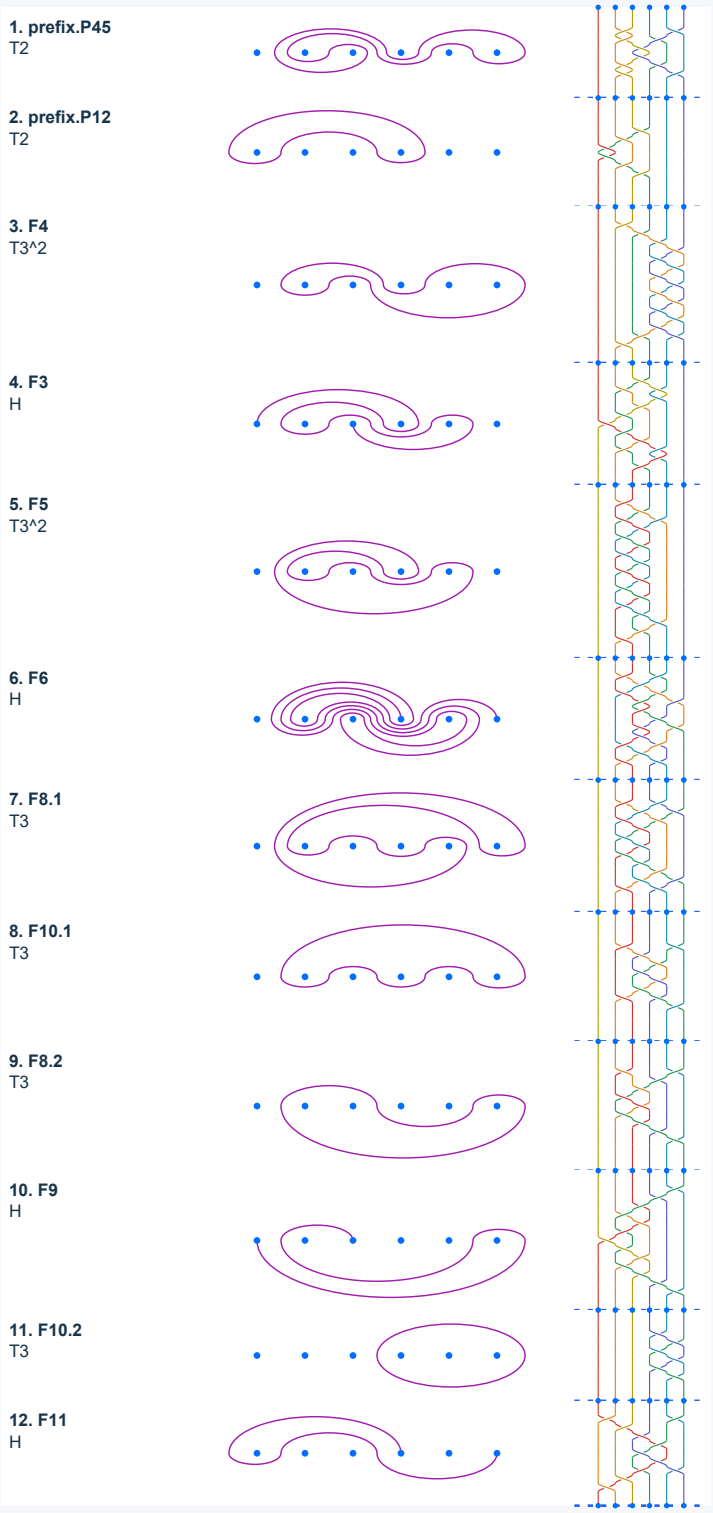
I - step 012

Global frame: fixed working point 2 becomes displayed point 6

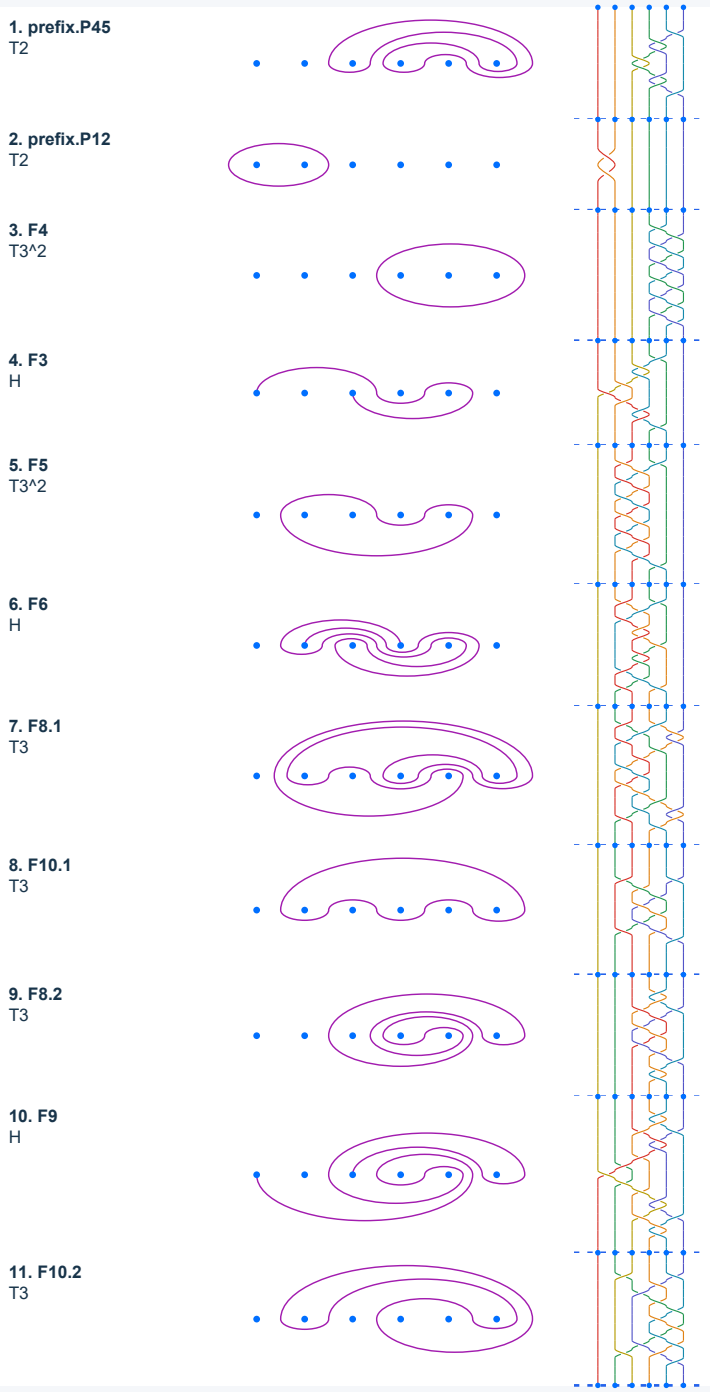
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



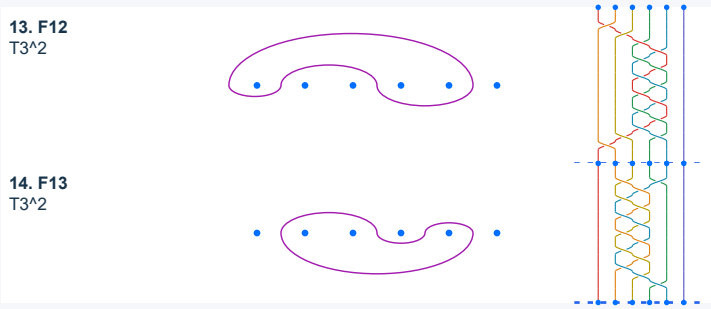
I - step 012 - continued

Global frame: fixed working point 2 becomes displayed point 6

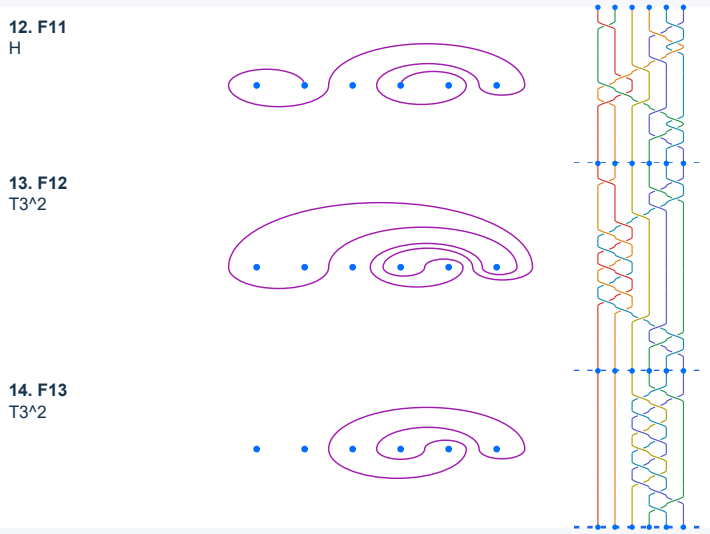
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



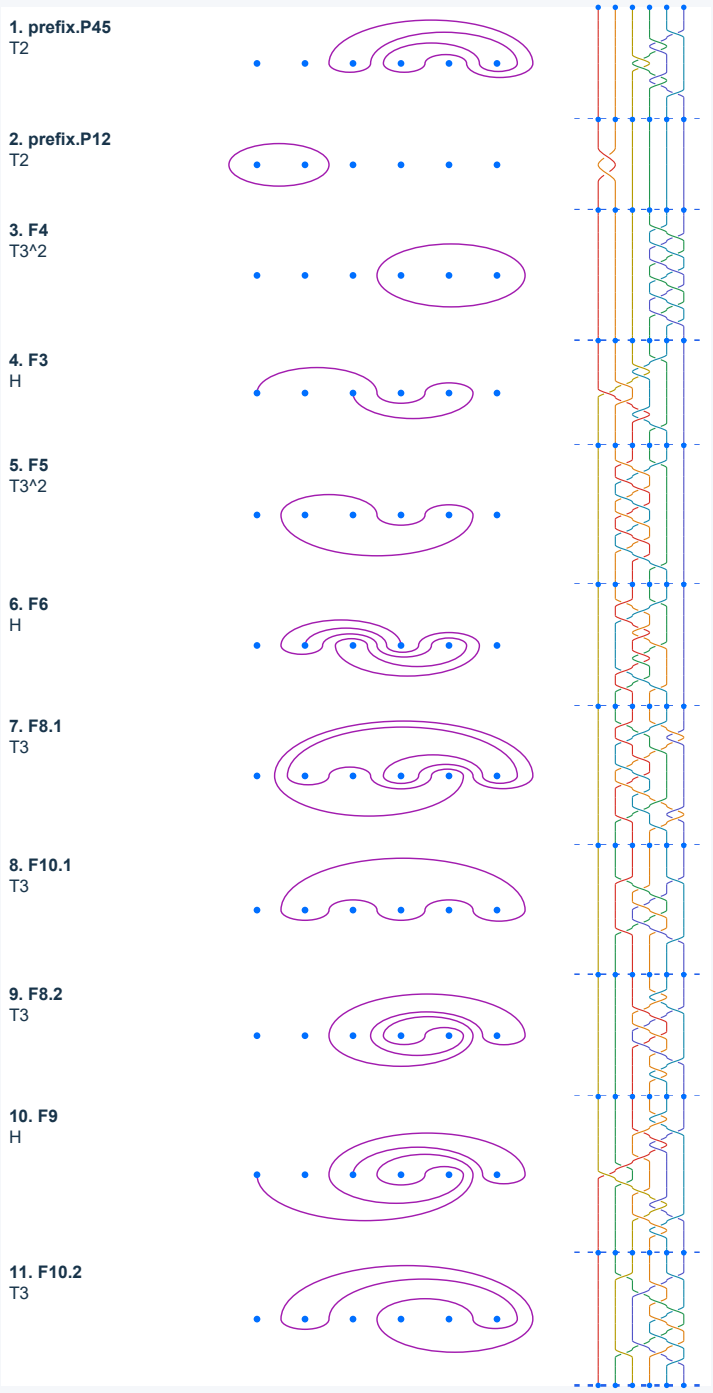
I - step 013

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

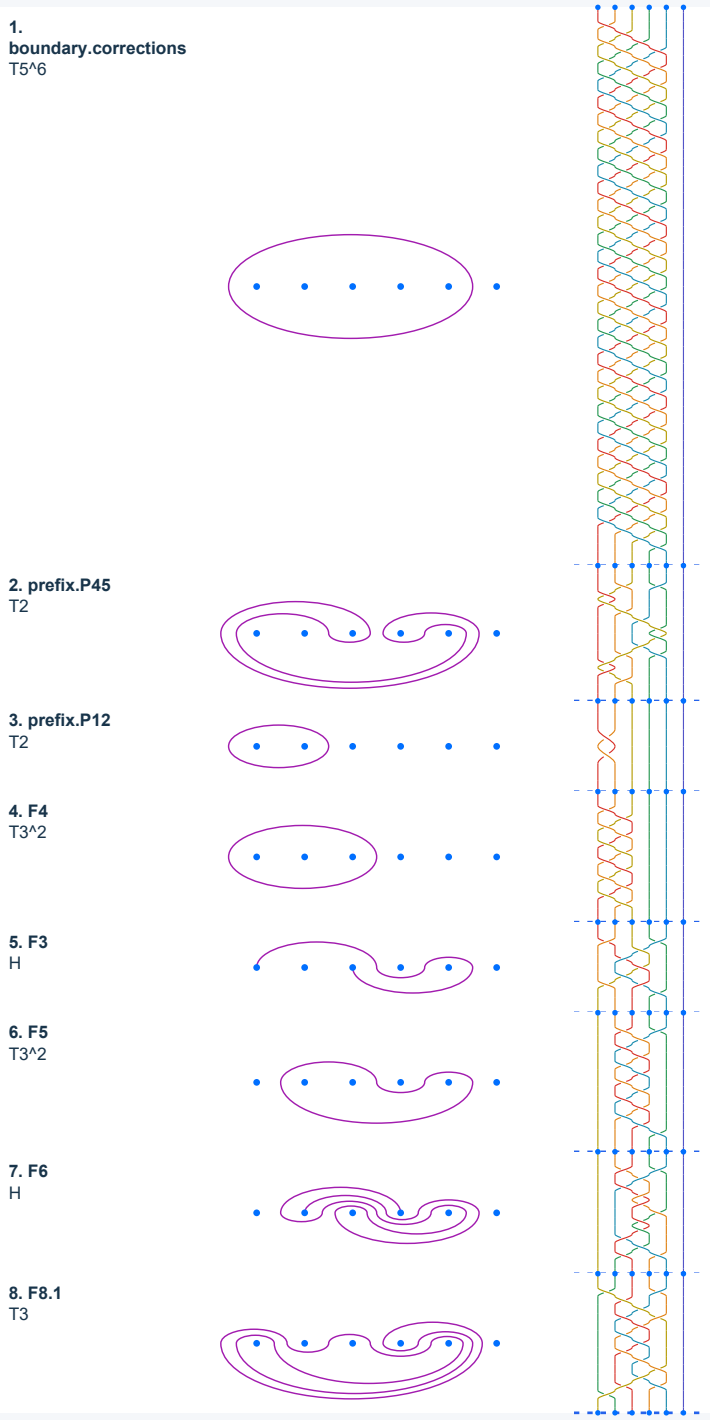
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



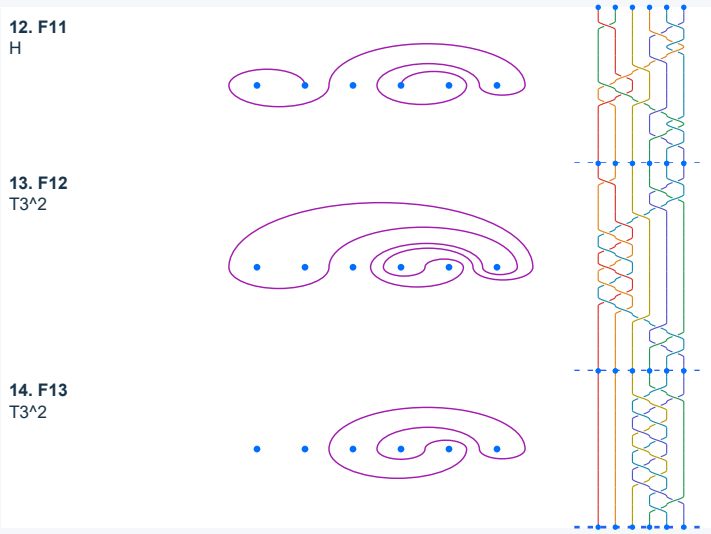
I - step 013 - continued

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

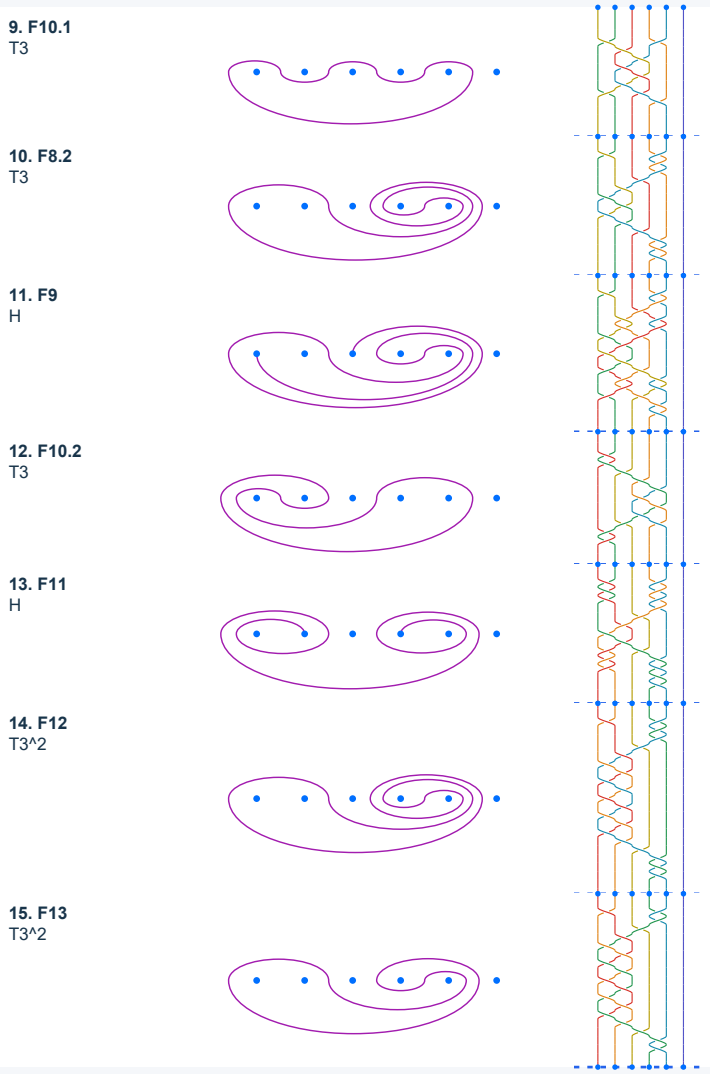
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



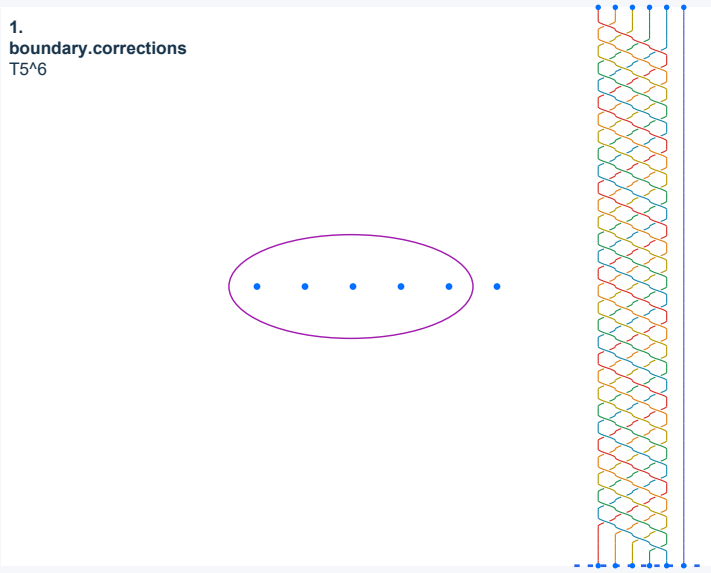
I - step 014

SPHERE MOVE: cap the SIX framing corrections

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

II. Existing roots remove all half twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

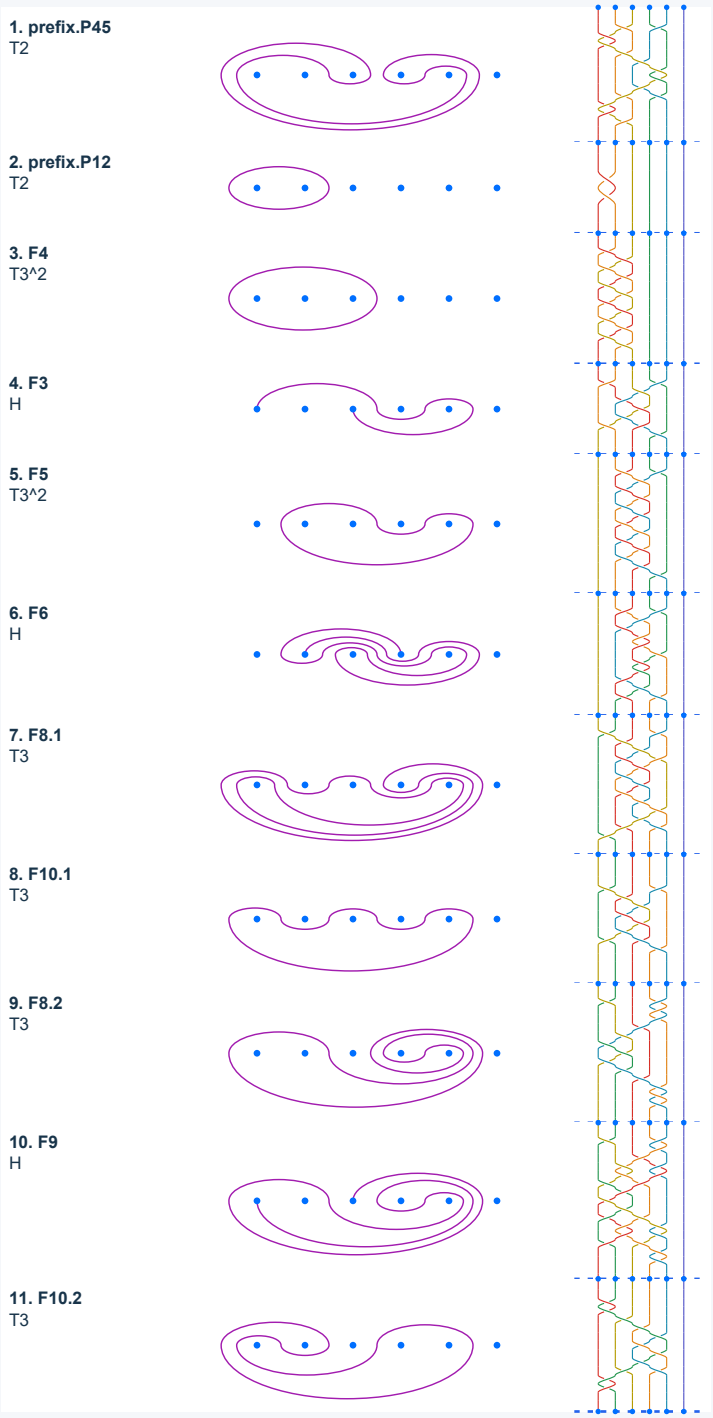
II - step 015

Cyclic start at the WHOLE original124 square

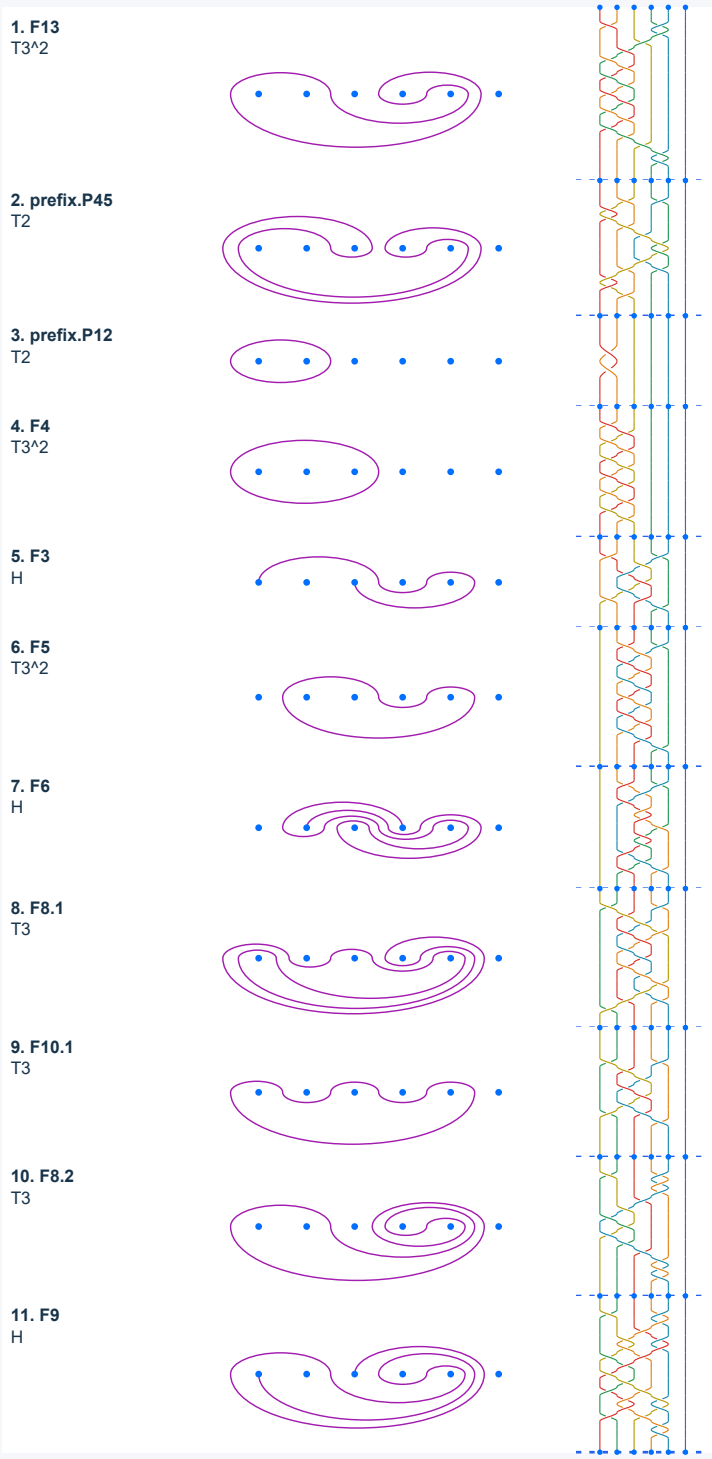
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



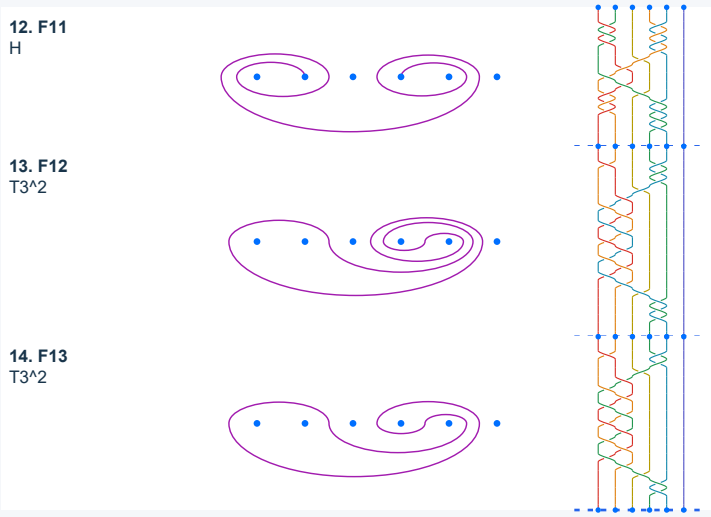
II - step 015 - continued

Cyclic start at the WHOLE original124 square

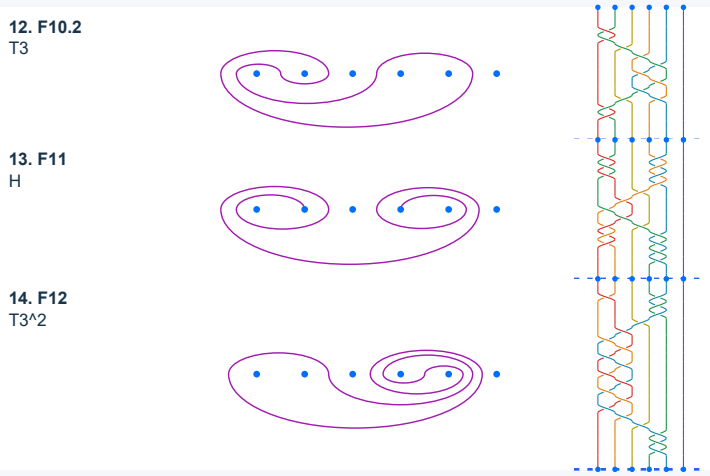
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



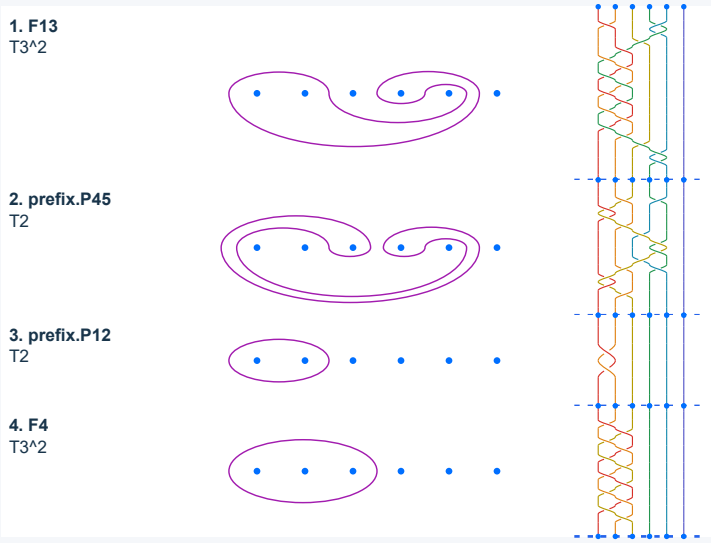
II - step 016

Powered cabled PC lantern keeps both root squares intact and absorbs12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 10 below. Only the changed interval is shown. Powered cabled PC lantern: both root squares enter intact and the actual inner12 is retained in its output square. All individual matches and the expanded local checks are certified.

Before



After



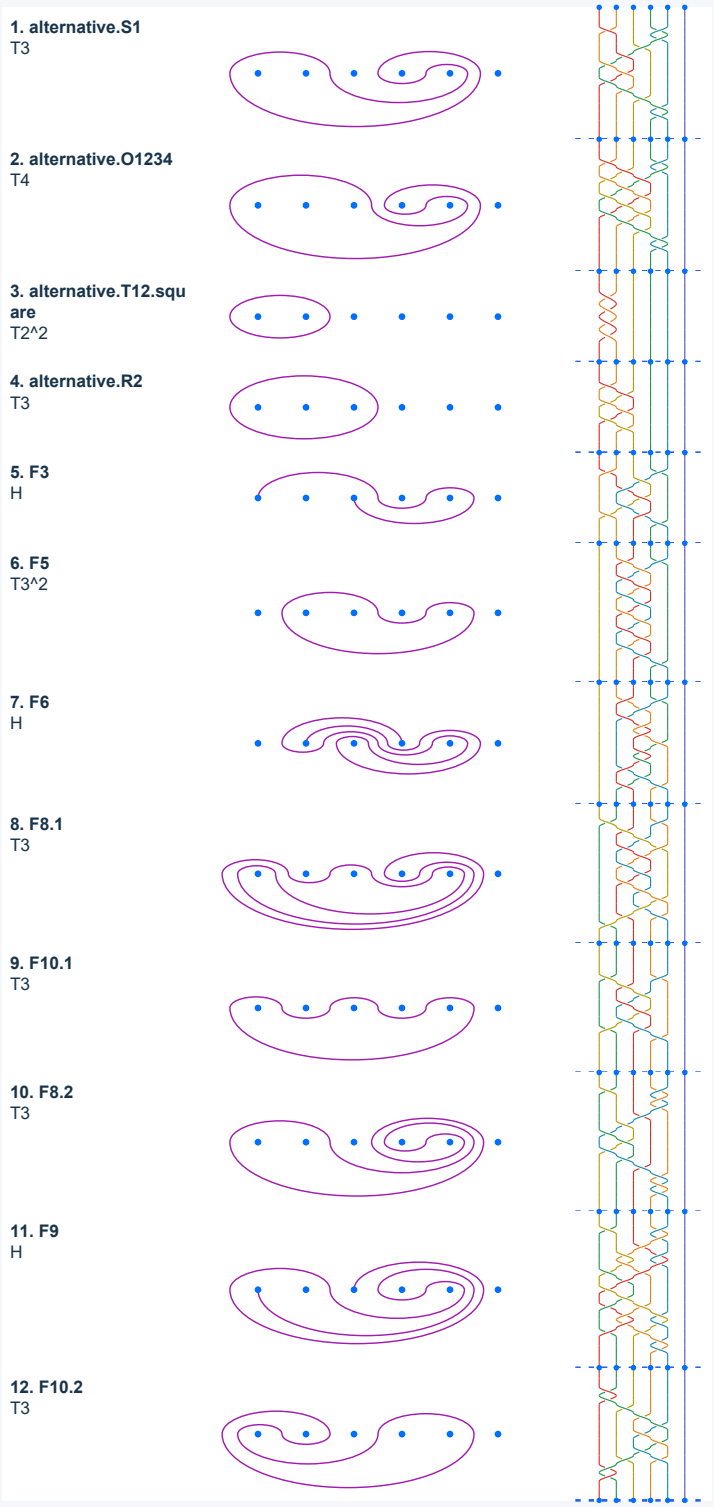
II - step 017

Cyclic start after the retained actual124 unit

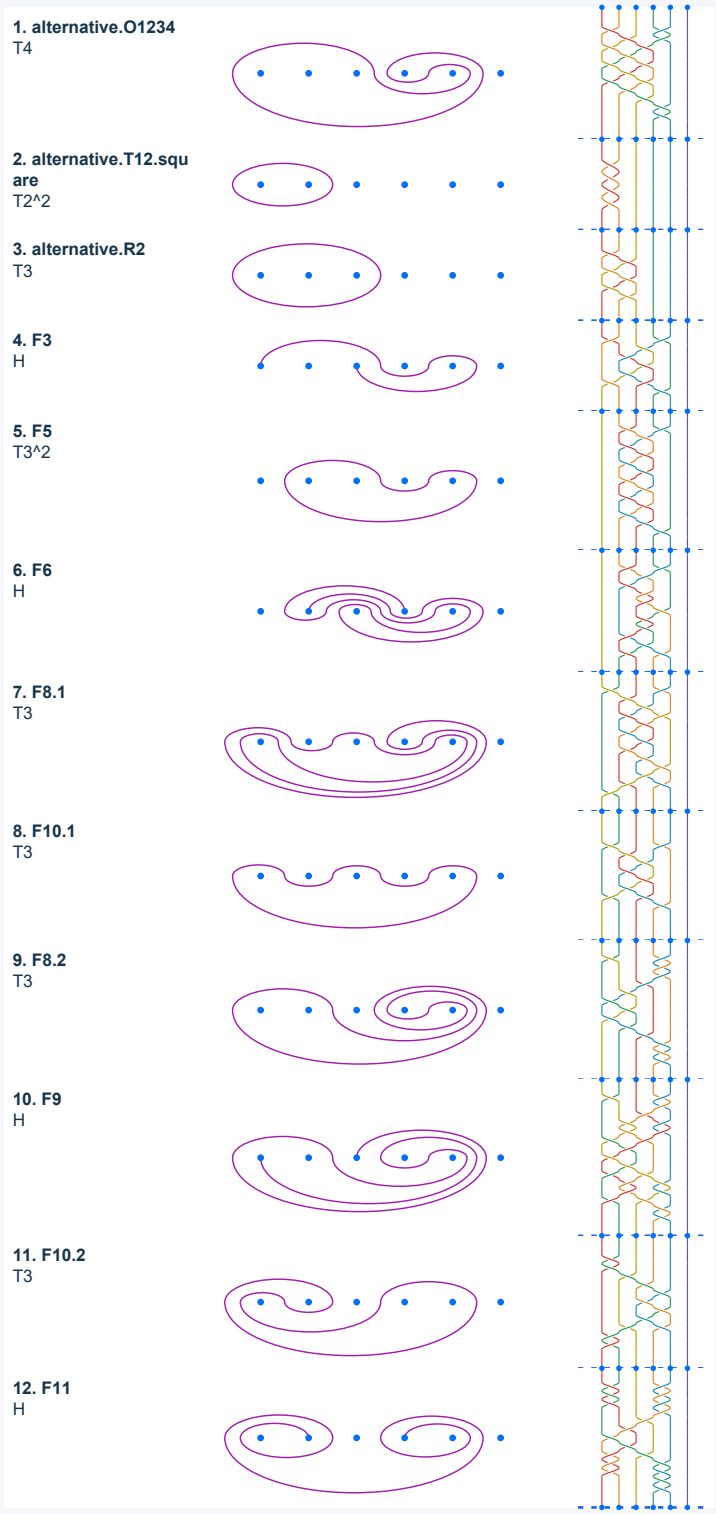
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



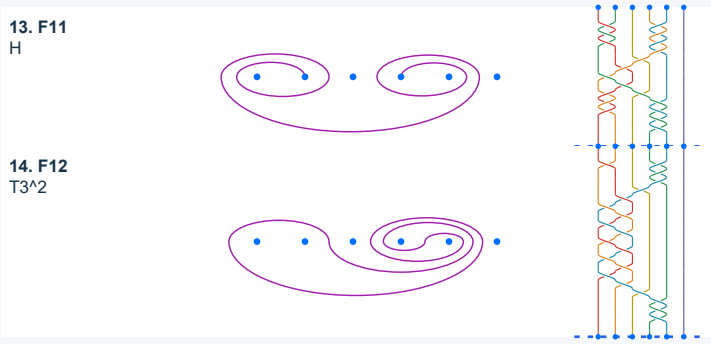
II - step 017 - continued

Cyclic start after the retained actual124 unit

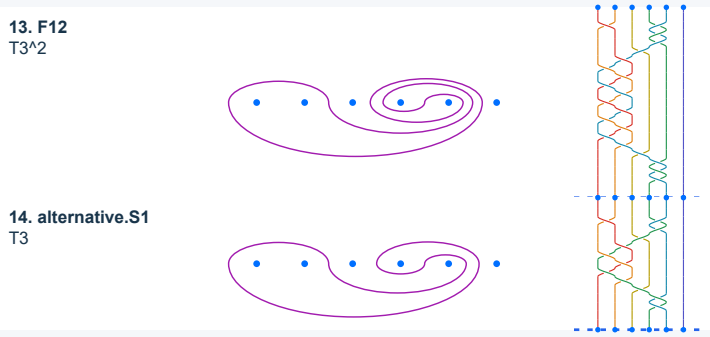
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



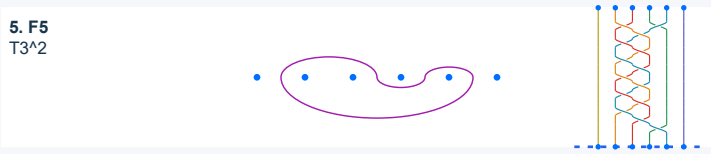
II - step 018

Split F5 into its two positive units

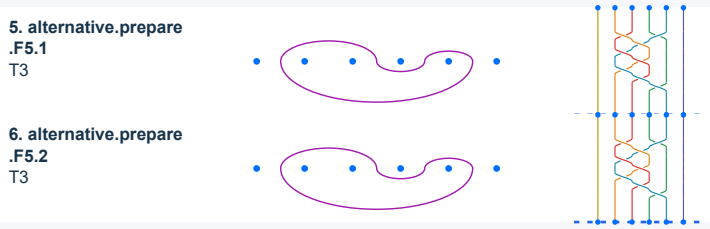
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



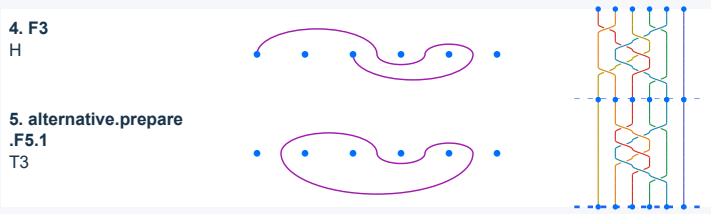
II - step 019

One 235 root up over F3

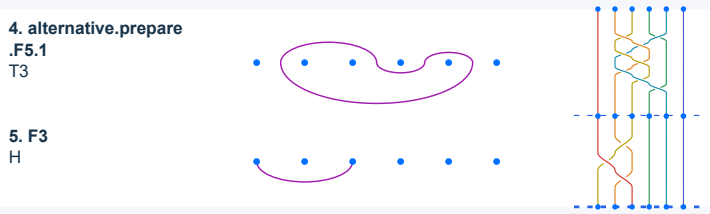
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



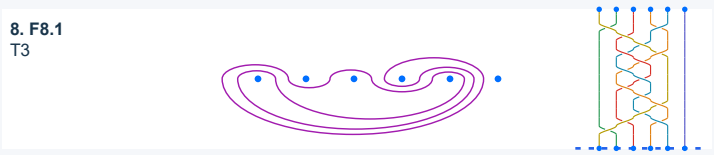
II - step 020

Original 345 daisy supplies actual P35

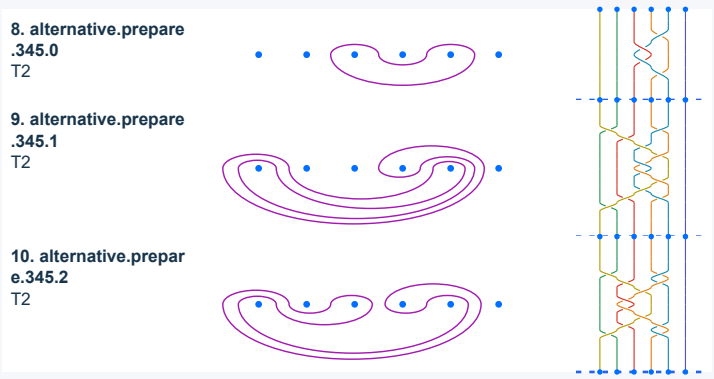
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



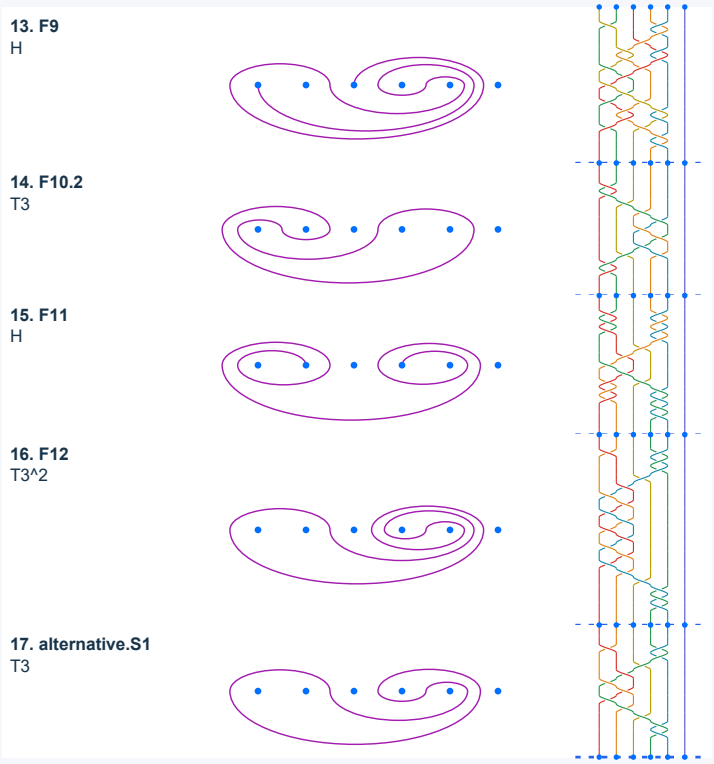
II - step 021

Remaining 124 root up over F9

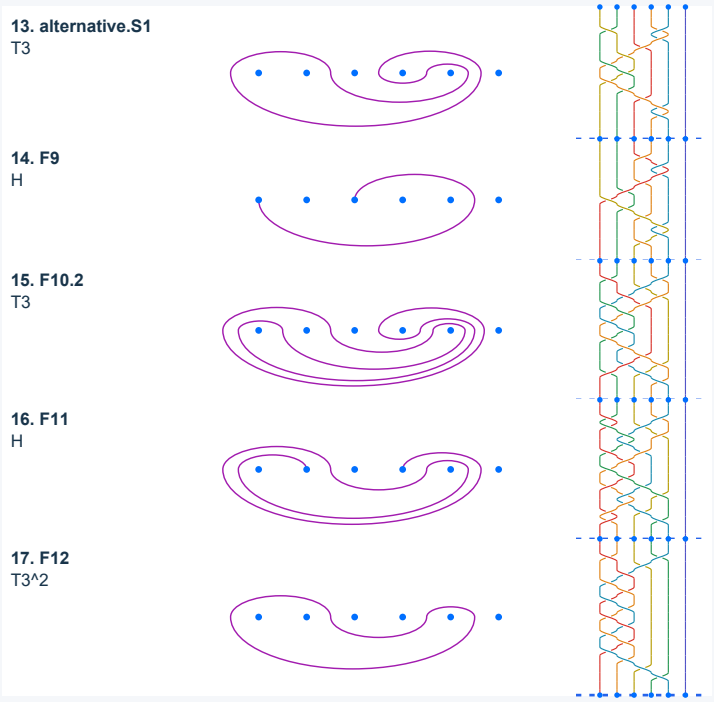
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



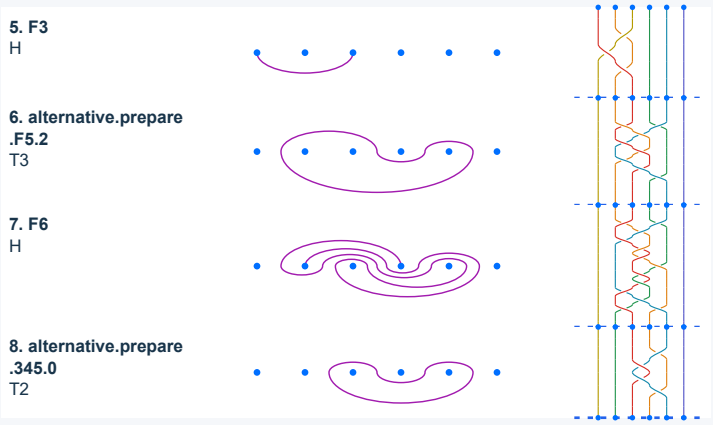
II - step 022

Carry P35 up over F3; a Q14 transport is still needed

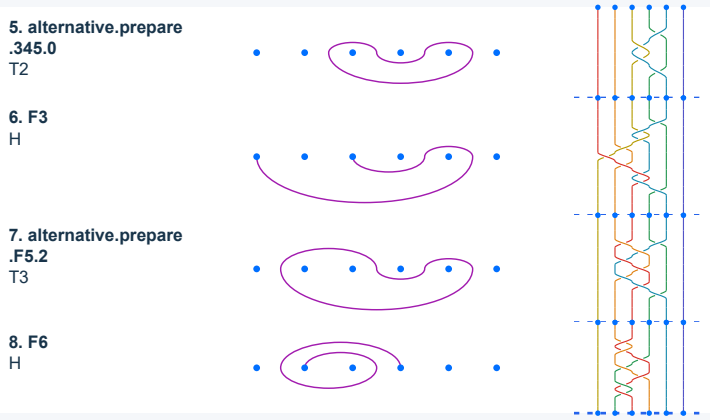
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



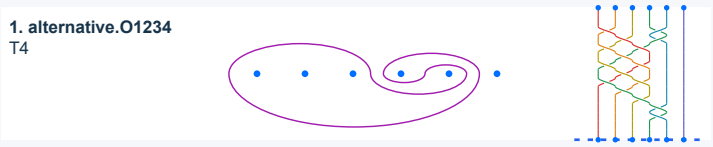
II - step 023

PC daisy lends its actual Q14 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown.

Before



After



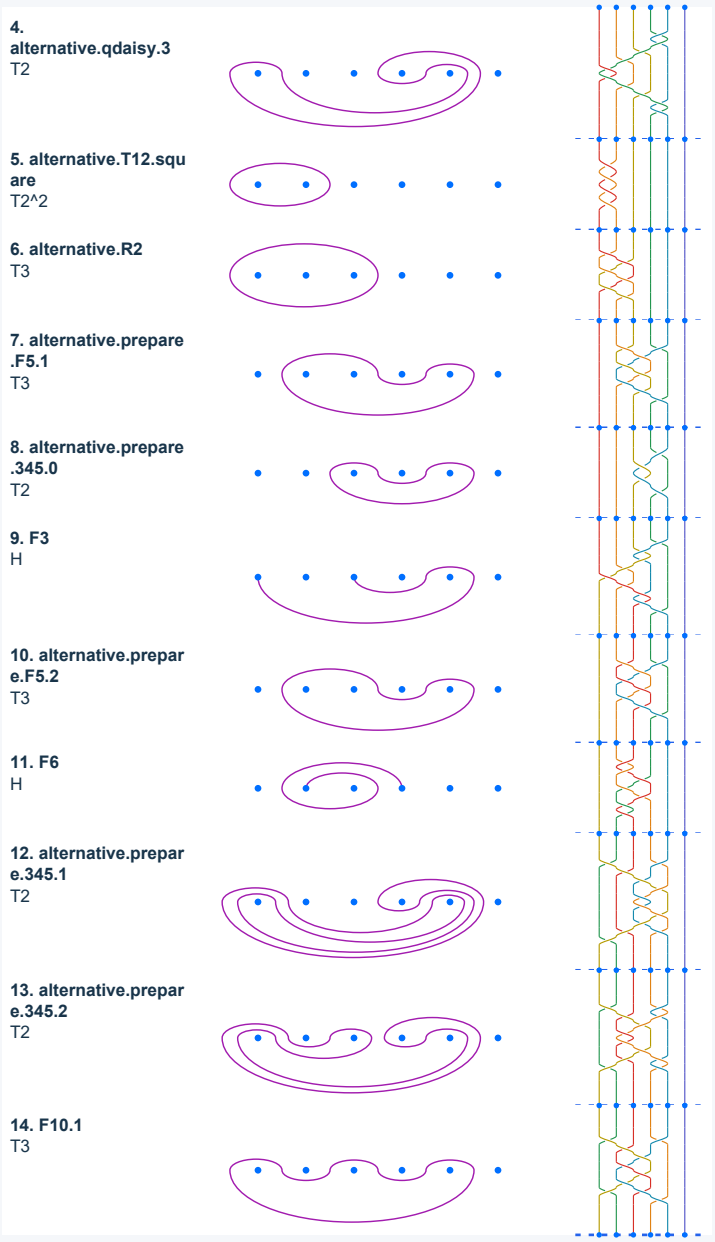
II - step 024

Carry Q14 down across F3 and the remaining 345 petals

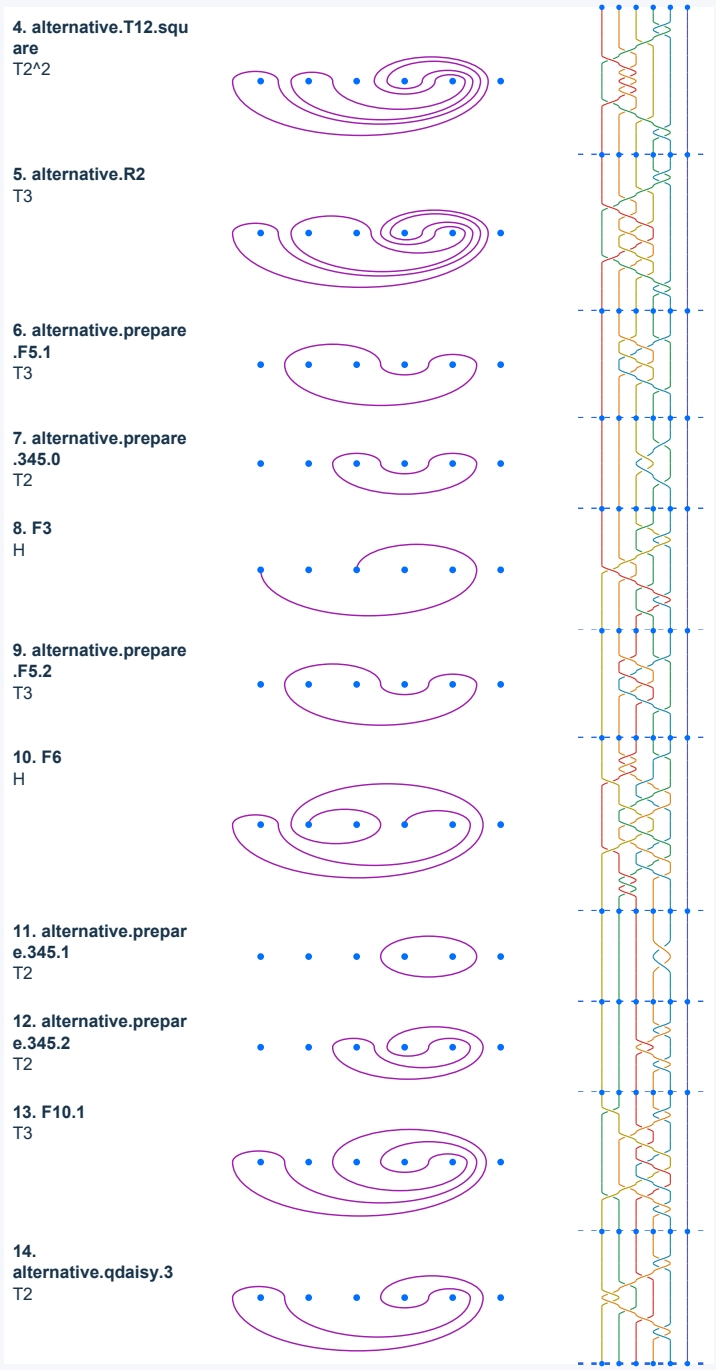
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 10.

Before



After



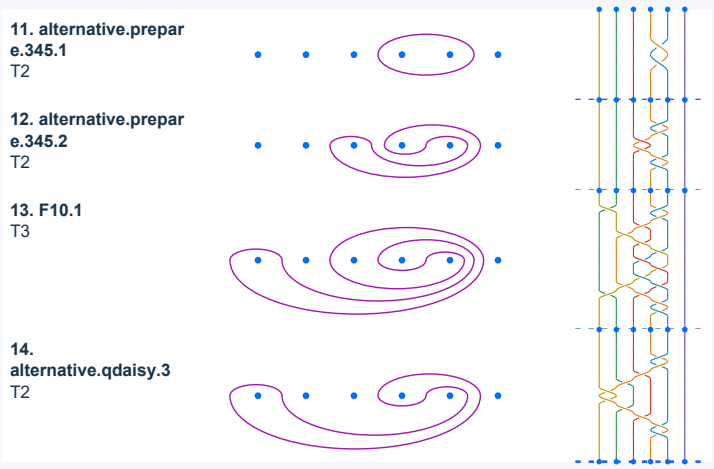
II - step 025

PC daisy contracts the ACTUAL transported petals to O1345

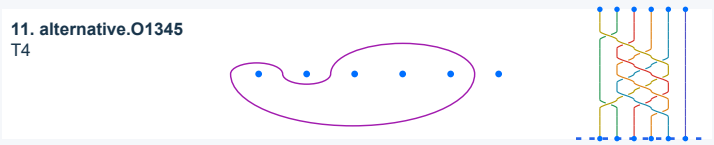
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



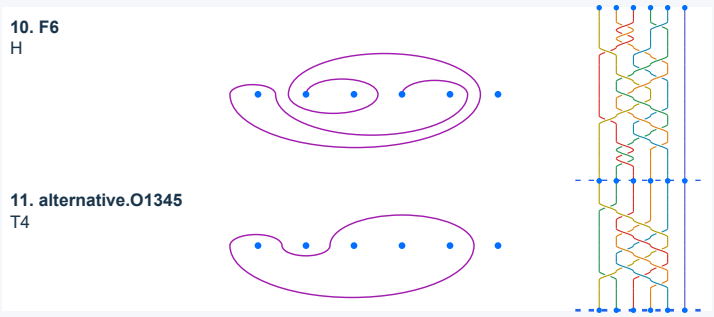
II - step 026

Carry O1345 up across F6 before the matching-half gather

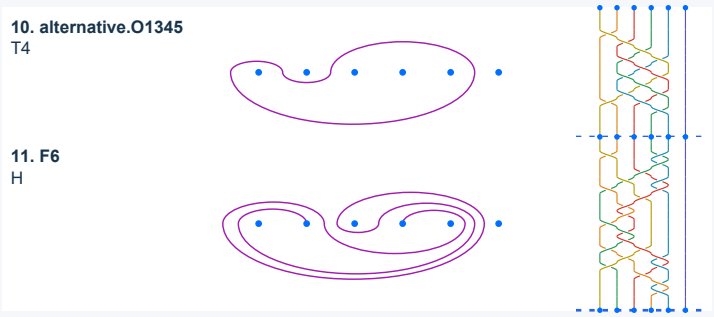
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



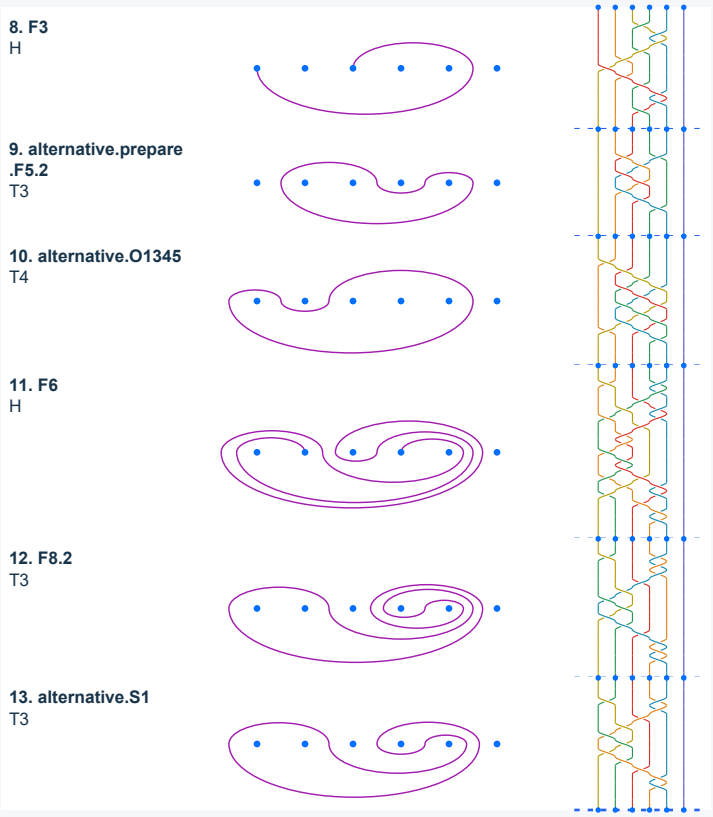
II - step 027

Carry the matched F3 down to F9; F6 unwinds

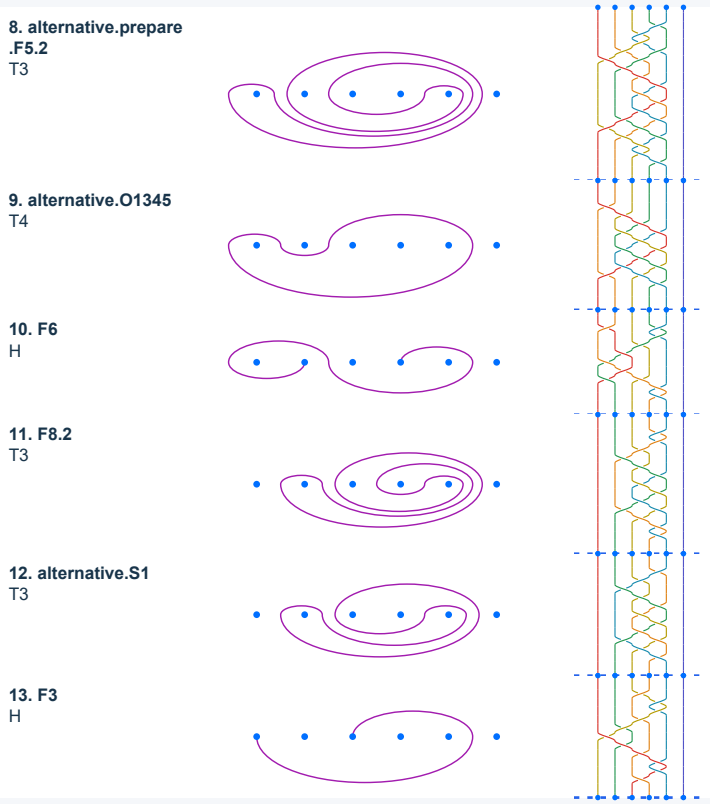
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



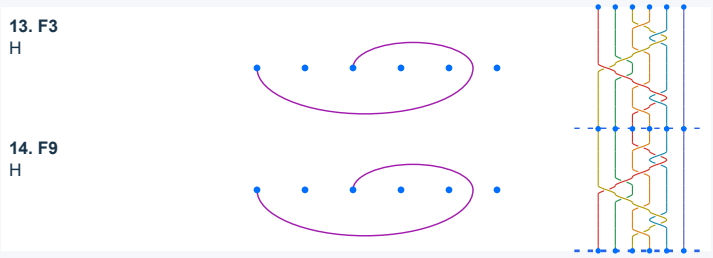
II - step 028

Combine the matching F3 and F9

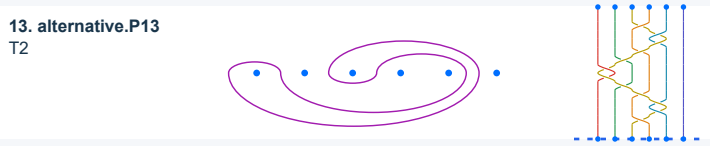
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



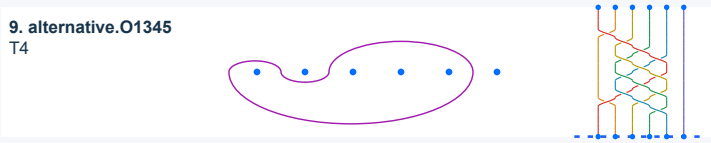
II - step 029

The O1345 daisy returns EXACTLY the borrowed Q14

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



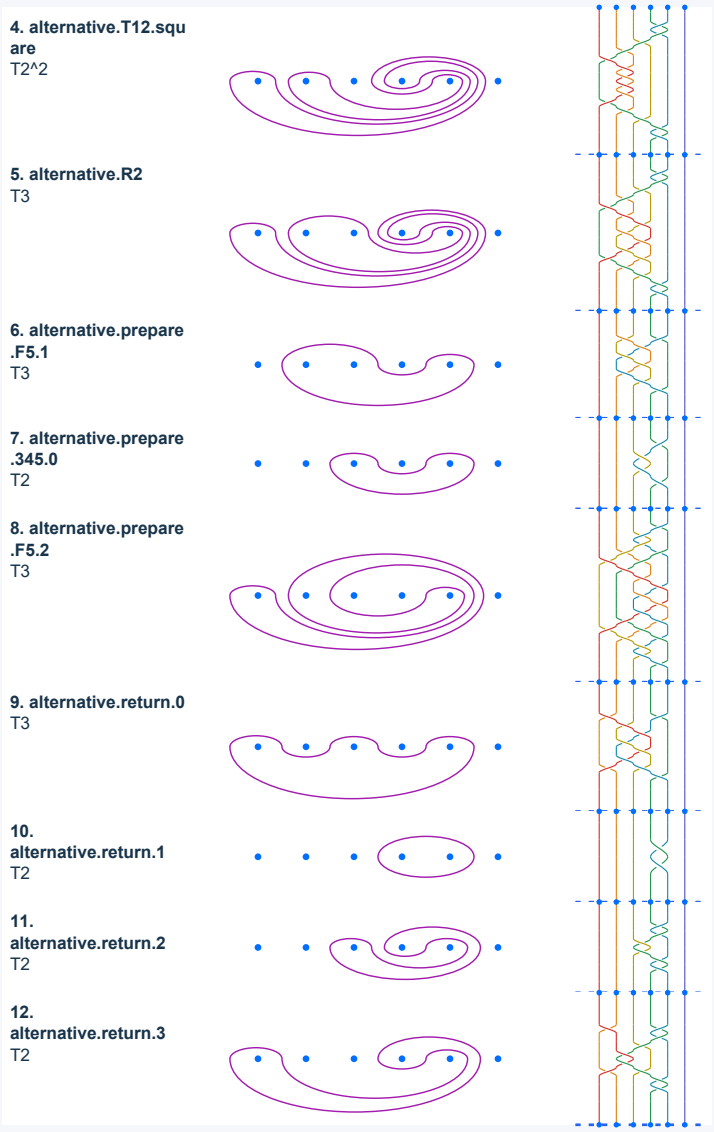
II - step 030

Return Q14 up to restore the round T12 square and T123

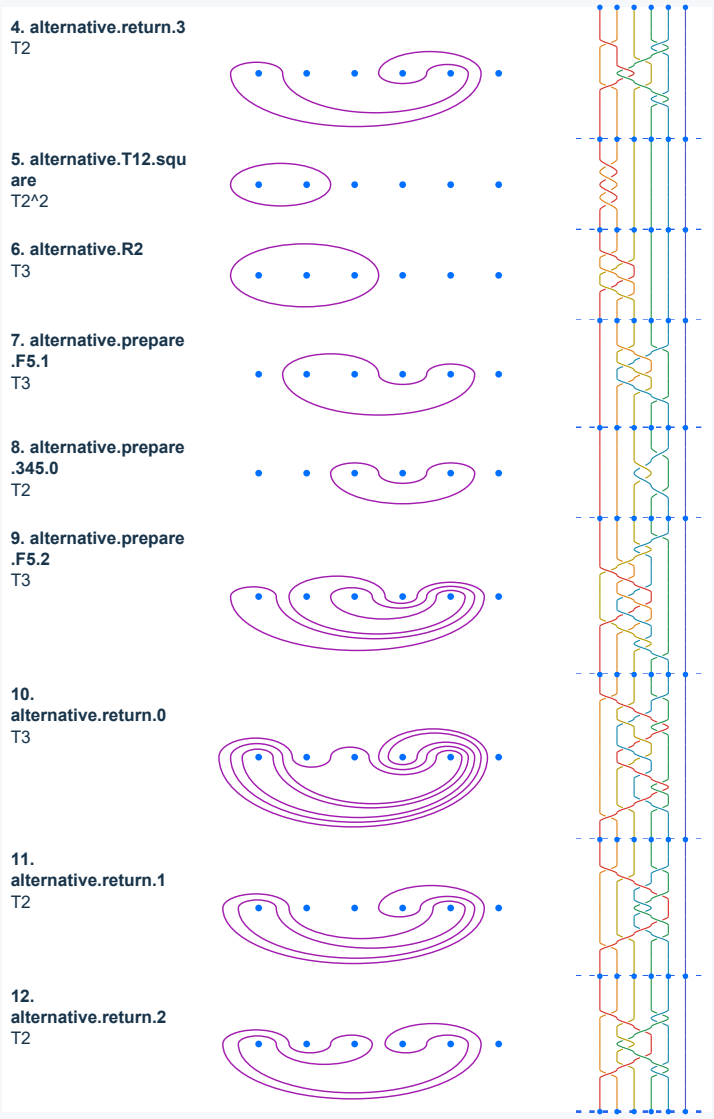
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



II - step 031

Close the original daisy after its actual Q14 returns

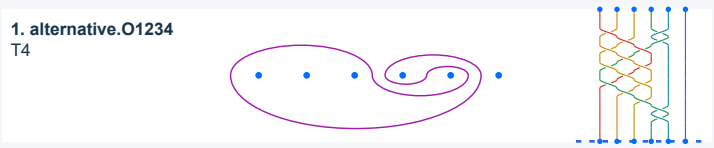
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown.

Before



After



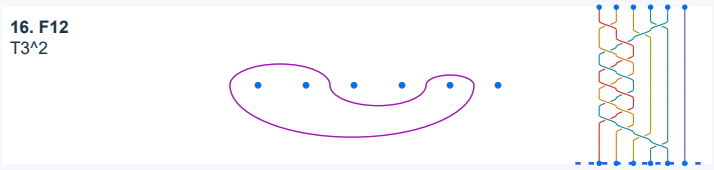
II - step 032

Partial PC daisy exposes the squared root petals EARLY

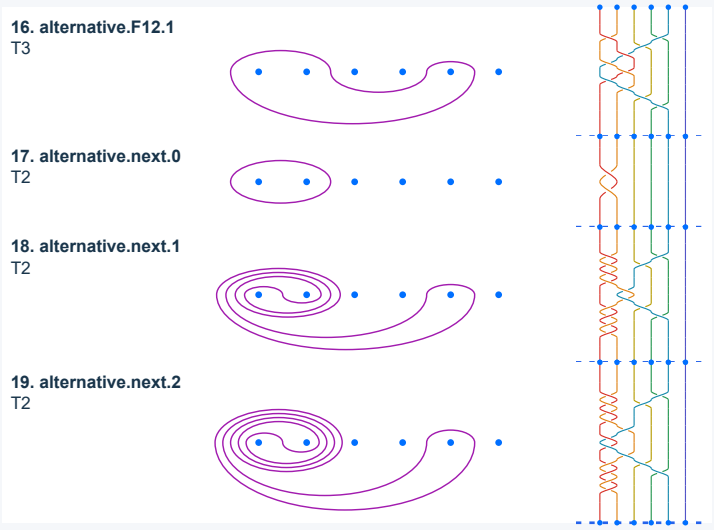
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown. Partial PC daisy: three early petals replace one donor unit; all five later square crossings are retained.

Before



After



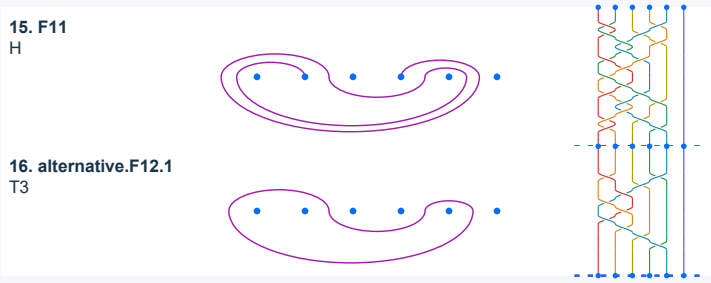
II - step 033

One C12 root shortens F11

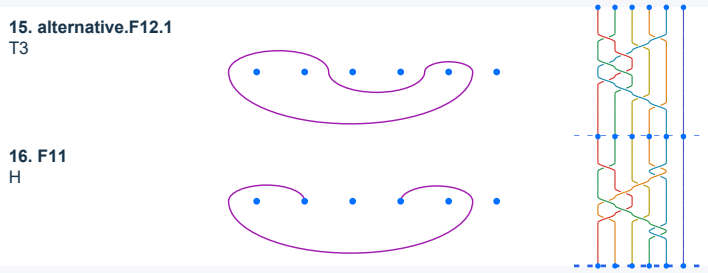
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



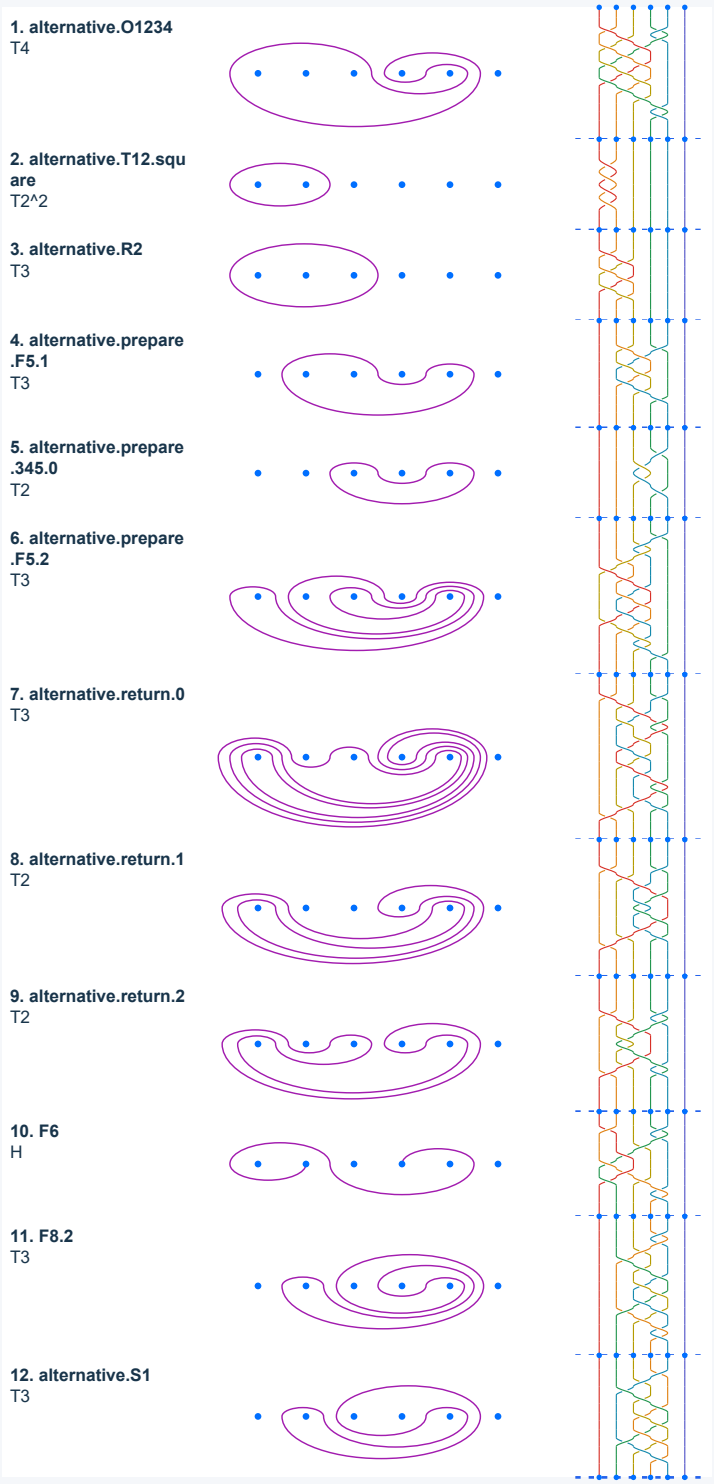
II - step 034

Cyclic start after BOTH the square and its commuting outer

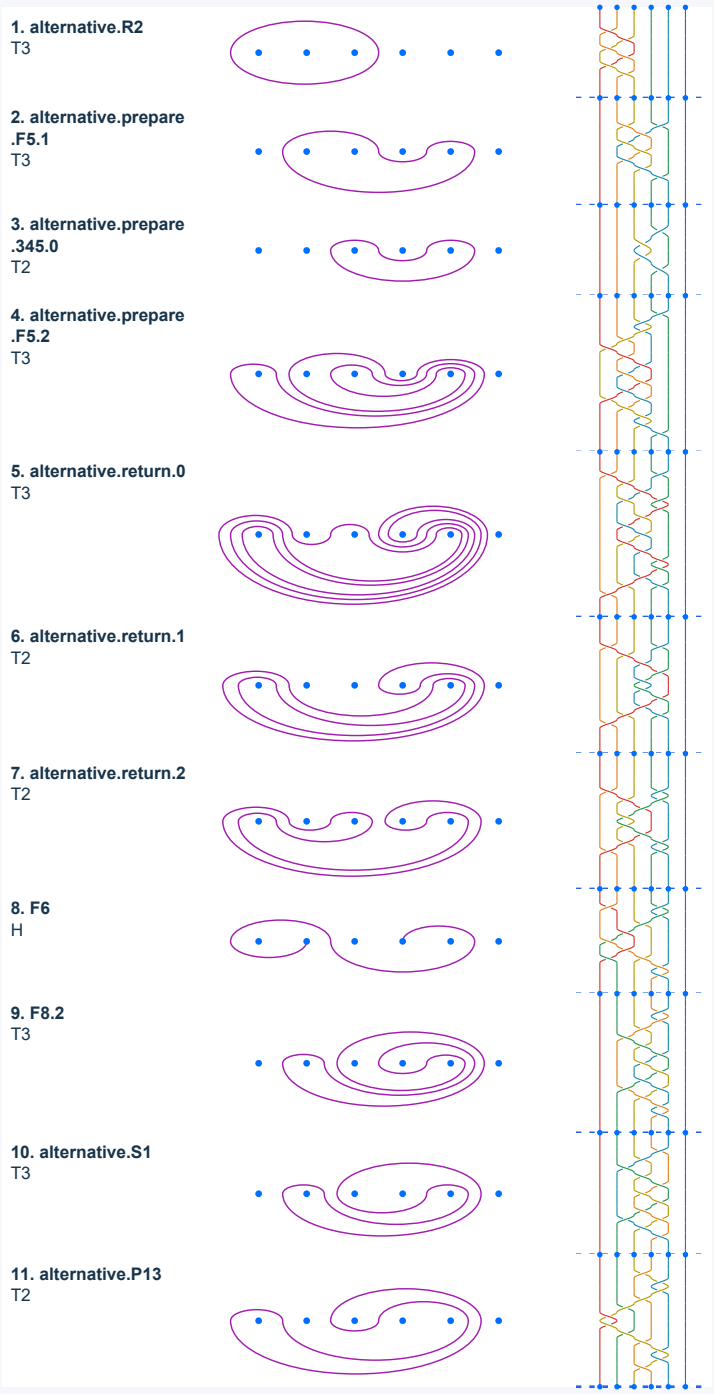
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



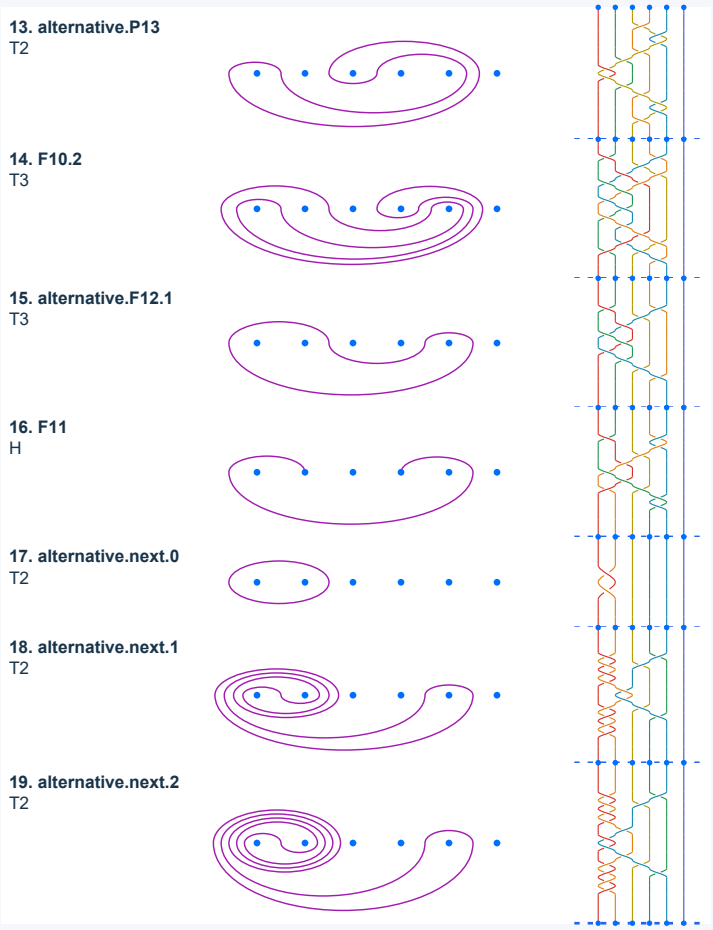
II - step 034 - continued

Cyclic start after BOTH the square and its commuting outer

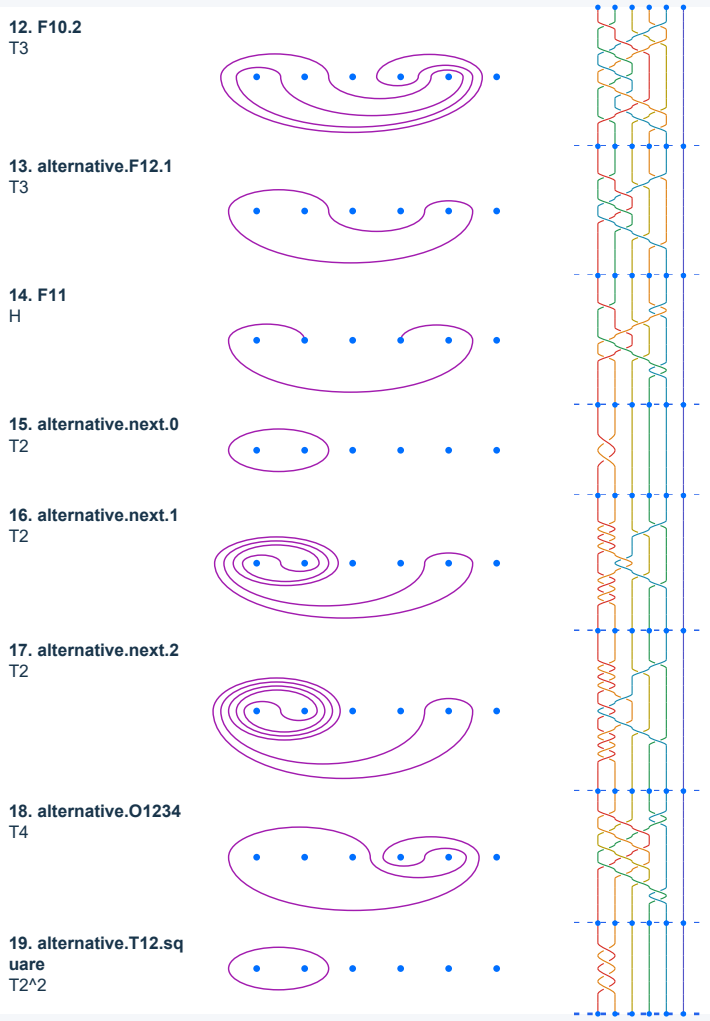
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



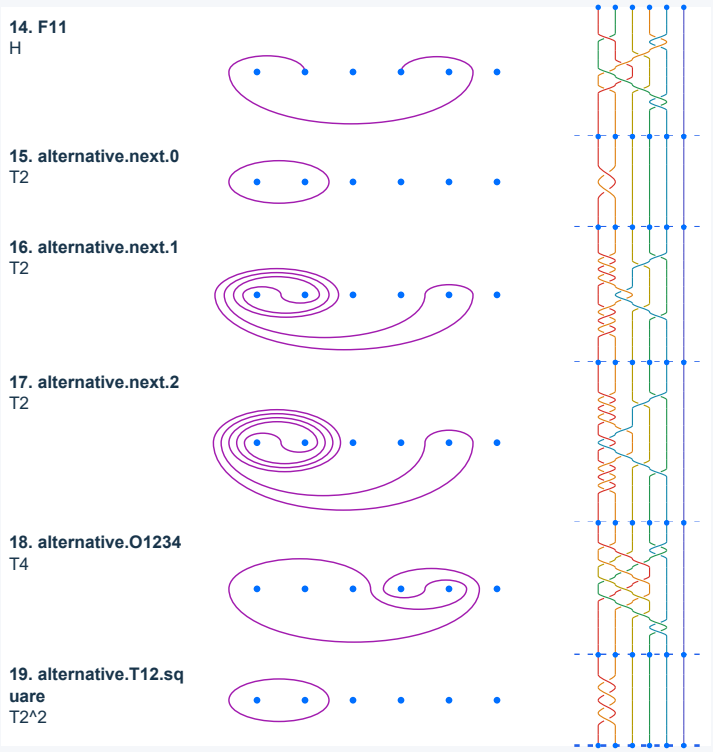
II - step 035

One short-side square drag carries ALL THREE early daisy petals

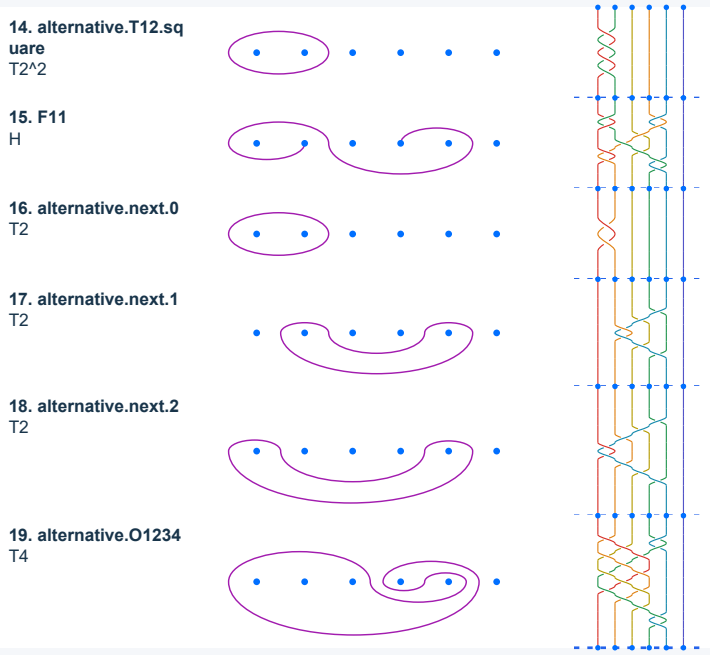
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



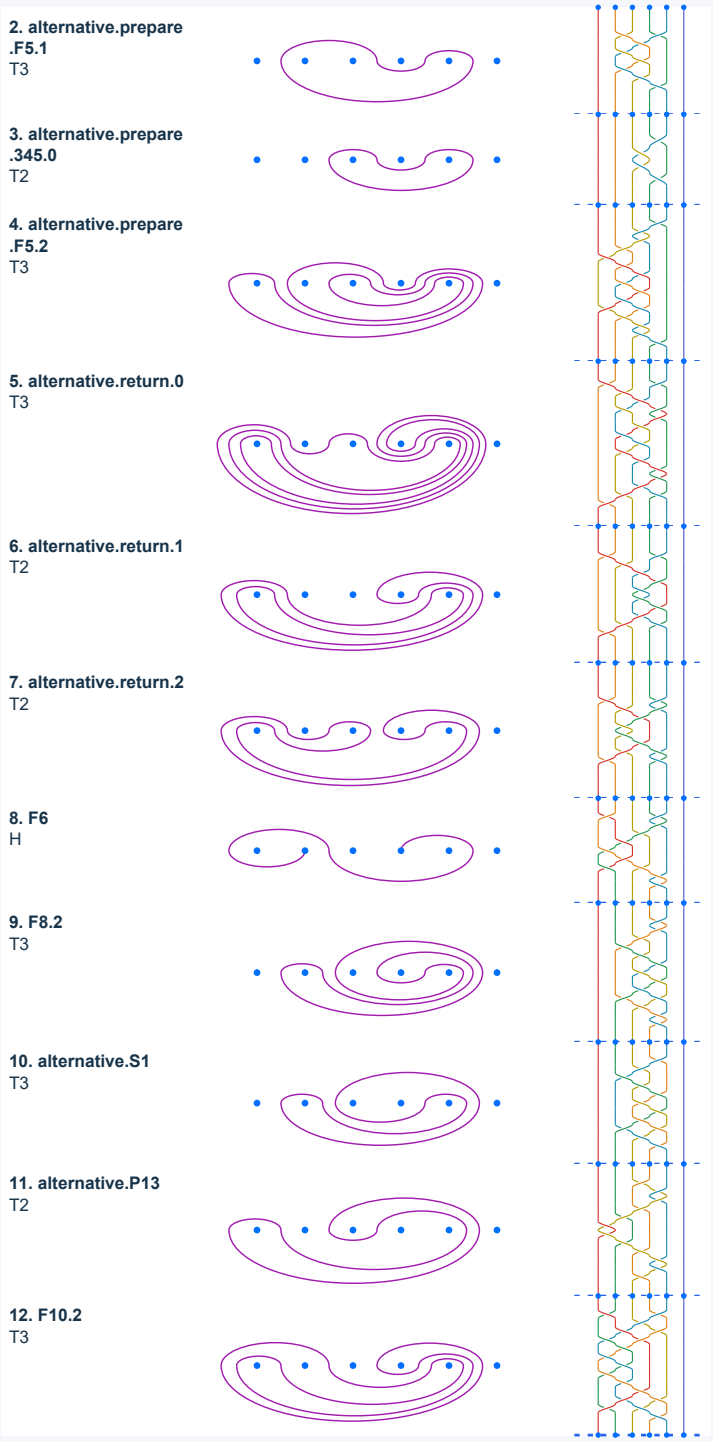
II - step 036

The same composed sigma4 common frame

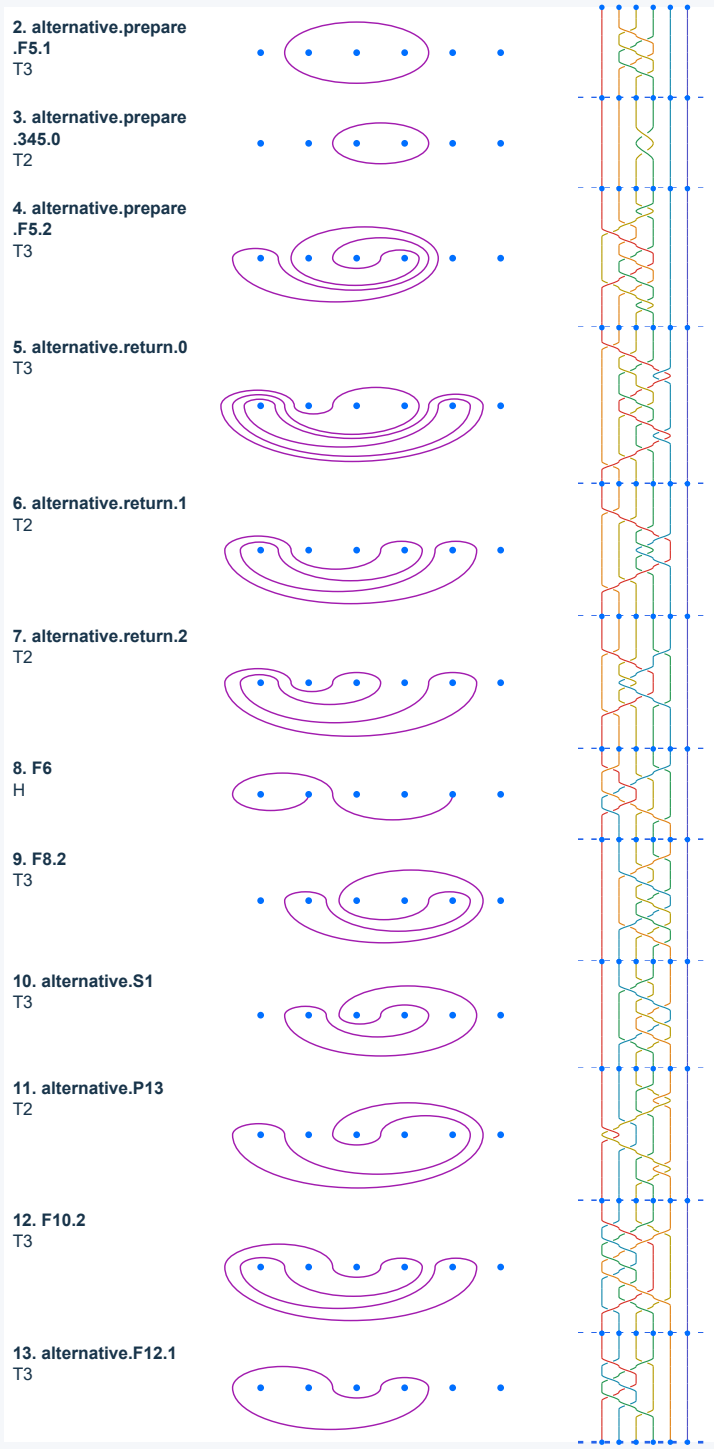
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



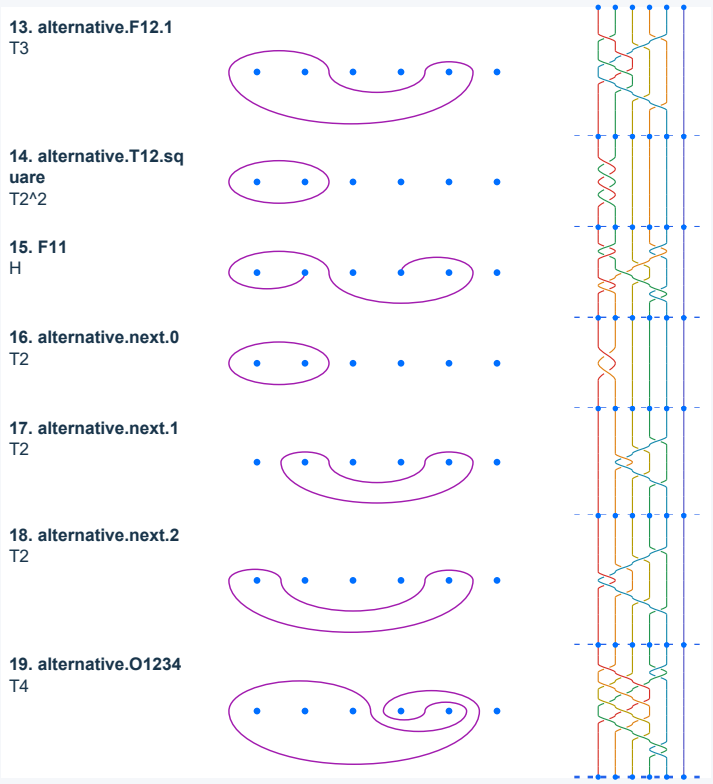
II - step 036 - continued

The same composed sigma4 common frame

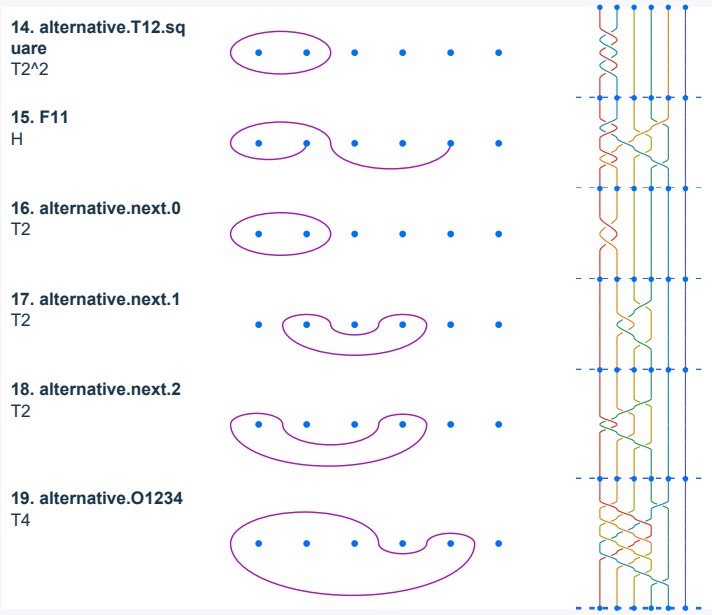
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



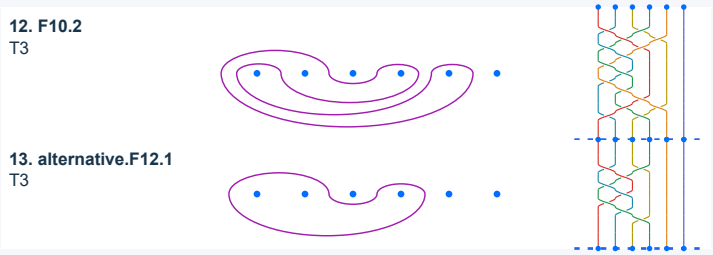
II - step 037

Reorder C12 before the matching-half gather

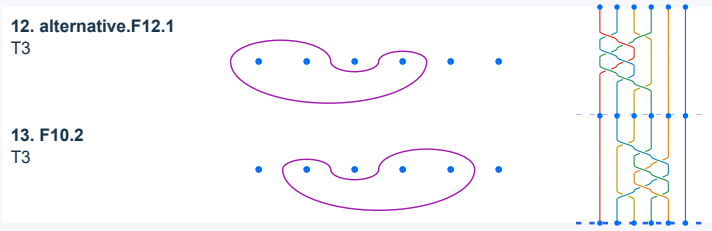
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



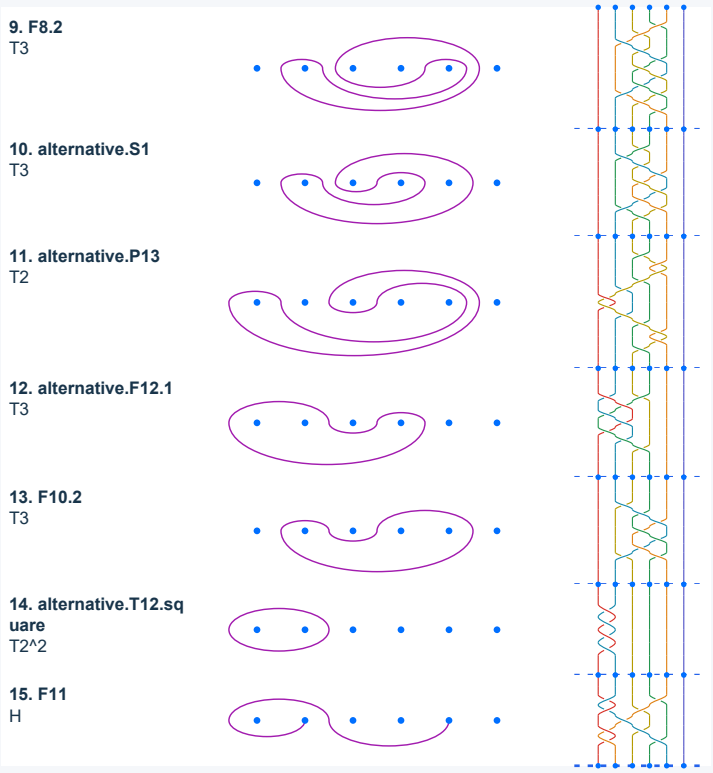
II - step 038

Gather the final matching original halves

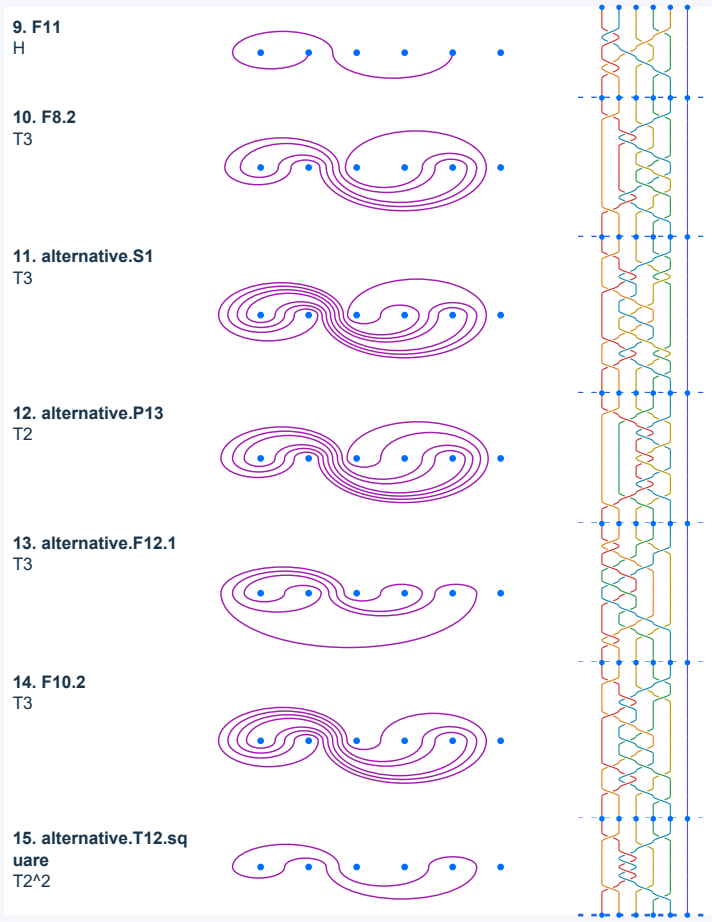
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



III. The first exterior and simpler inside daisy

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

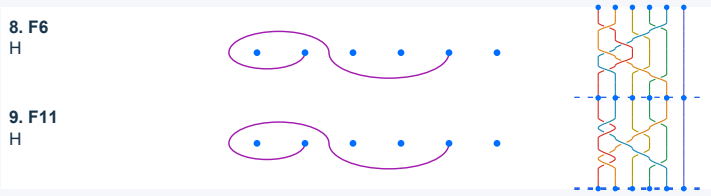
III - step 039

Combine the matching F6 and F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 10 below. Only the changed interval is shown.

Before



After



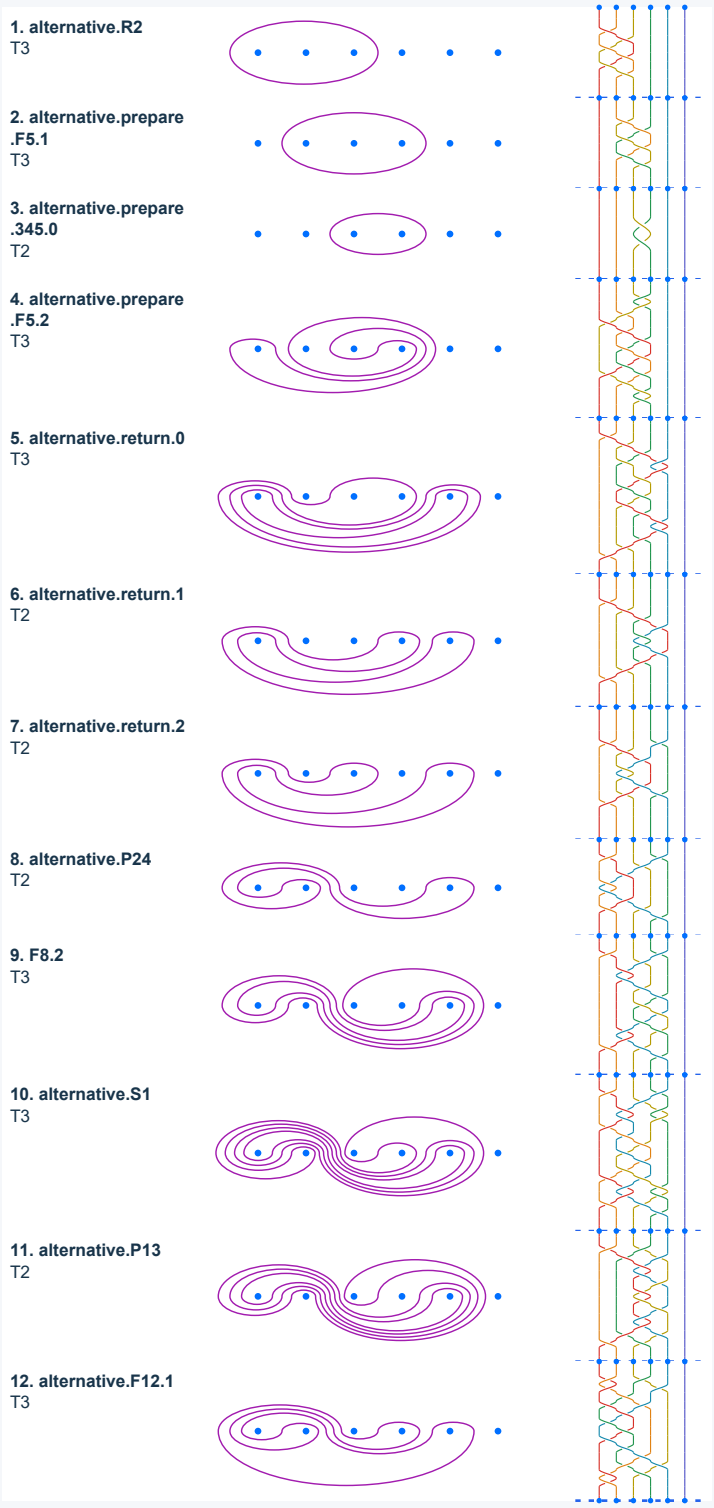
III - step 040

Cyclic start at Q15 REJOINS the unchanged route

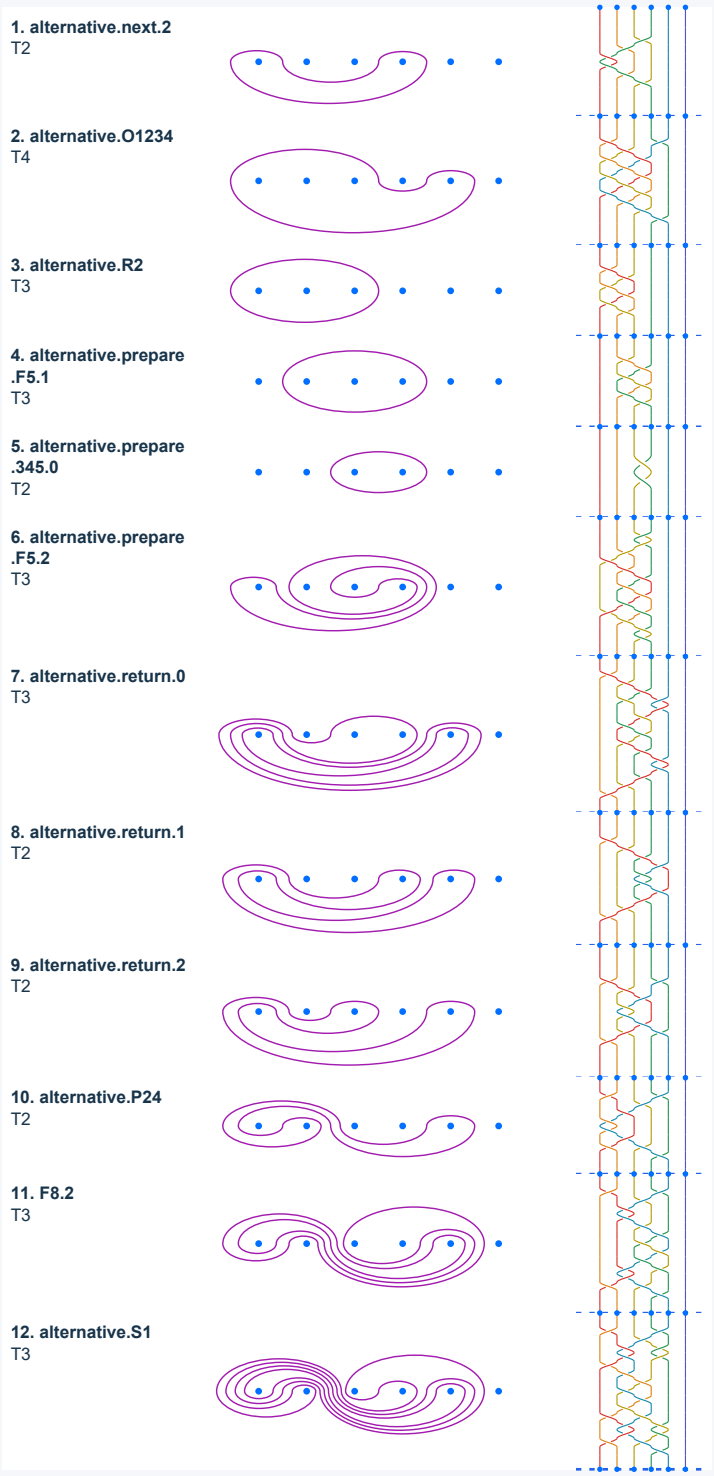
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



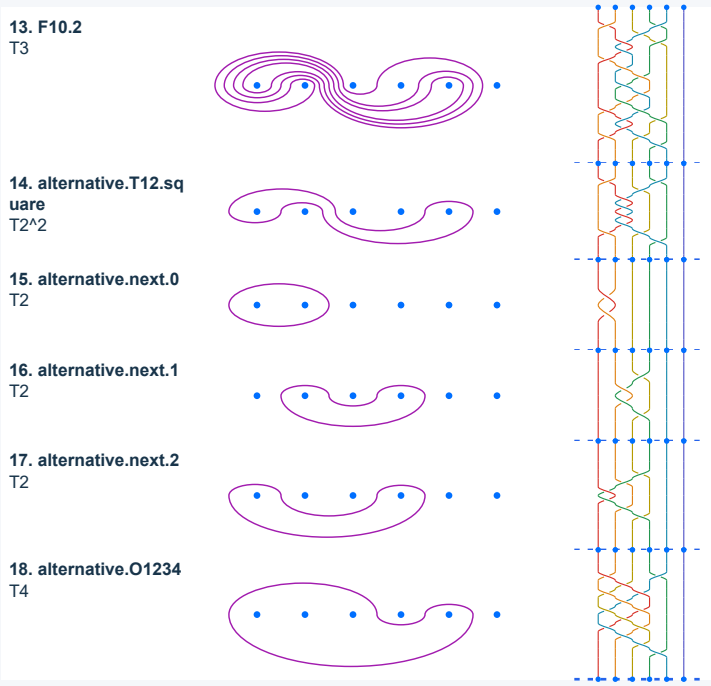
III - step 040 - continued

Cyclic start at Q15 REJOINS the unchanged route

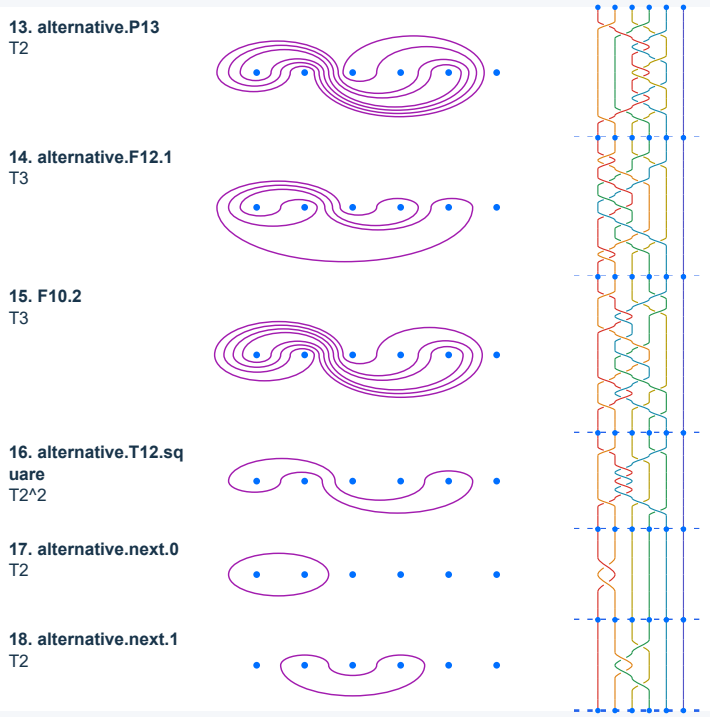
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



III - step 041

Carry Q15 through the combined four-point prefix and the two roots

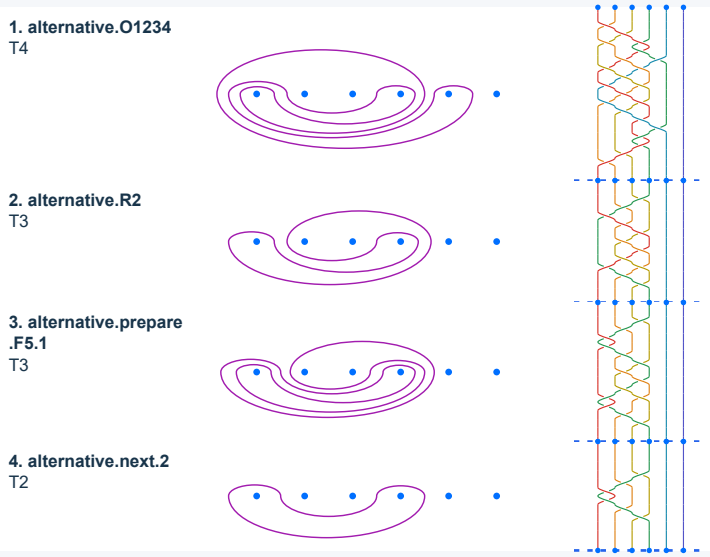
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



III - step 042

PC cluster lantern gives the second four-point twist

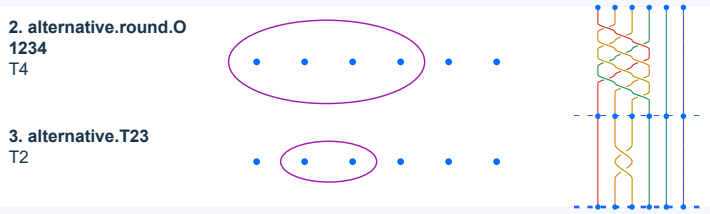
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



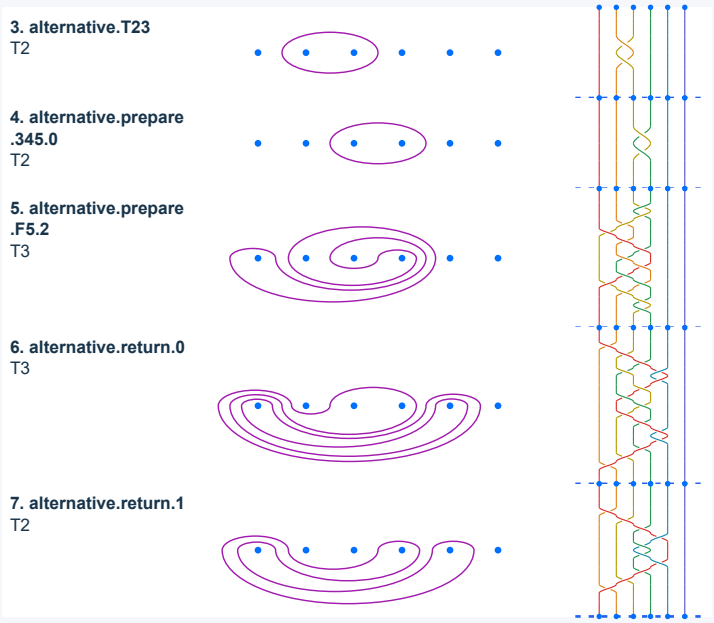
III - step 043

Gather the ACTUAL missing 45 petal by the short side

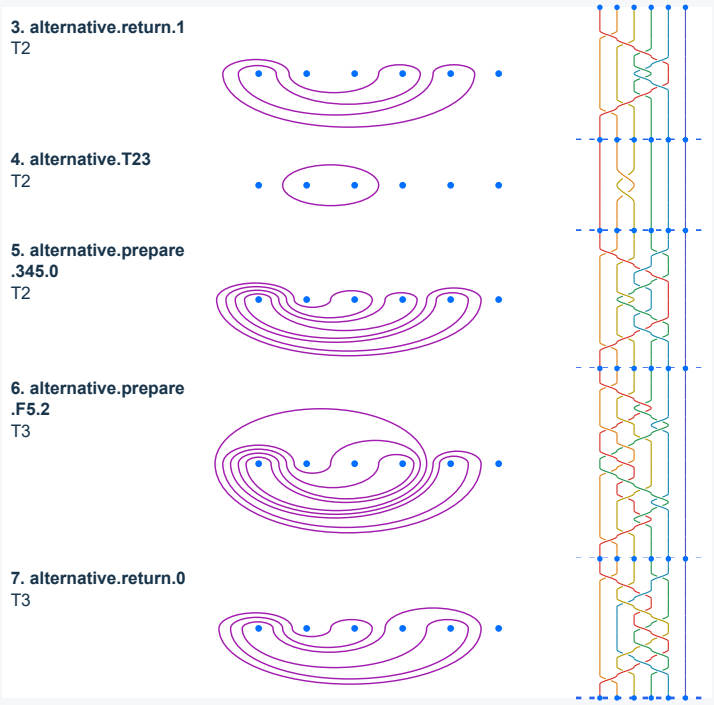
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



III - step 044

Cabled PC lantern on the ACTUAL two four-point curves and supplied 45 petal

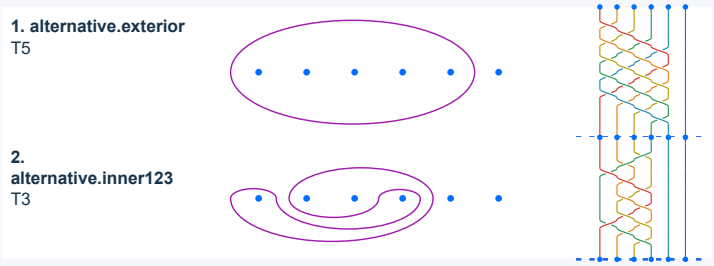
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown.

Before



After



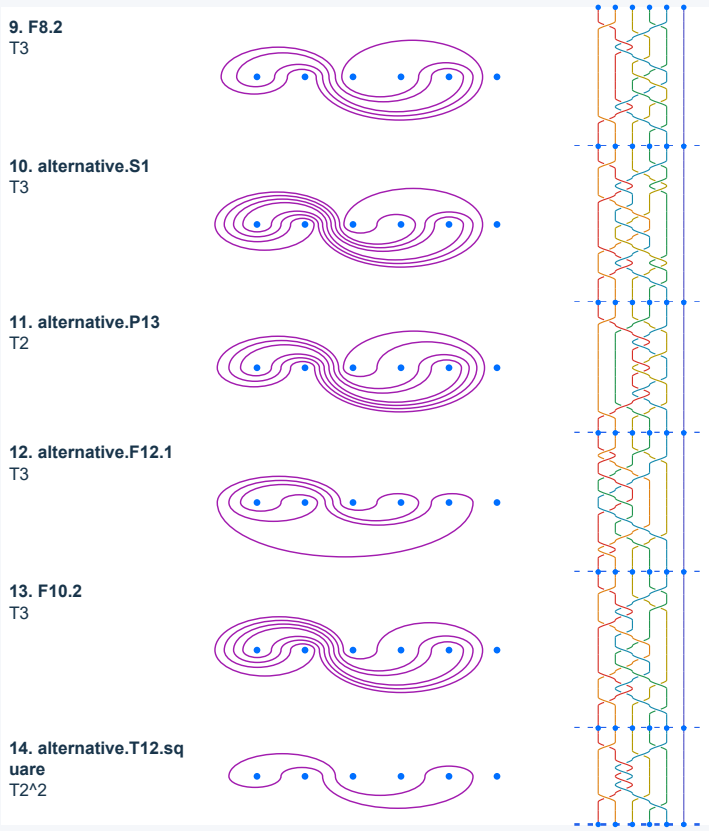
III - step 045

Unwind five actual residual curves with the fixed positive square

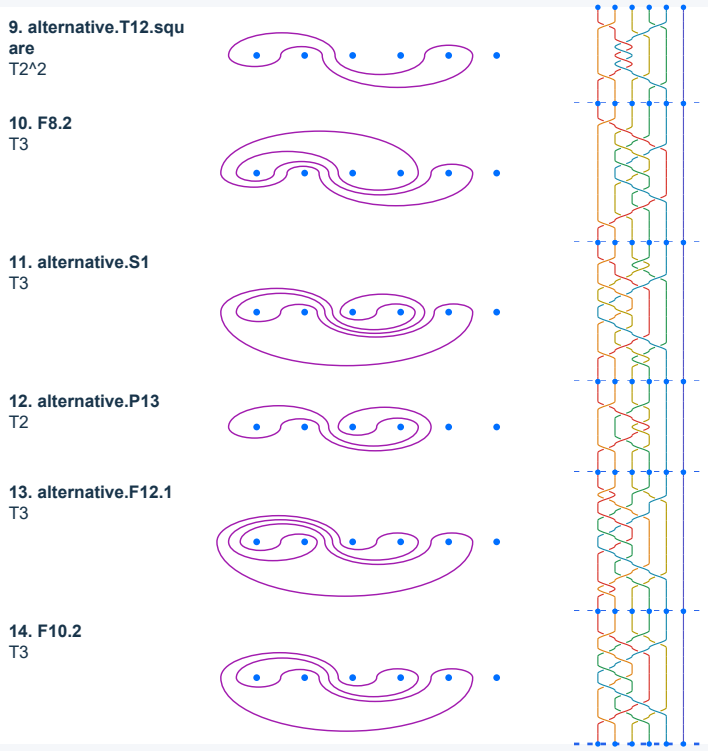
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



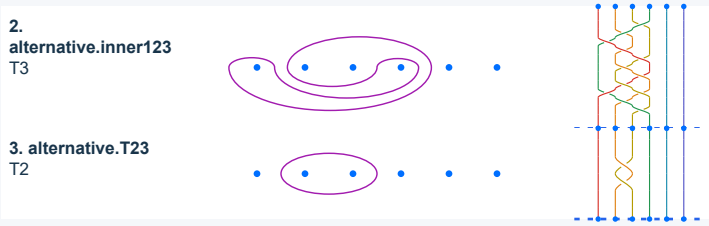
III - step 046

Commute the 123 root past T23 to its actual 34 and 124 companions

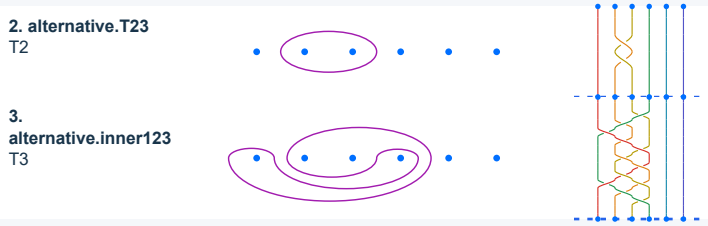
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



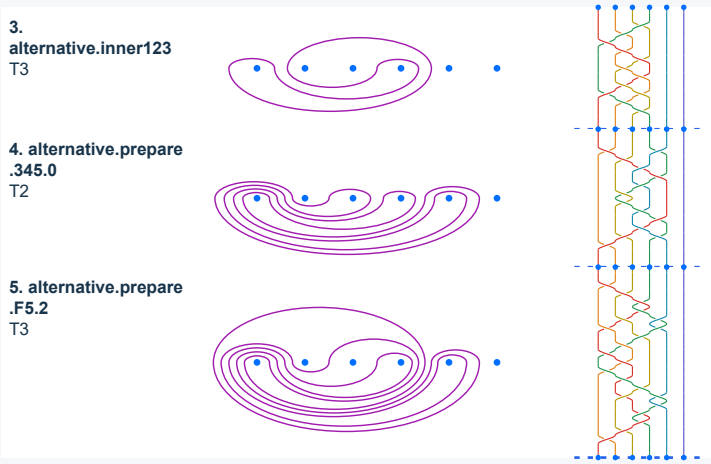
III - step 047

Cabled PC lantern contracts the ACTUAL 123, 34 and 124 curves

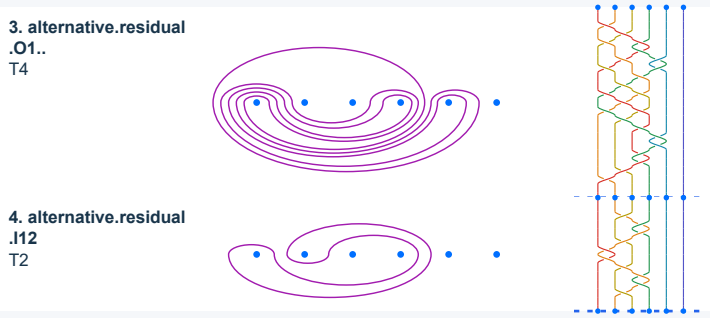
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

Before



After



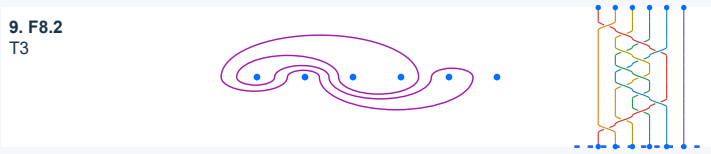
III - step 048

The actual 345 daisy lends its round 34 petal

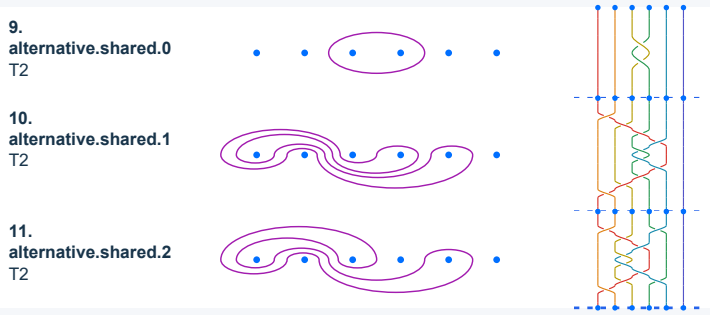
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



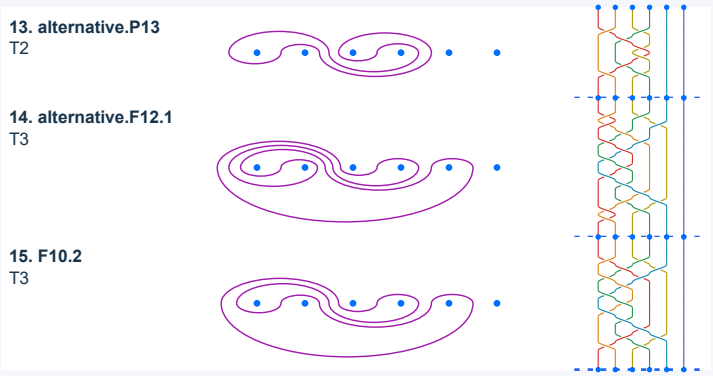
III - step 049

Carry the 245 root up across the 145 root and the 13 pair

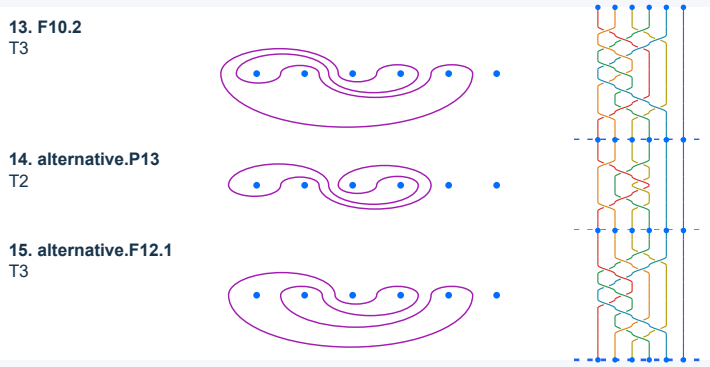
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



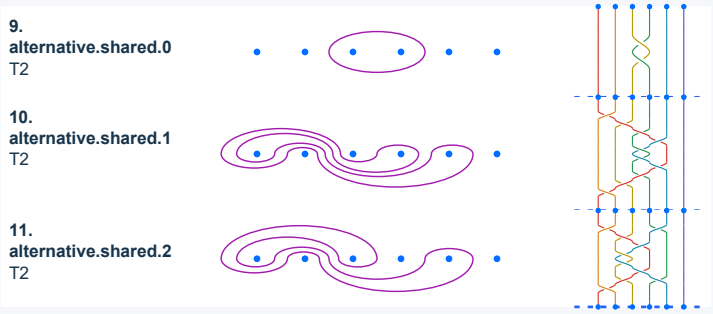
III - step 050

Carry round 34 down through its own two remaining petals

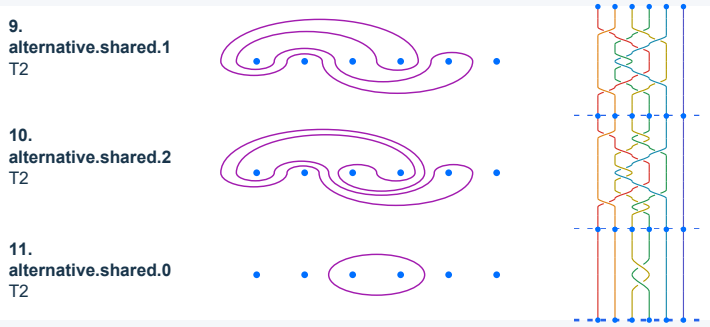
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



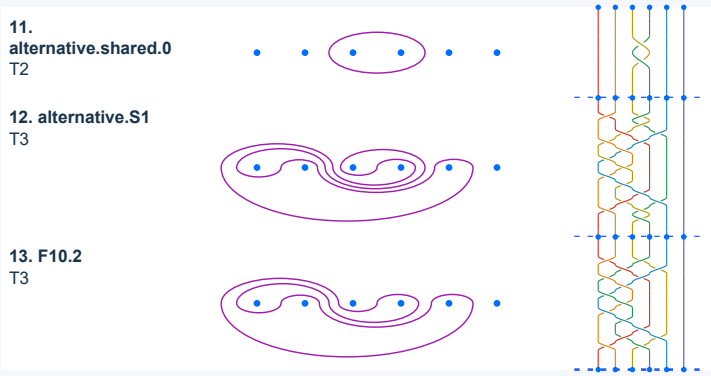
III - step 051

Cabled PC lantern closes the actual 34, 235 and 245 cluster

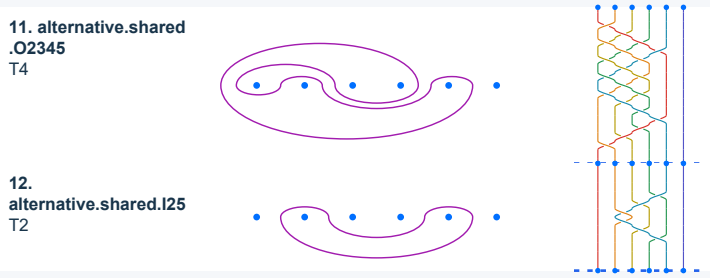
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



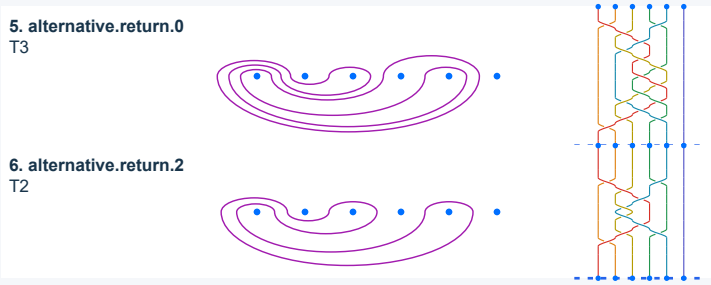
III - step 052

The actual 35 pair unwinds the 134 root to a short chart

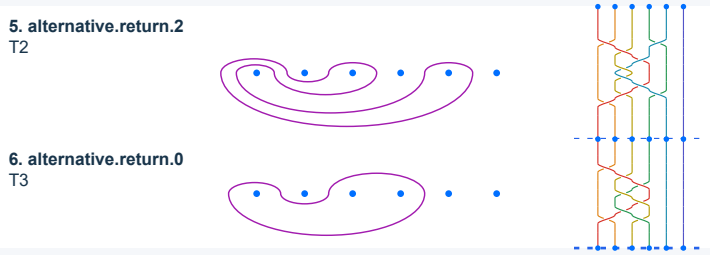
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



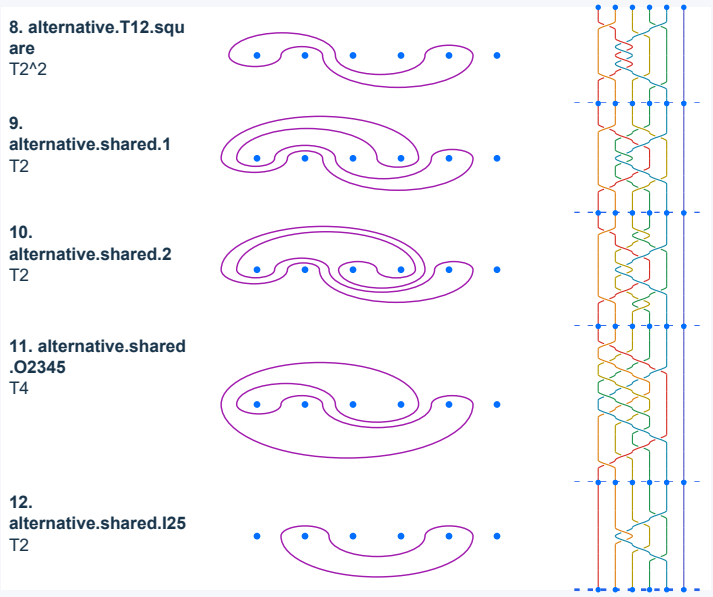
III - step 053

Carry the shared 25 pair up; unwind the square and its two petals

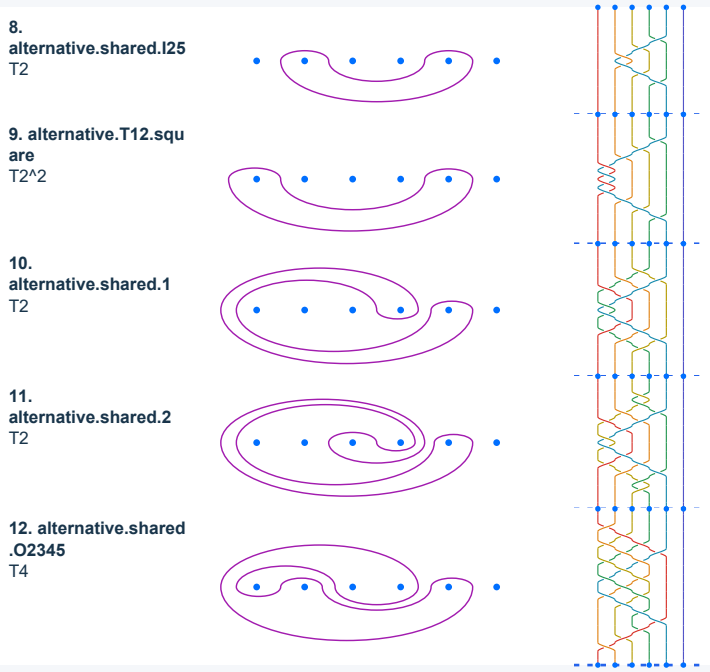
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



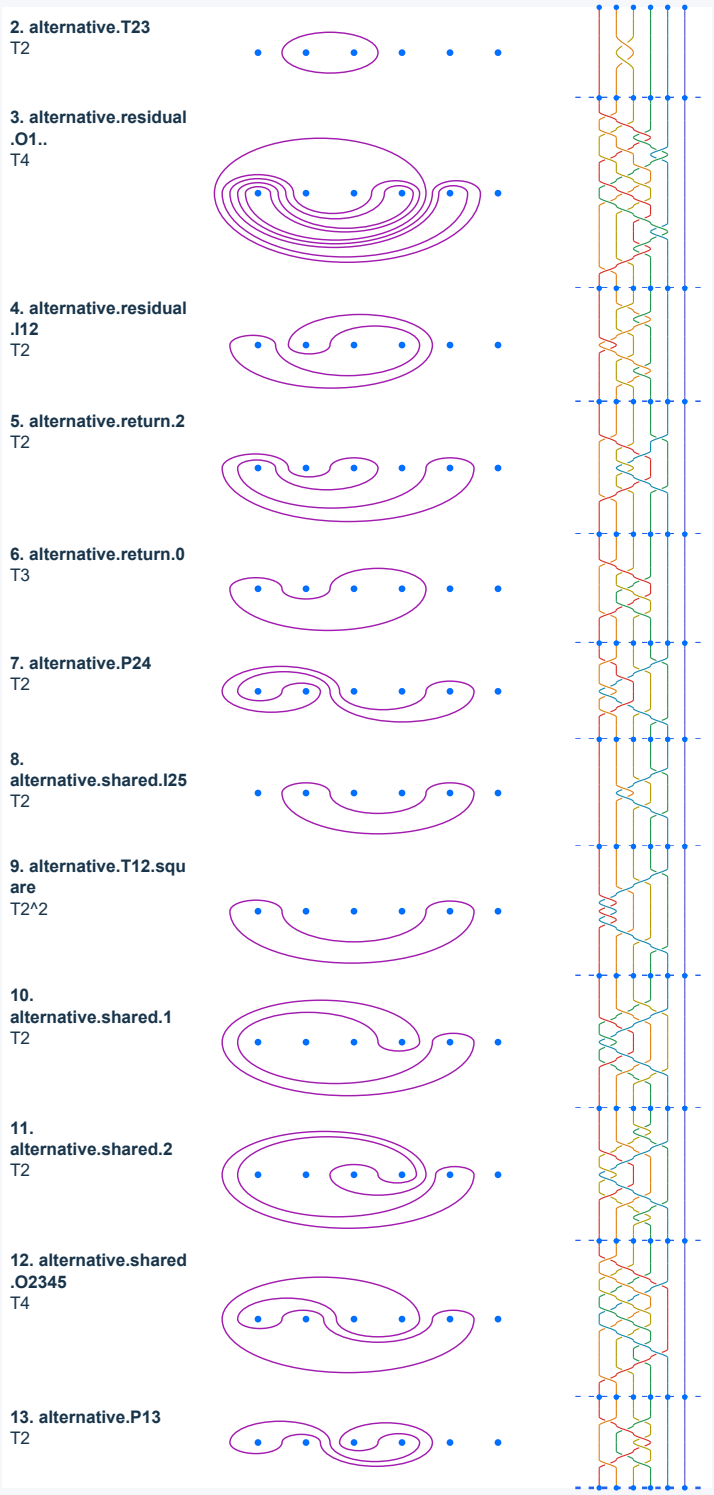
III - step 054

Common disk frame keeps the inside-four daisy readable

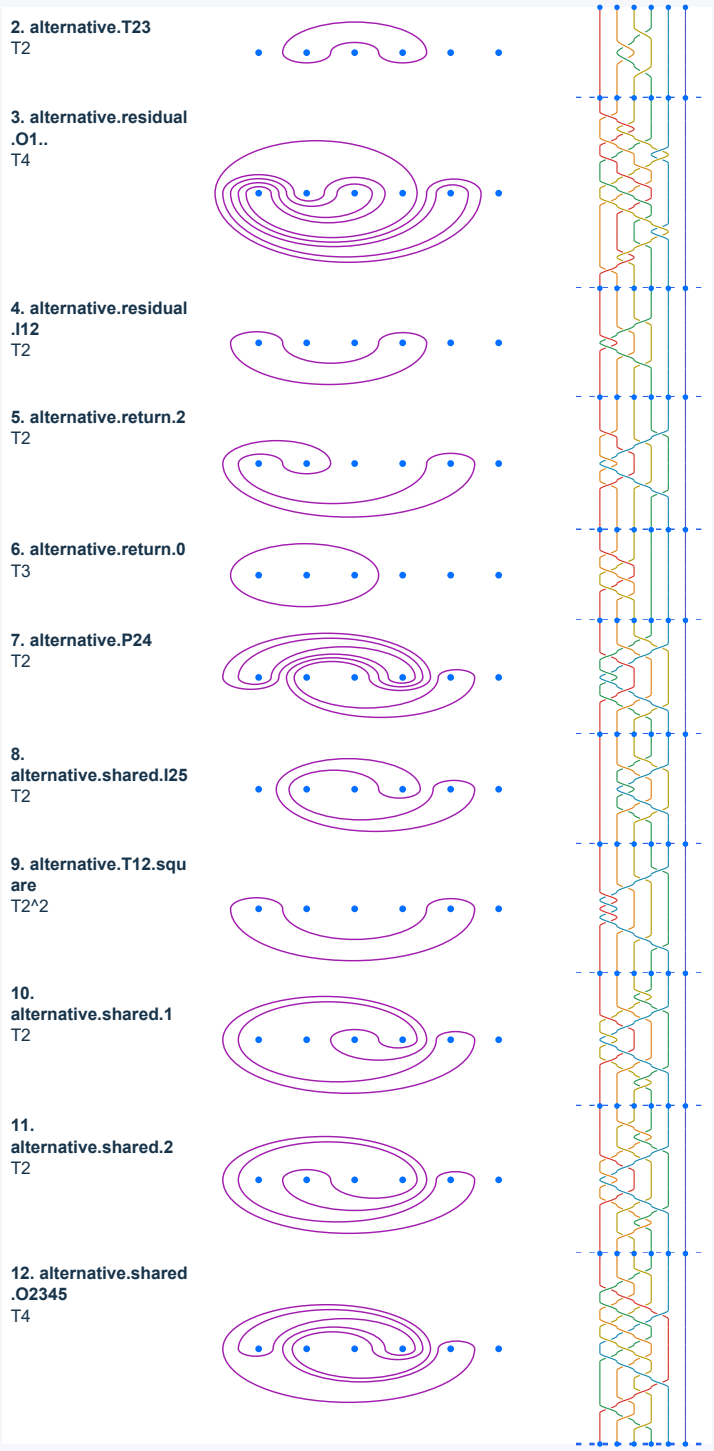
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



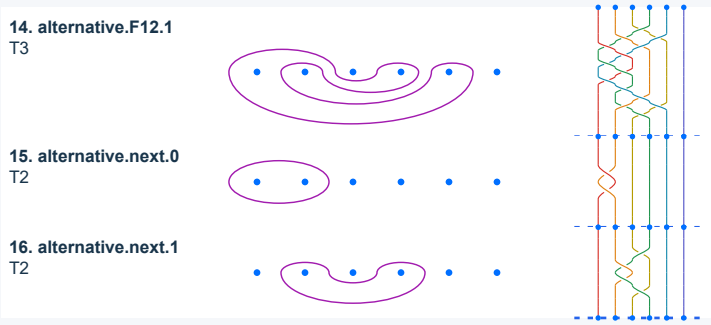
III - step 054 - continued

Common disk frame keeps the inside-four daisy readable

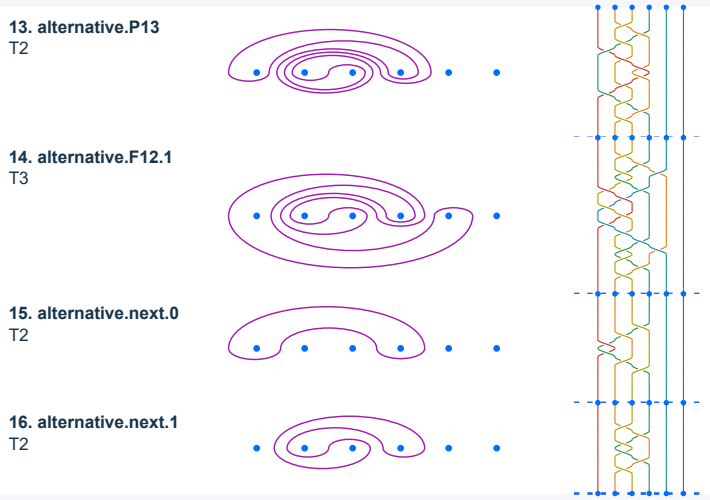
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



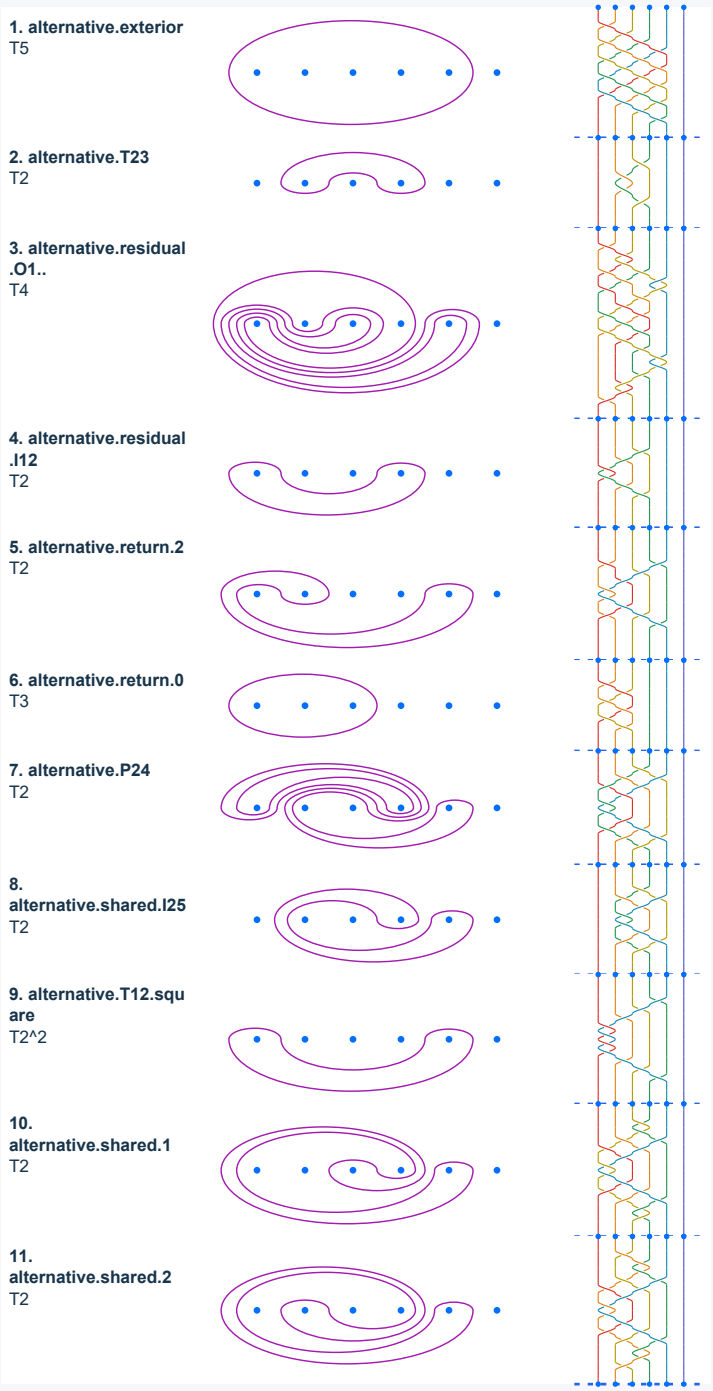
III - step 055

Choose the cyclic start at the actual three-point root

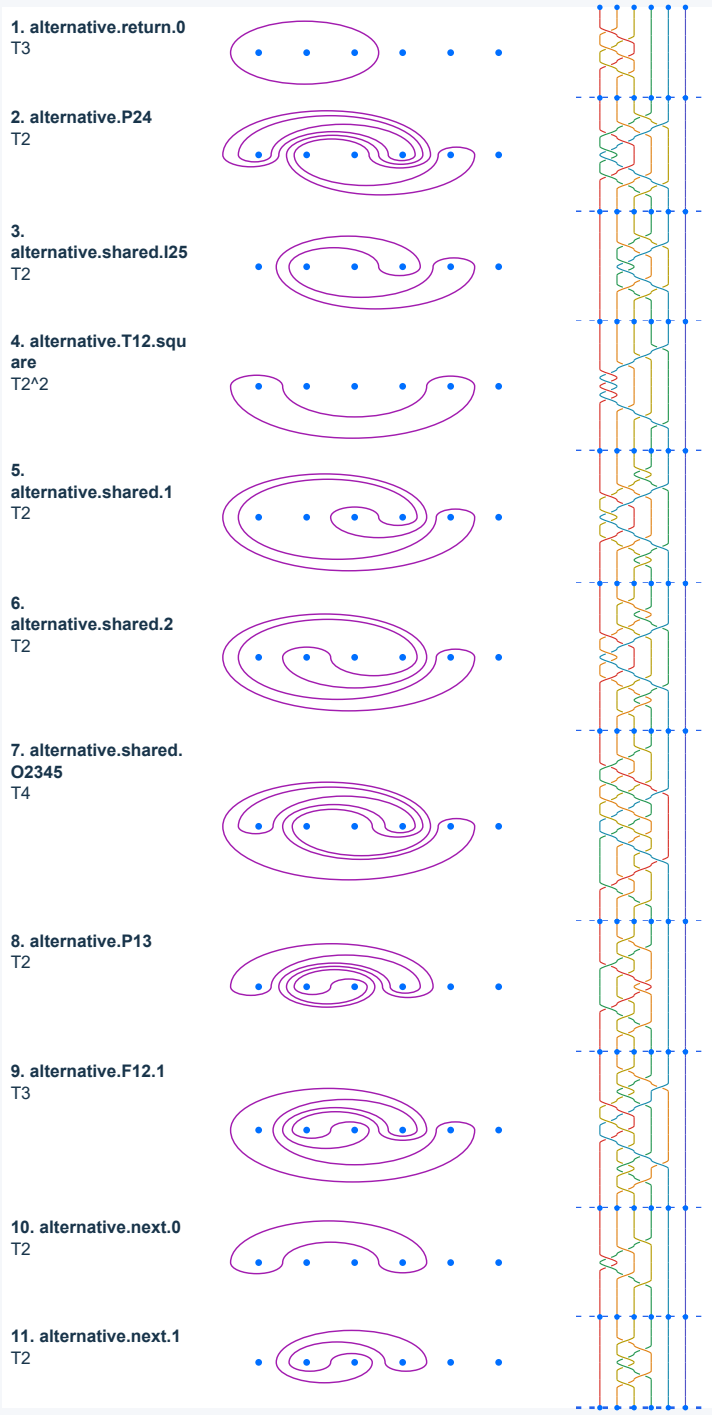
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



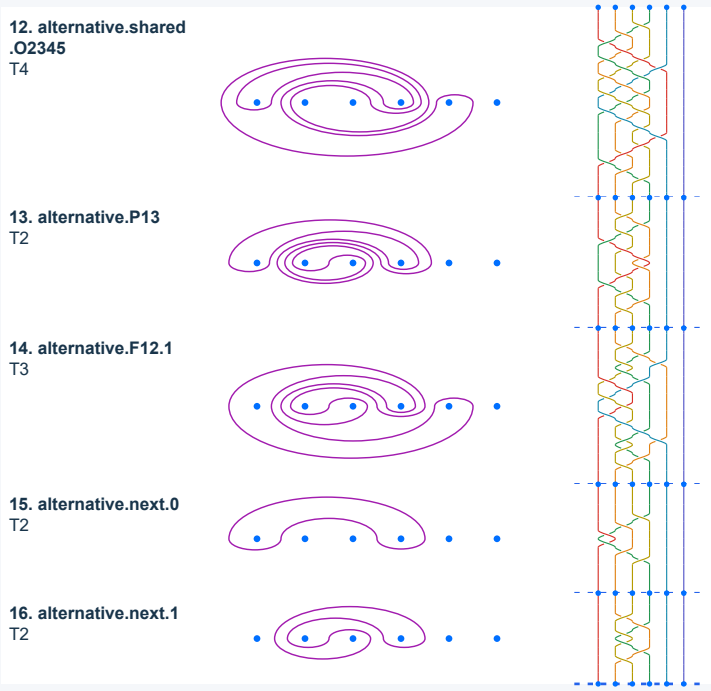
III - step 055 - continued

Choose the cyclic start at the actual three-point root

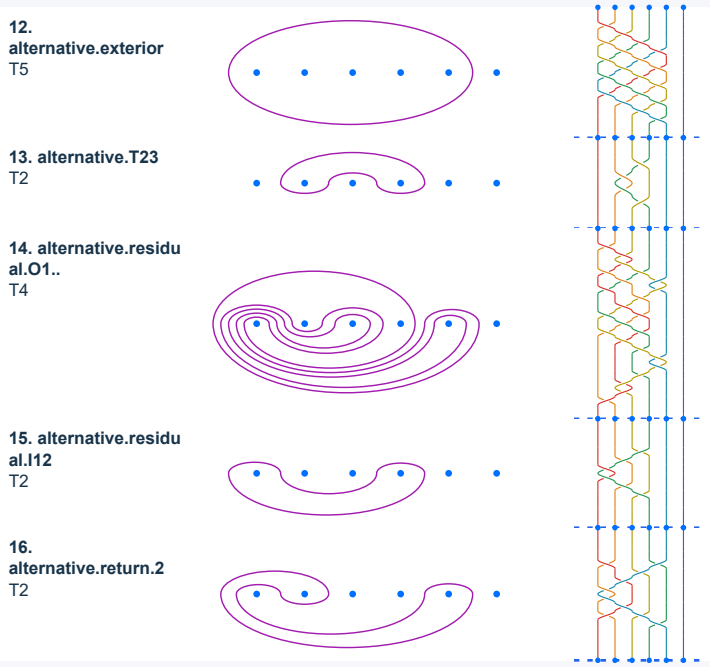
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



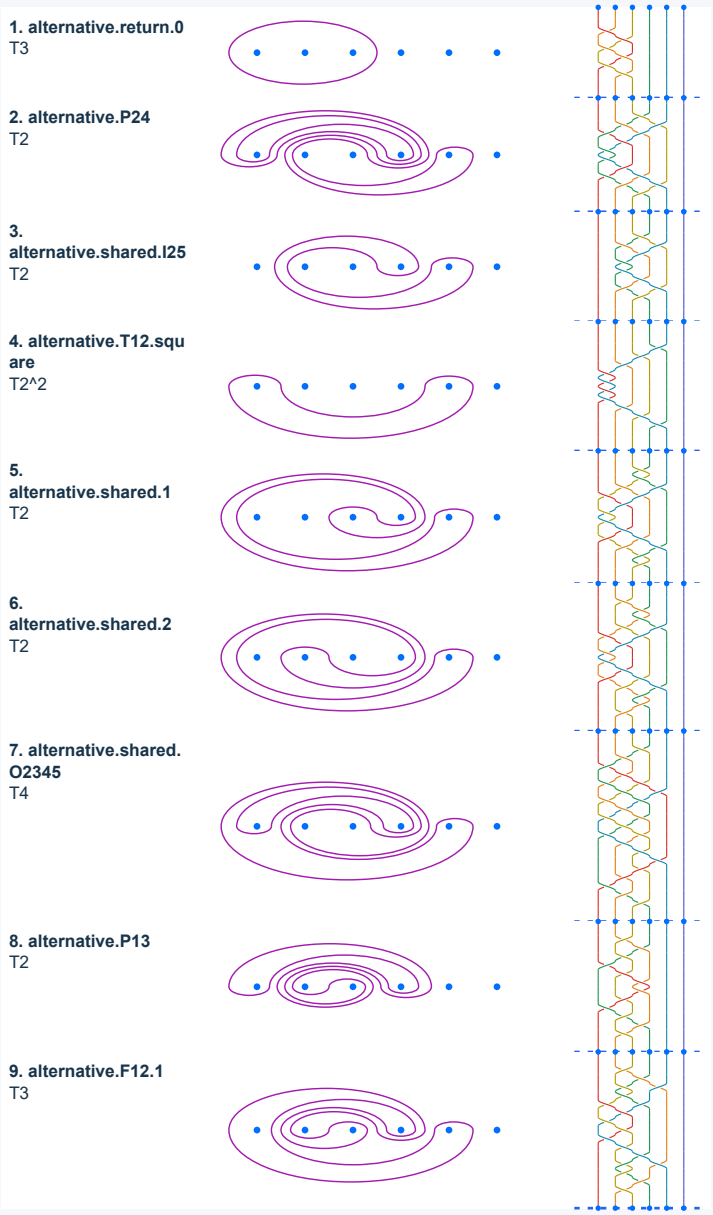
III - step 056

Carry the three-point root down through the eight residual factors

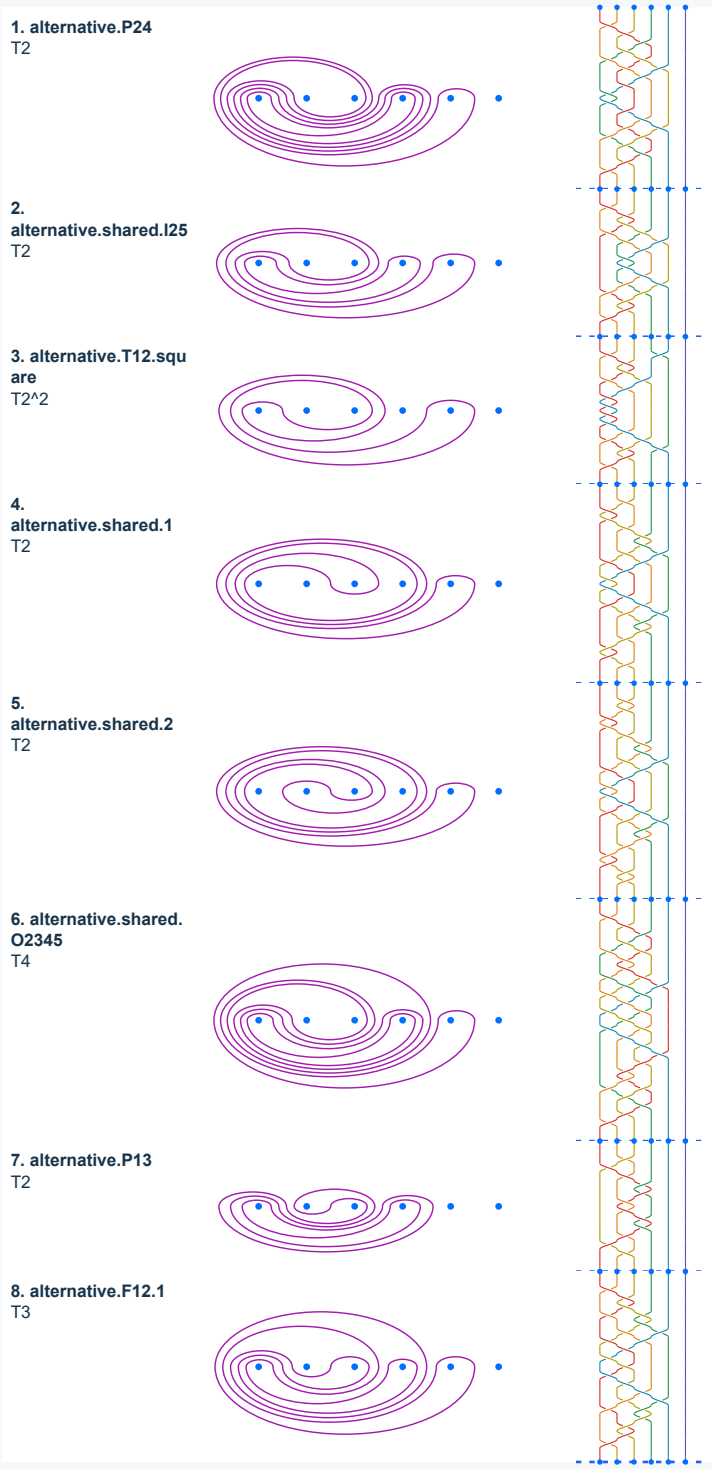
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



III - step 056 - continued

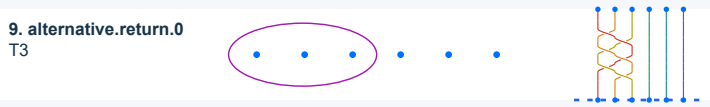
Carry the three-point root down through the eight residual factors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before

After



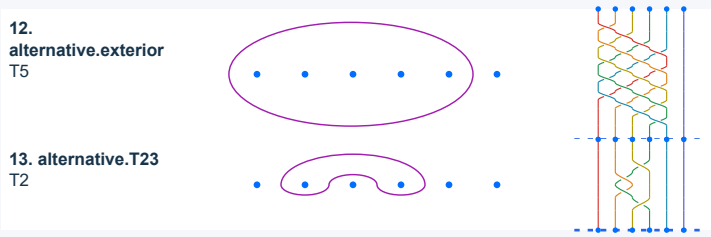
III - step 057

Commute the final pair petal past the visible exterior

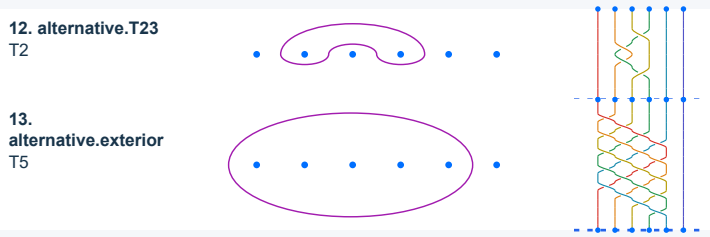
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



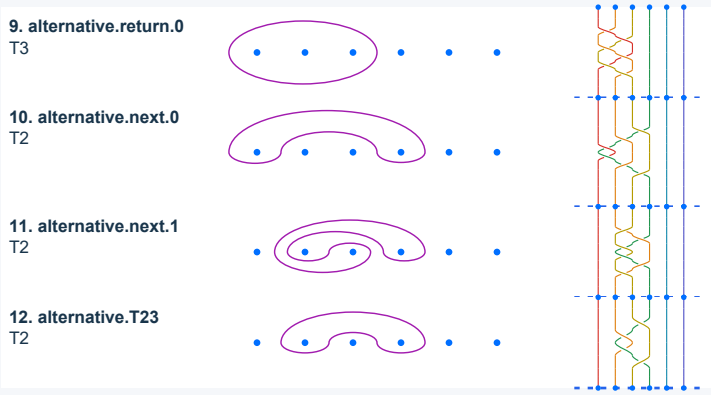
III - step 058

Transported PC daisy closes the actual root and its three pair petals

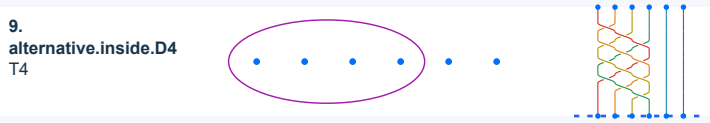
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



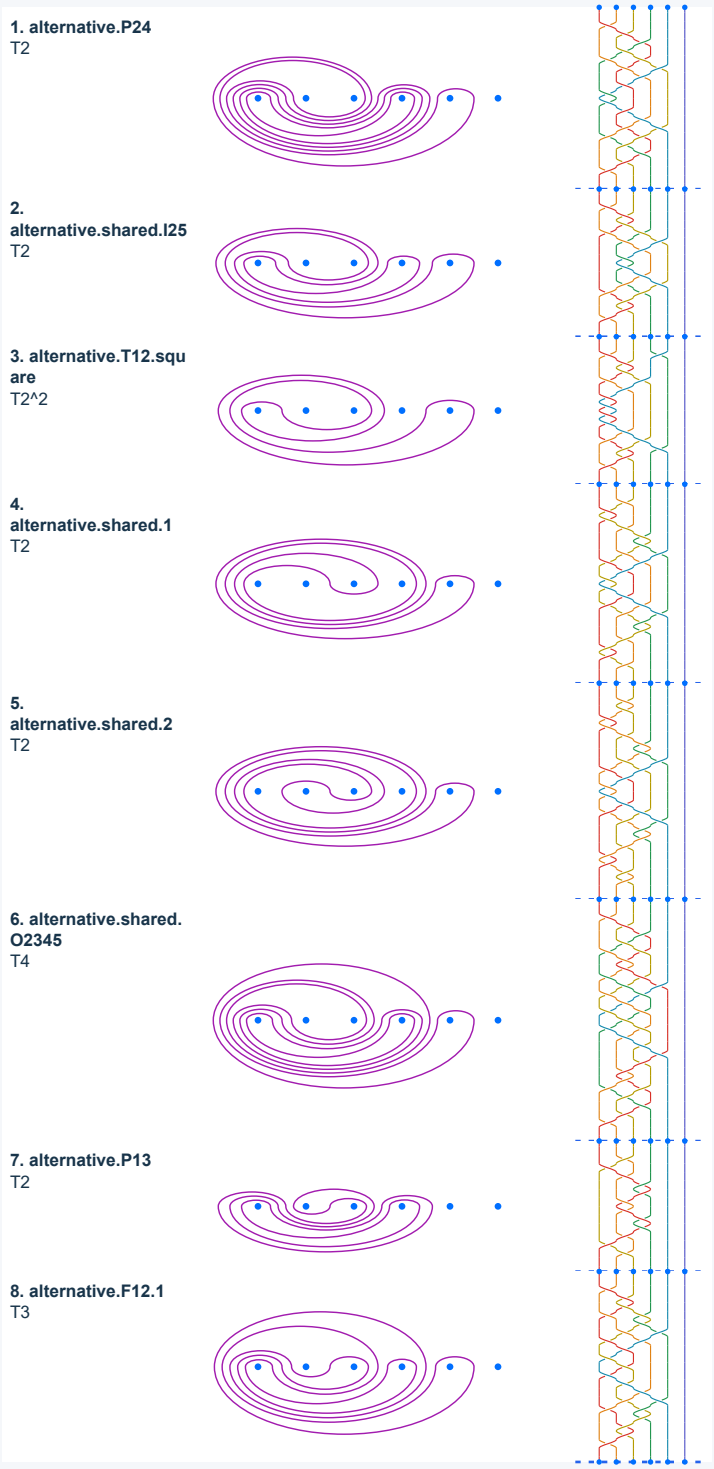
III - step 059

Carry the new round four-point twist up to unwind the residual block

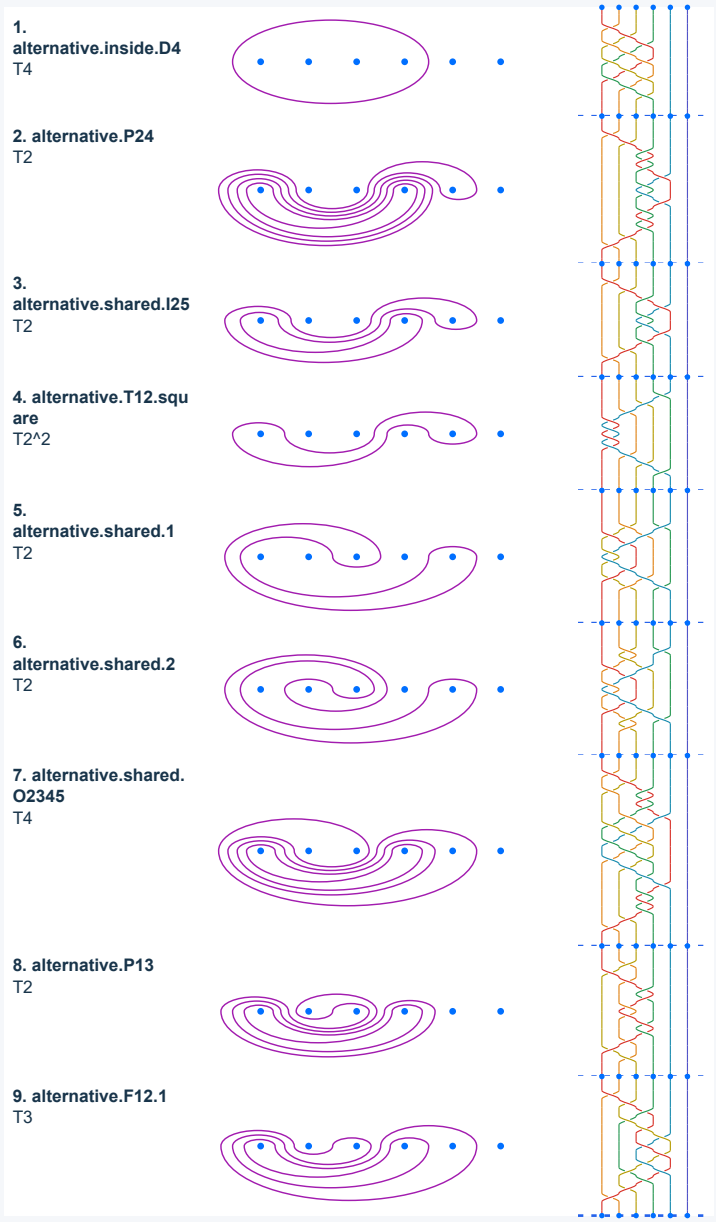
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before



After



III - step 059 - continued

Carry the new round four-point twist up to unwind the residual block

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

Before

9.
alternative.inside.D4
T4



After

IV. Reach the second exterior twist

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

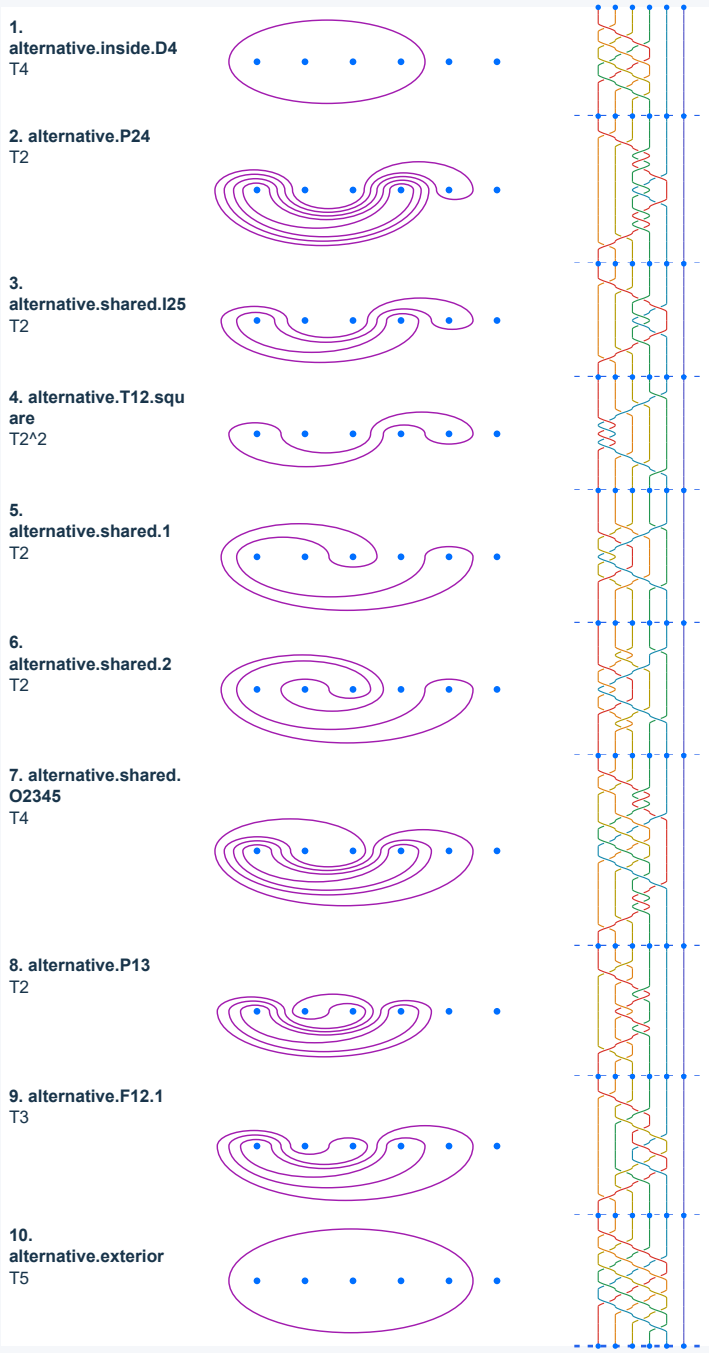
IV - step 060

ONE positive T45 lantern chart replaces the two former frames

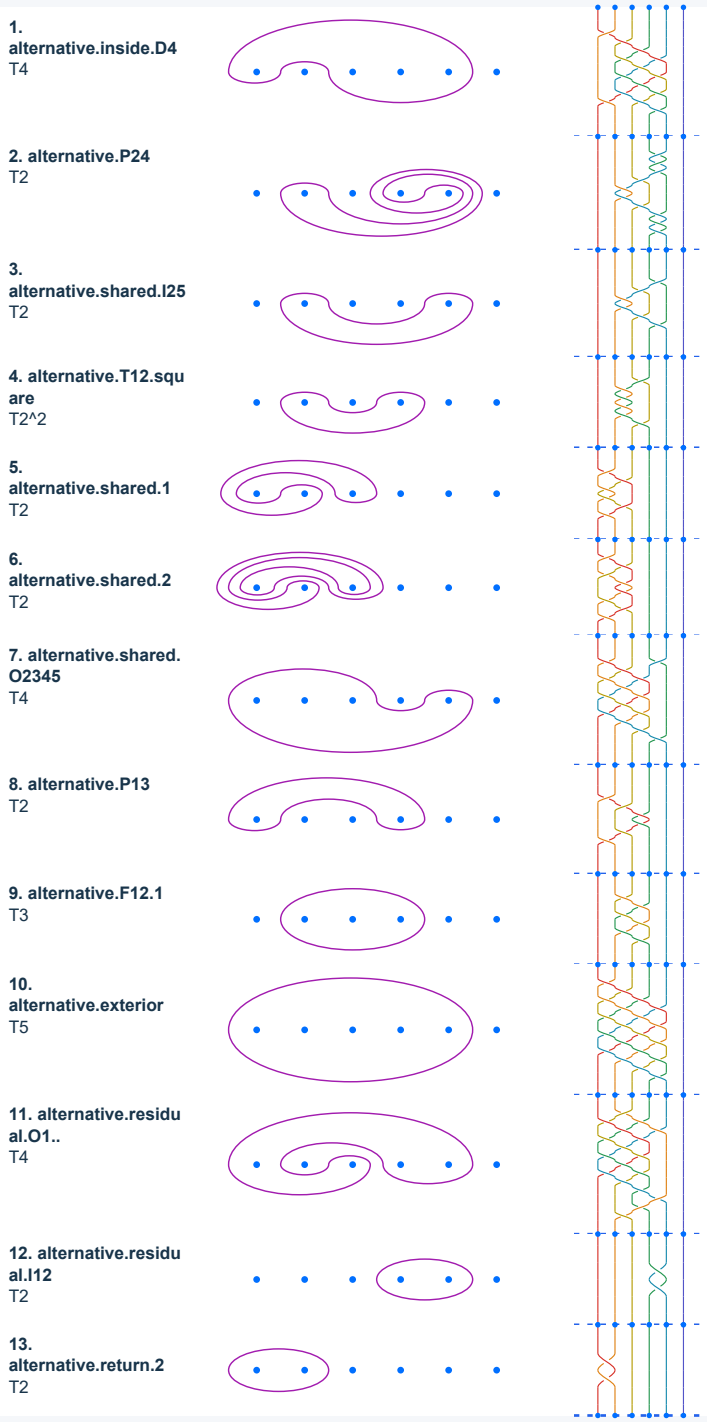
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



IV - step 060 - continued

ONE positive T45 lantern chart replaces the two former frames

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

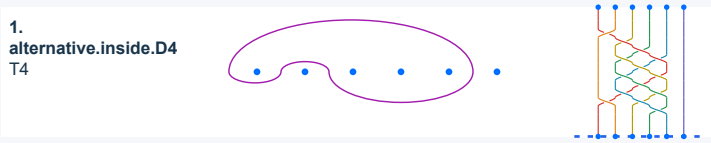
IV - step 061

The four-point PC daisy lends its actual three-point root

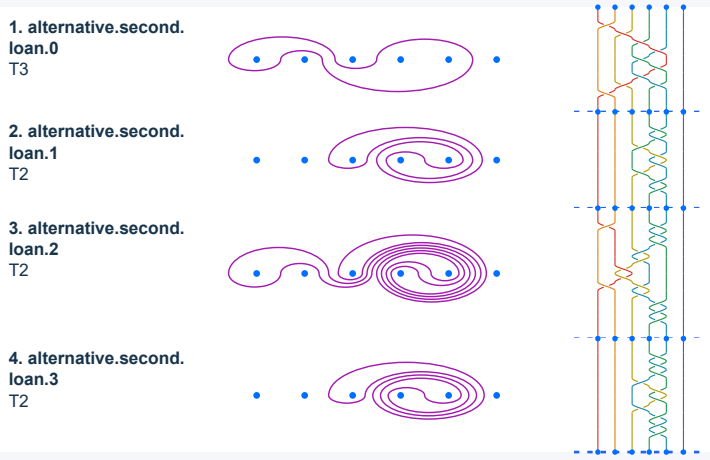
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



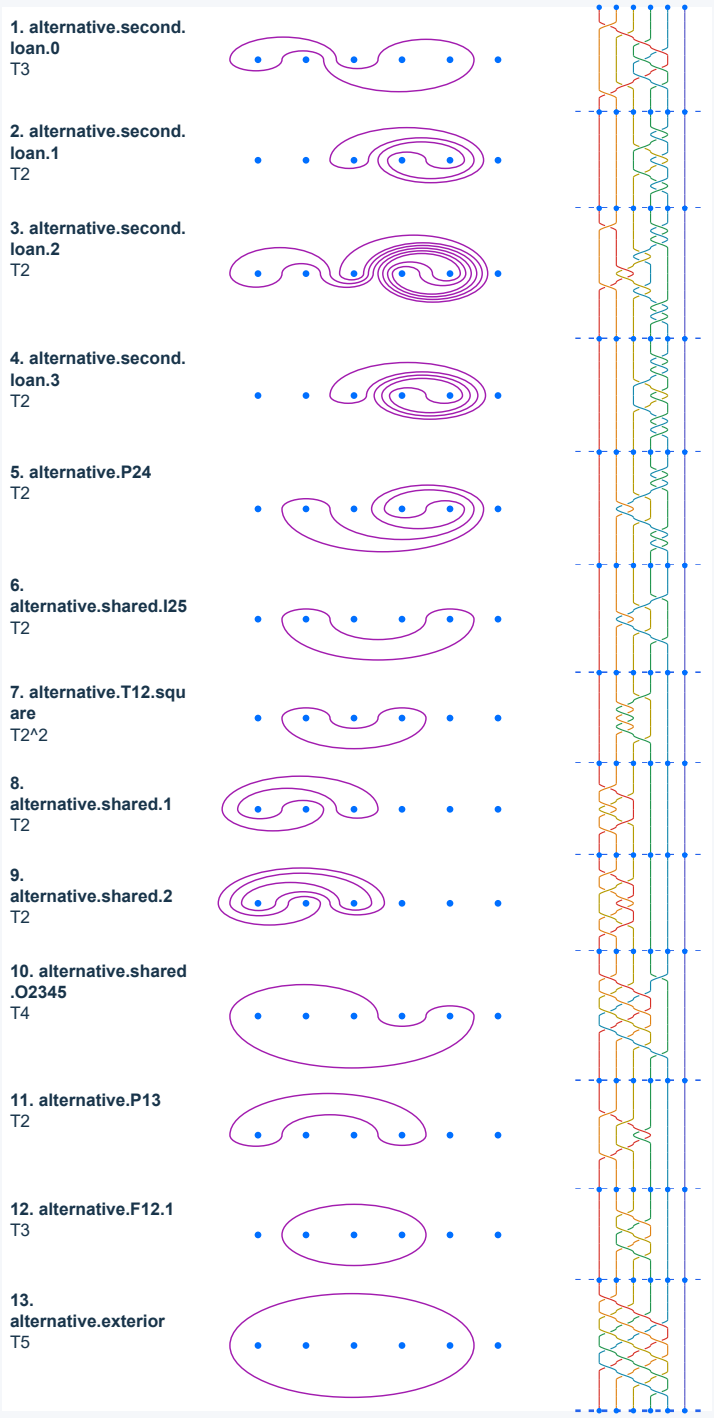
IV - step 062

Cyclic start at the actual pair that transports the borrowed root

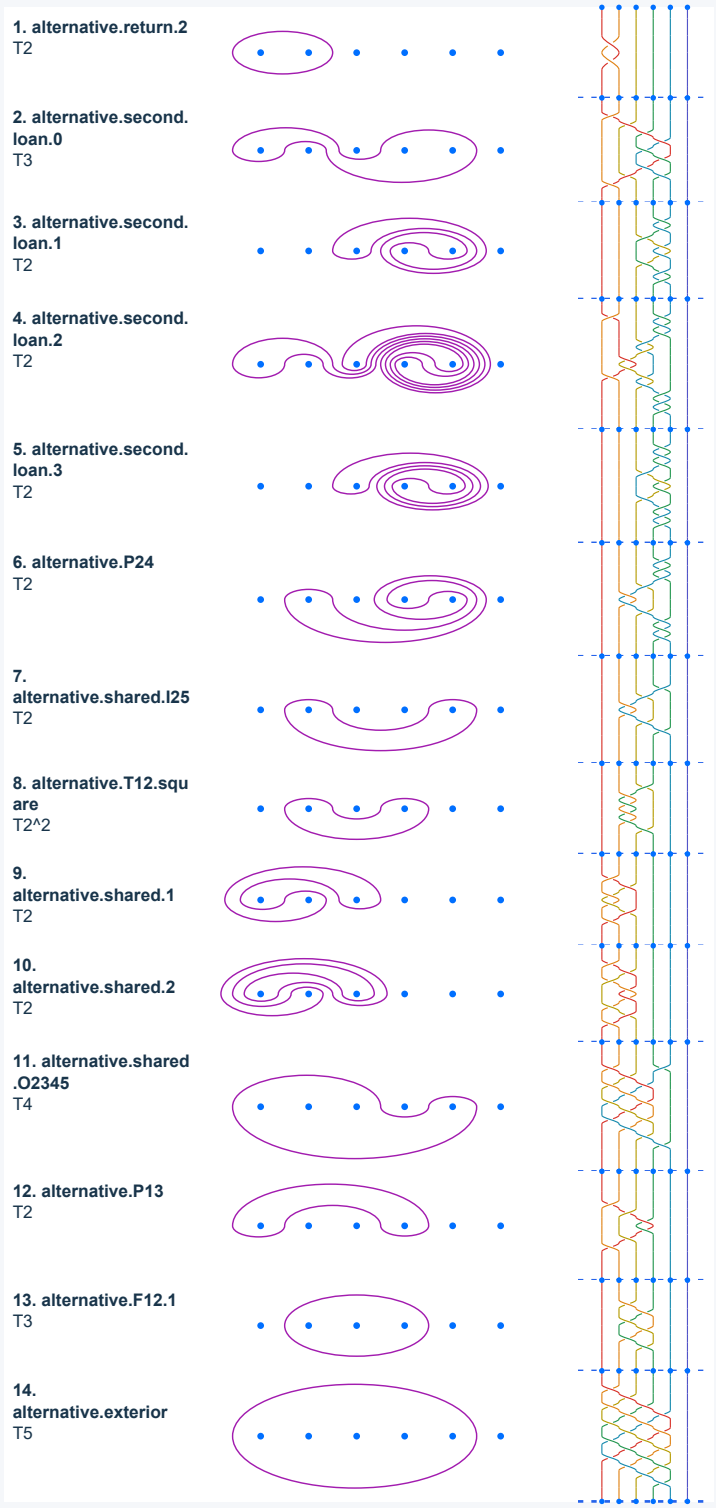
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



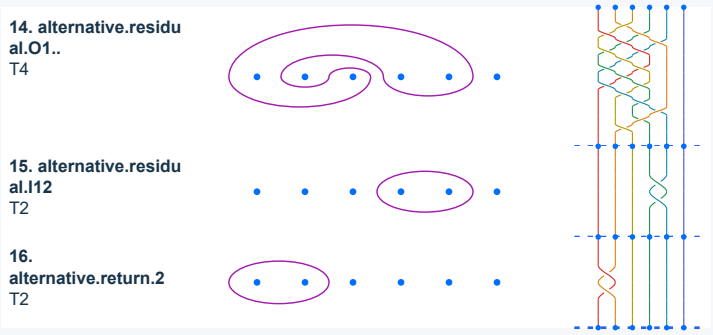
IV - step 062 - continued

Cyclic start at the actual pair that transports the borrowed root

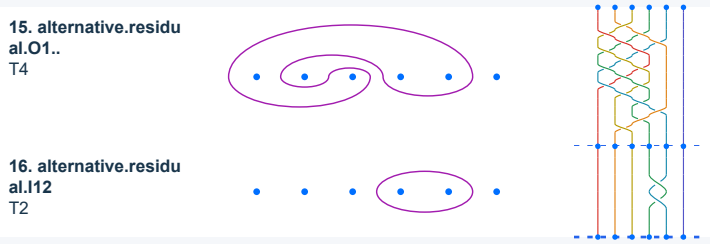
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



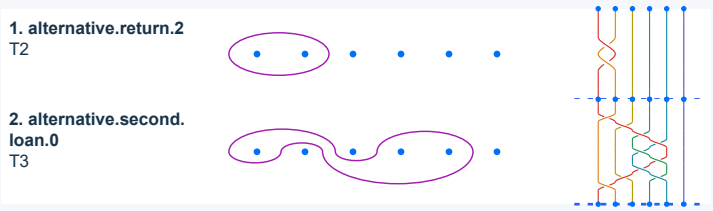
IV - step 063

The actual pair carries the borrowed root to the exact missing class

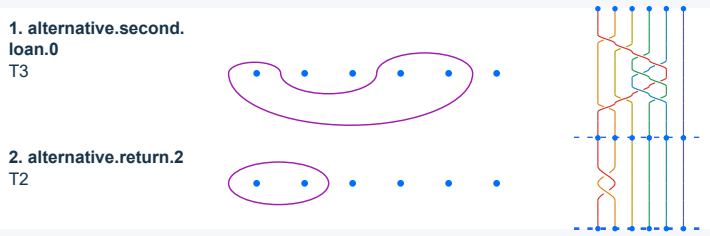
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



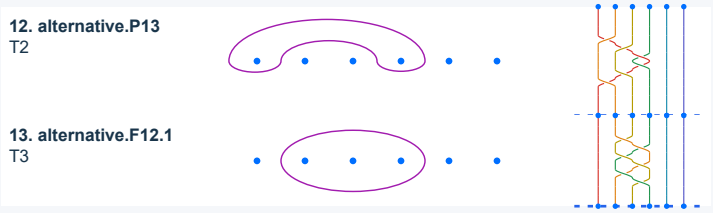
IV - step 064

Bring the three-point curve up across one neighboring pair

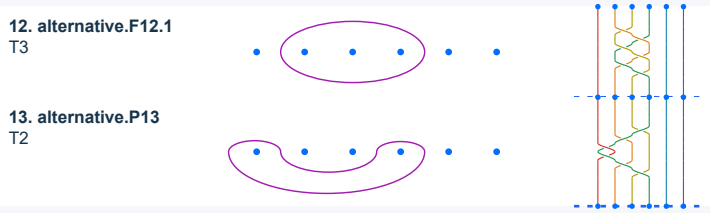
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



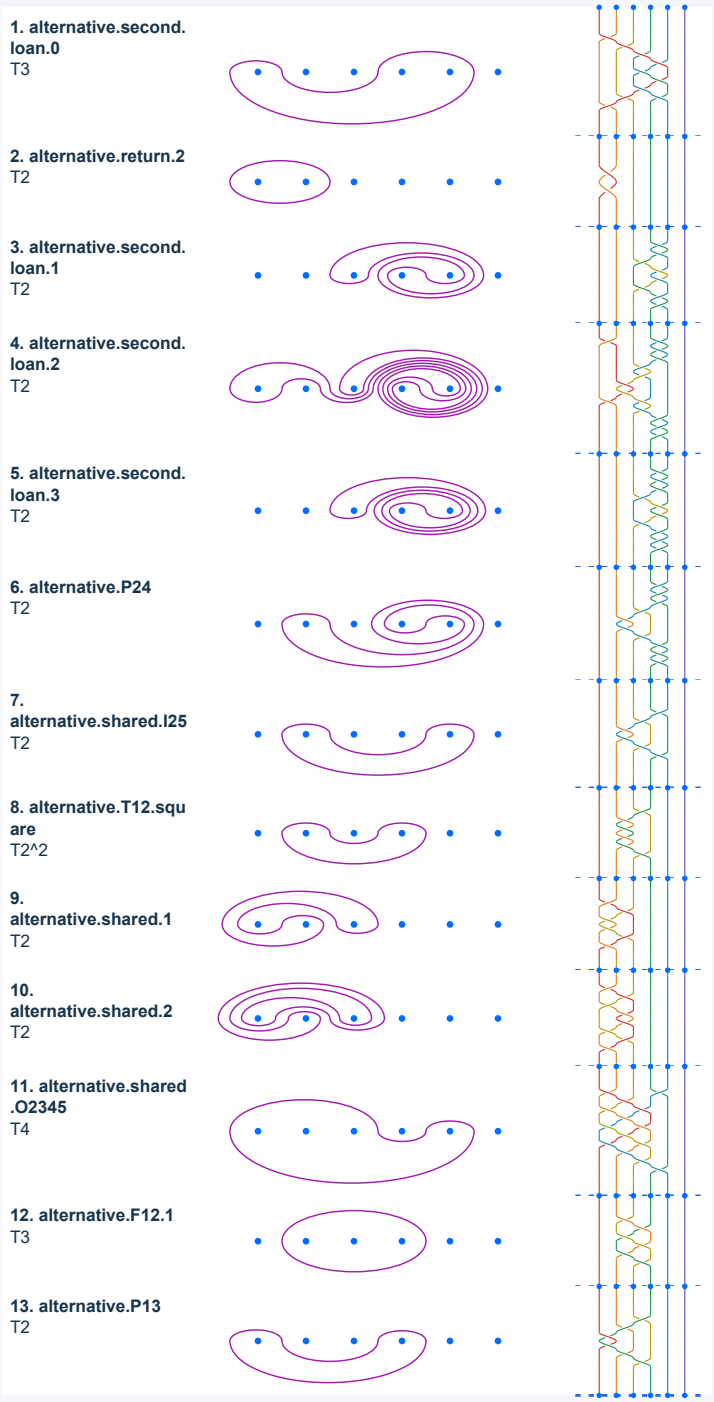
IV - step 065

Cyclic start at the matched four-point curve

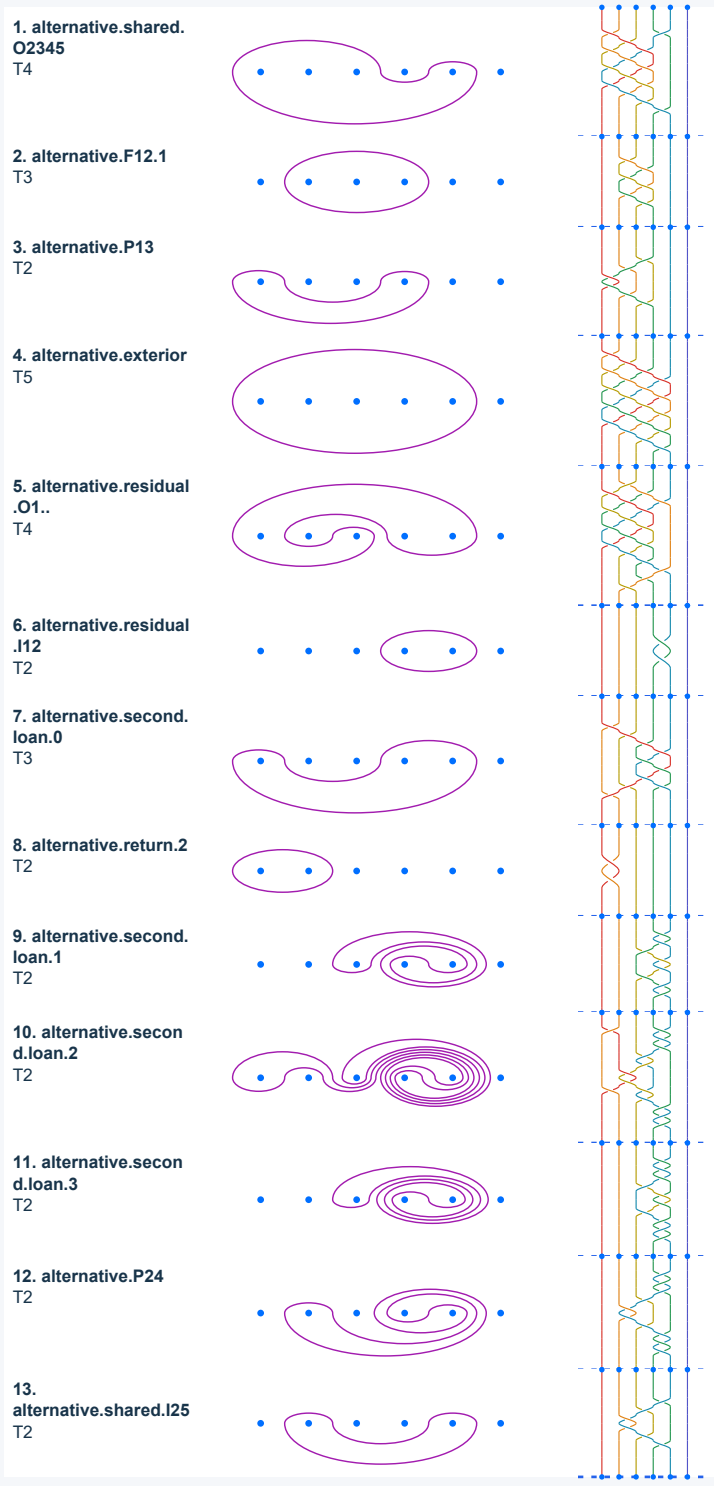
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



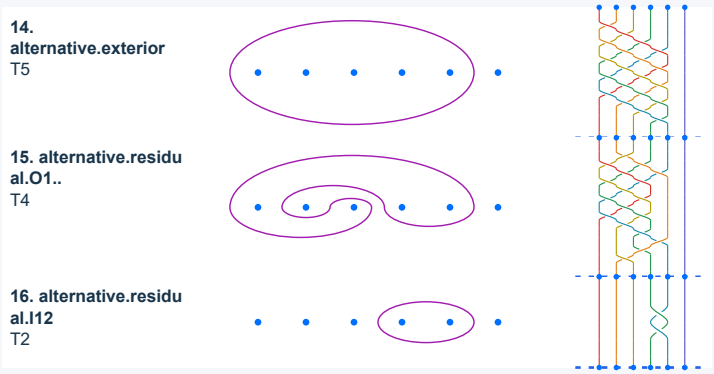
IV - step 065 - continued

Cyclic start at the matched four-point curve

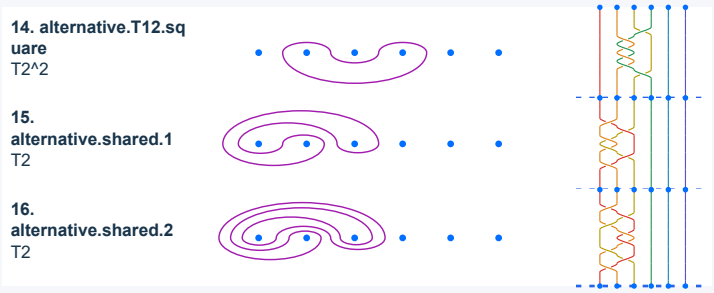
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



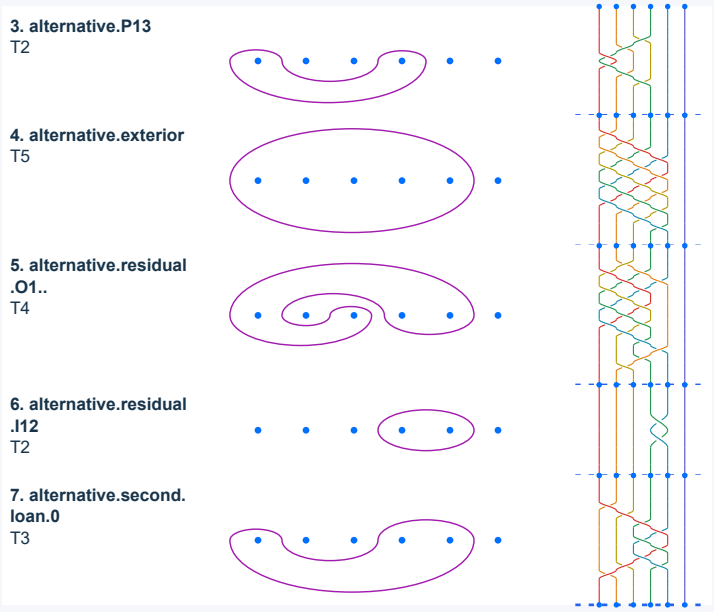
IV - step 066

Gather the borrowed root up the four-neighbor short side

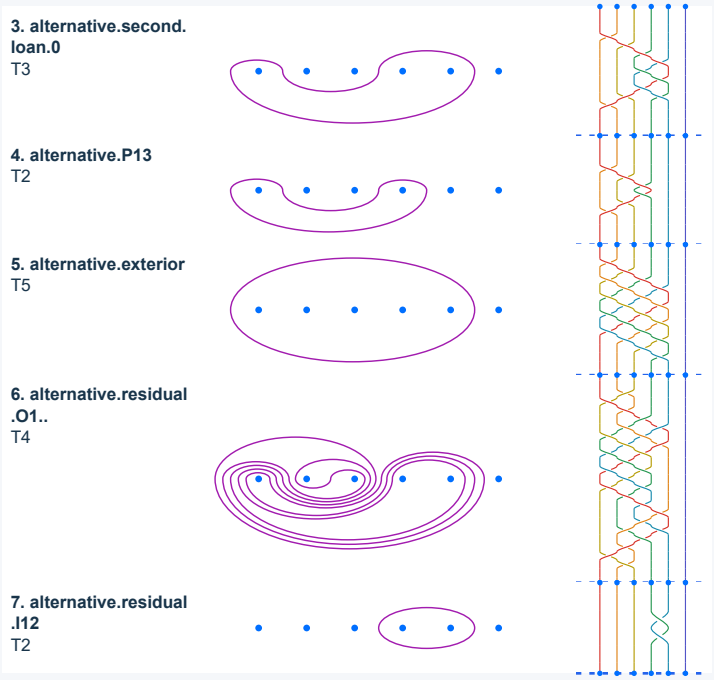
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



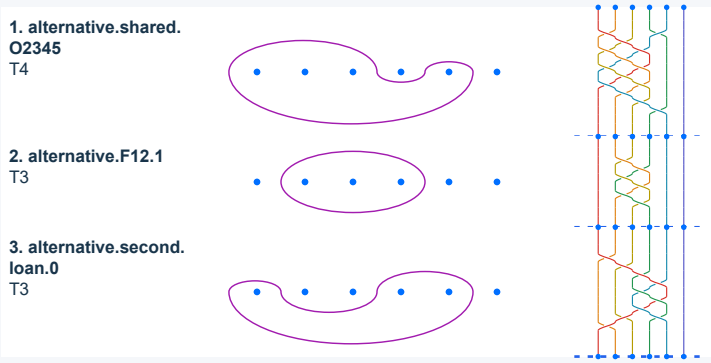
IV - step 067

Cabled PC lantern reaches a SECOND visible exterior twist

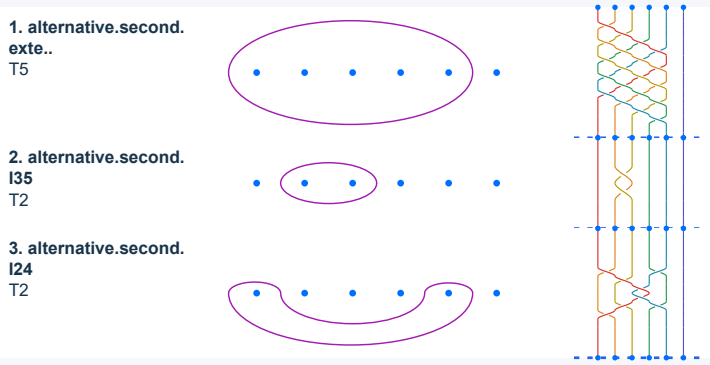
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



V. Return the loan as an expanded rose

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Return the loan as an expanded rose

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. alternative.second.exte..
T5

2. alternative.second.l35
T2

3. alternative.second.l24
T2

4. alternative.P13
T2

5. alternative.exterior
T5

6. alternative.residual.O1..
T4

7. alternative.residual.l12
T2

8. alternative.return.2
T2

9. alternative.second.loan.1
T2

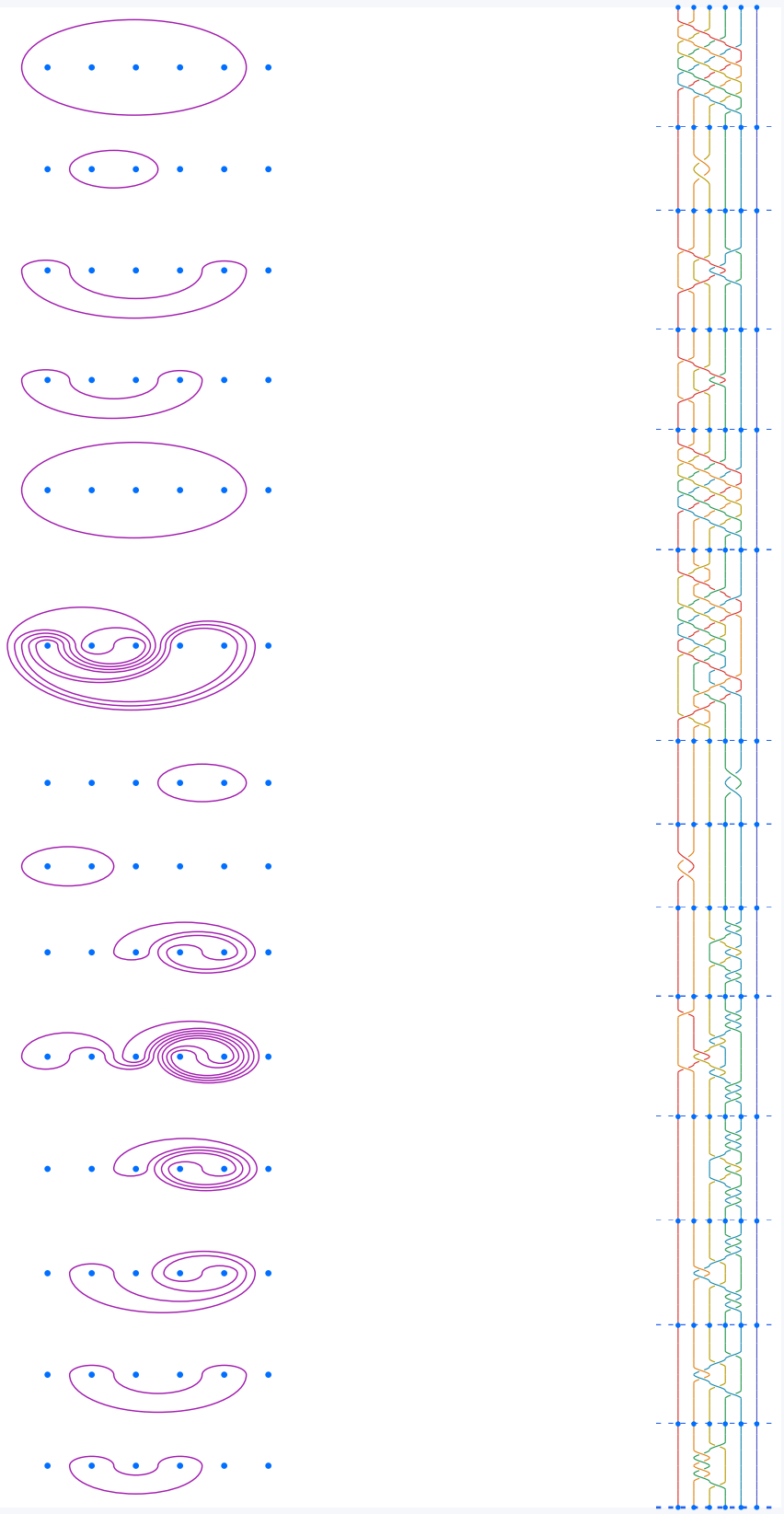
10. alternative.second.loan.2
T2

11. alternative.second.loan.3
T2

12. alternative.P24
T2

13. alternative.shared.l25
T2

14. alternative.T12.square
T2^2

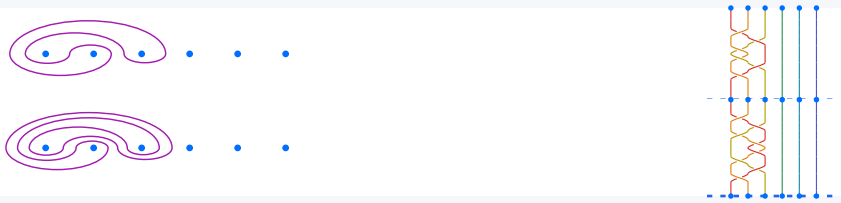


Complete factorization before Return the loan as an expanded rose - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

15. alternative.shared.1
T2

16. alternative.shared.2
T2



V - step 068

Commute the exterior out of the expanded loan

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 11 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



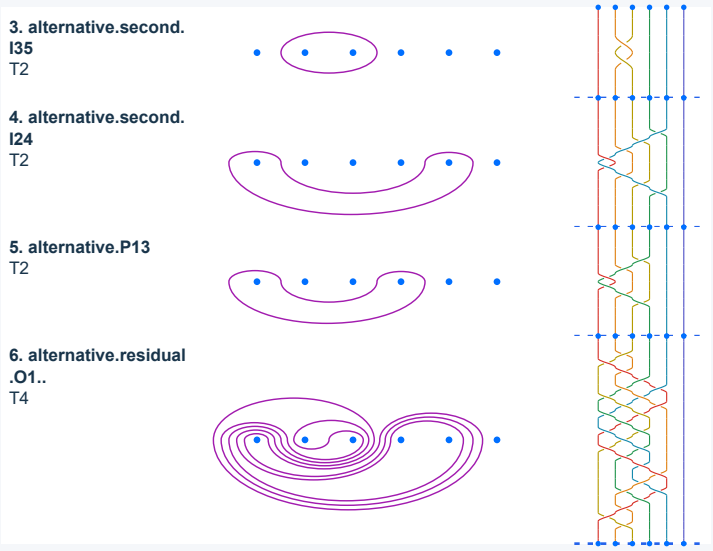
V - step 069

Inverse Hurwitz transport of the SAME outer past three petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 3. Inverse transport: the same outer moves upward and is conjugated at each crossed petal.

Before



After



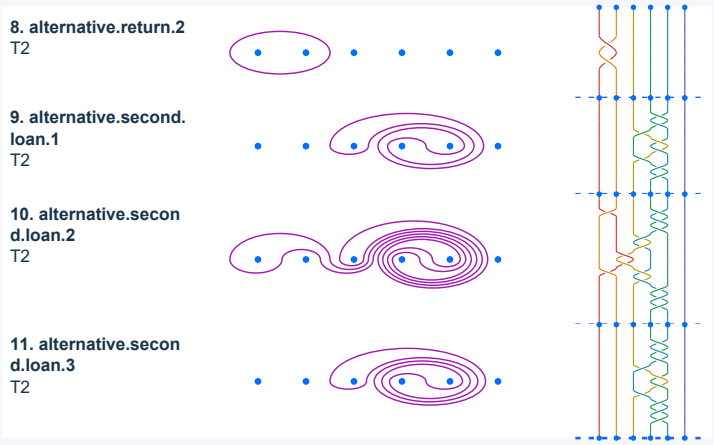
V - step 070

Return the12 petal through the three actual loan petals

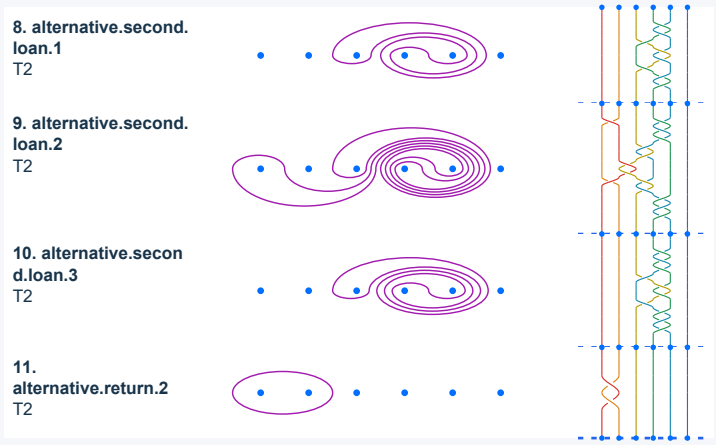
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



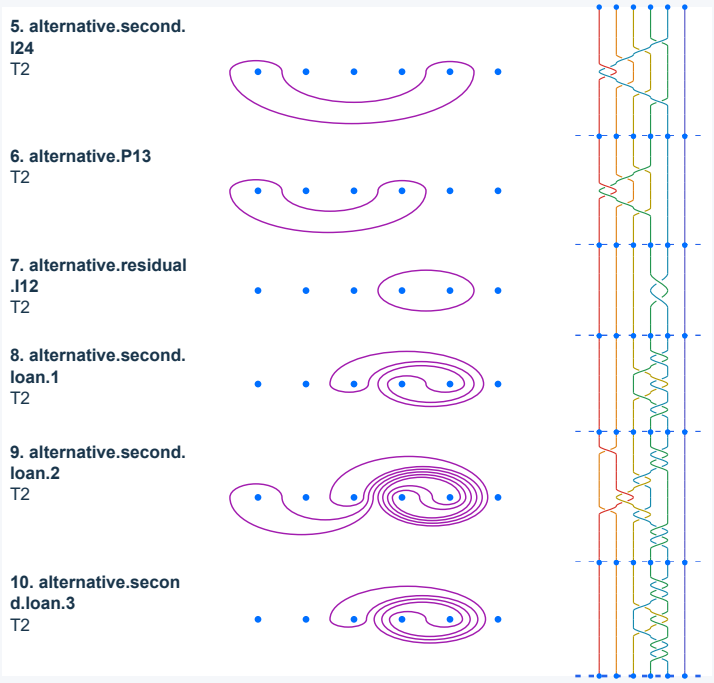
V - step 071

Expanded PC rose returns the loan WITHOUT a petal reorder

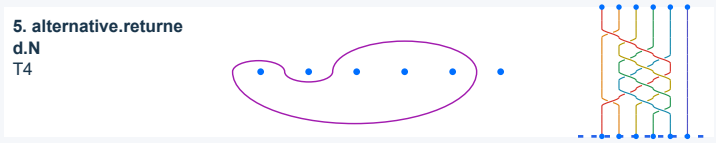
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



VI. First daisy with a doubled petal

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

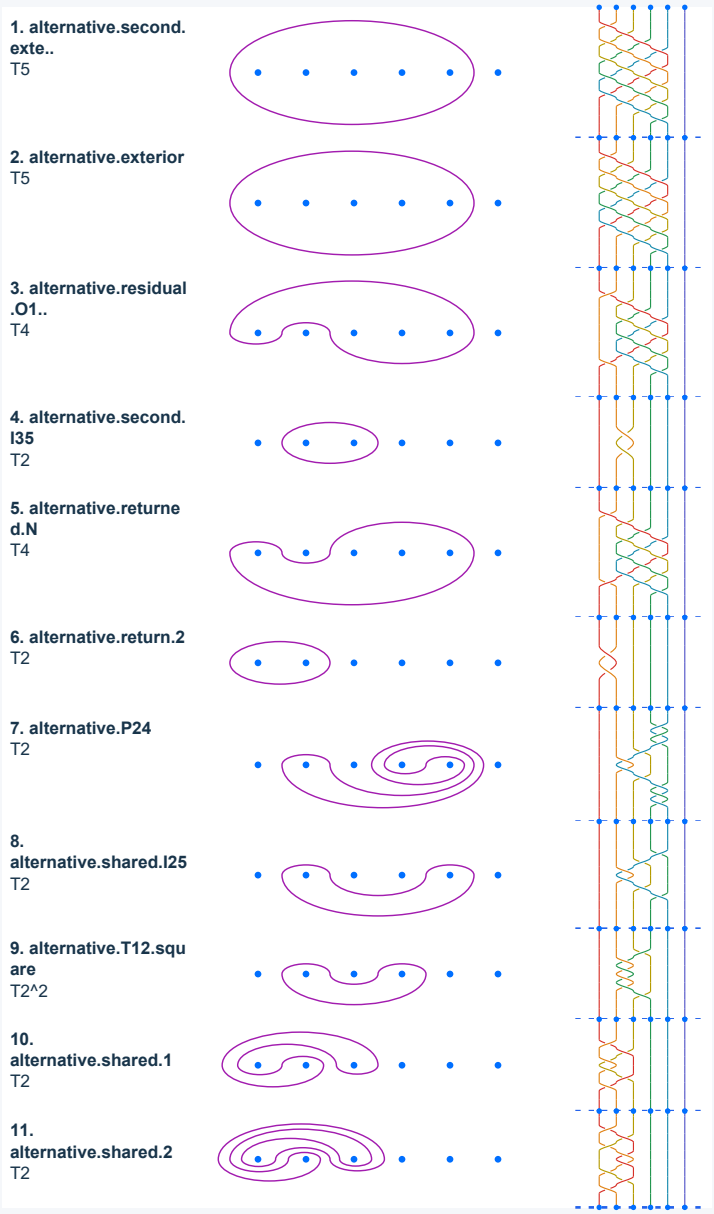
VI - step 072

Cyclic start BEFORE gathering the first daisy

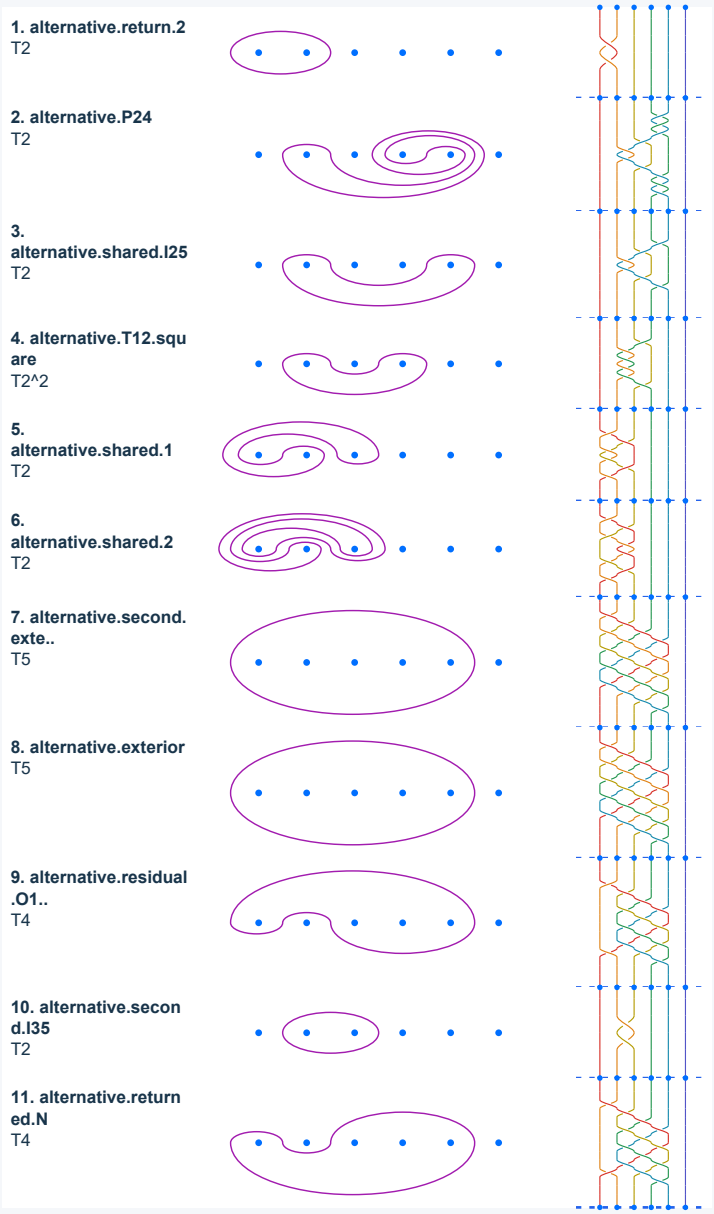
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



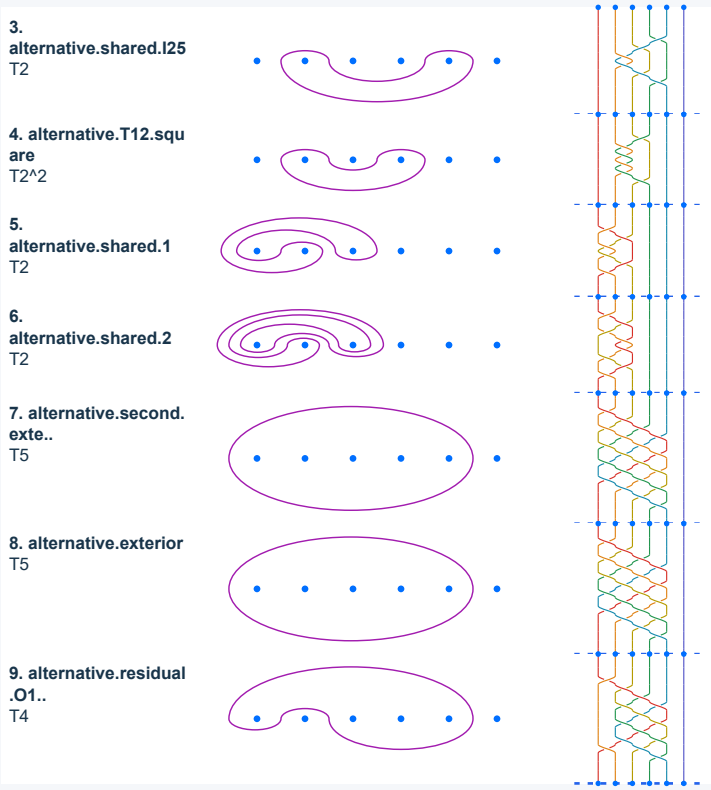
VI - step 073

Carry the SAME outer UP through its first-daisy petals

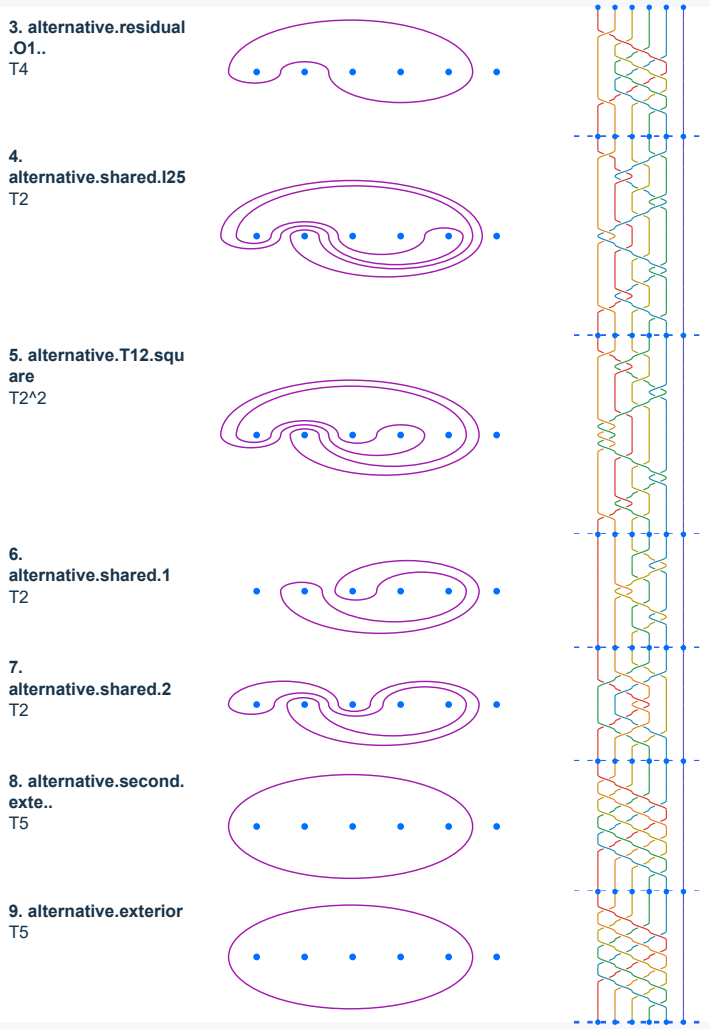
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



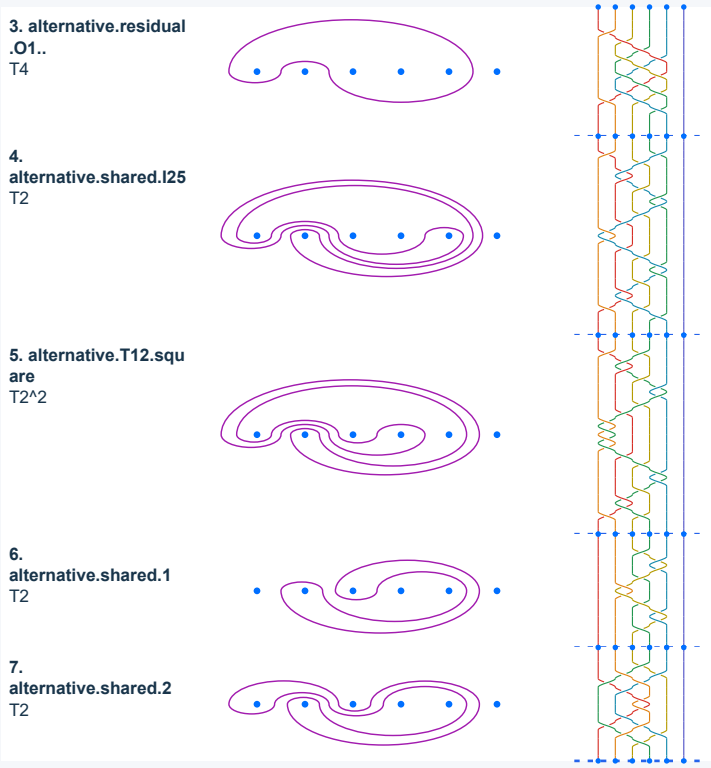
VI - step 074

PC daisy extracts the exterior DIRECTLY from the squared petal

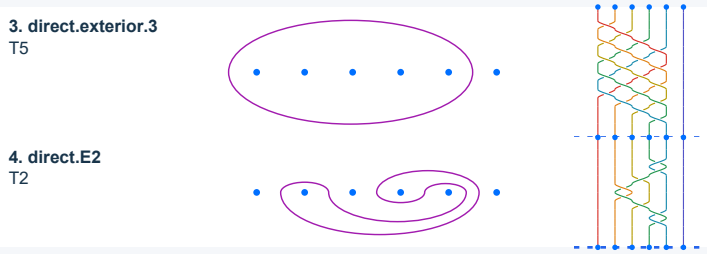
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 4 below. Only the changed interval is shown. Repeated-petal PC daisy: one squared input remains intact; the spare positive twist is retained. The full split/two-crossing expansion is certified.

Before



After



VII. Close the second daisy in cyclic order

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

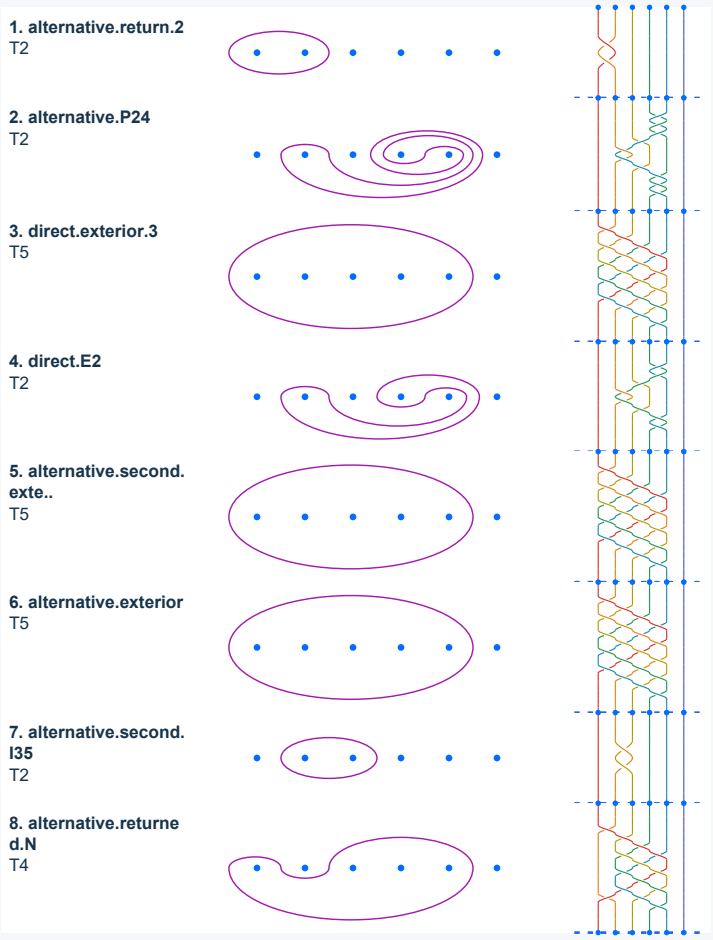
VII - step 075

Cyclic start at the NEW visible exterior

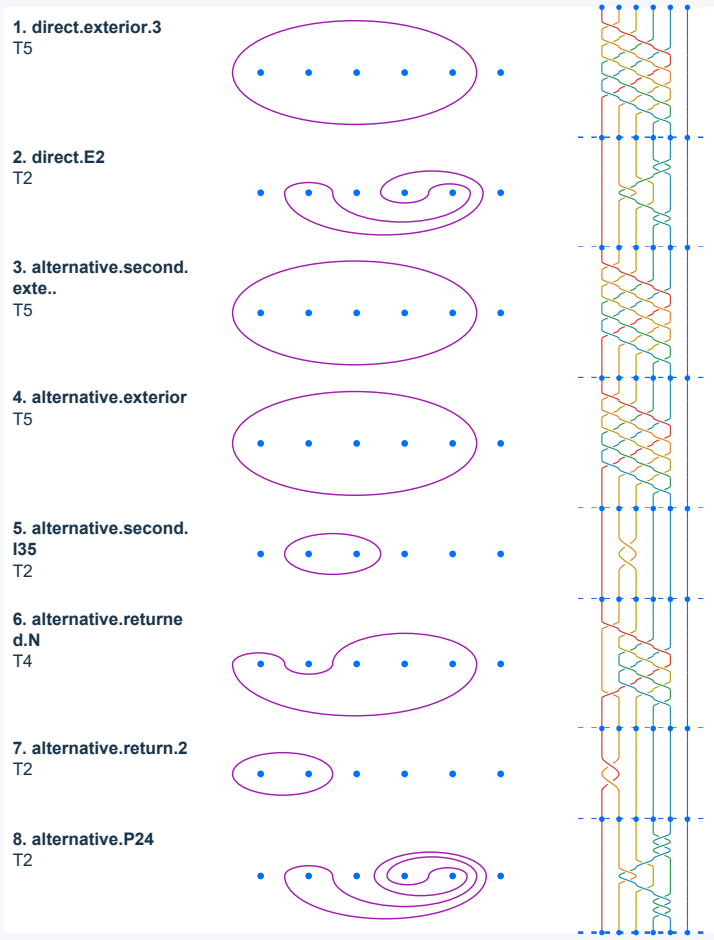
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



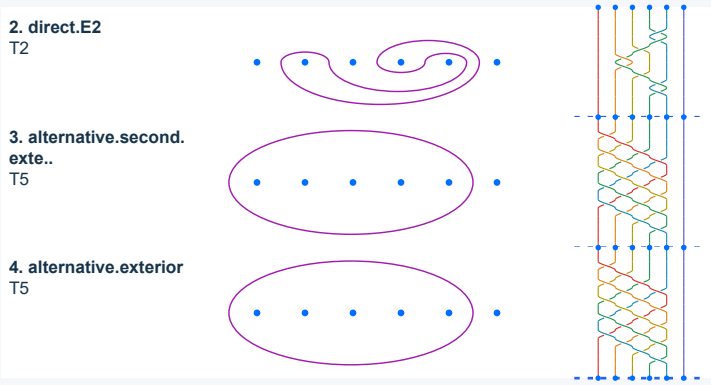
VII - step 076

The extra petal passes DOWN past two central exteriors

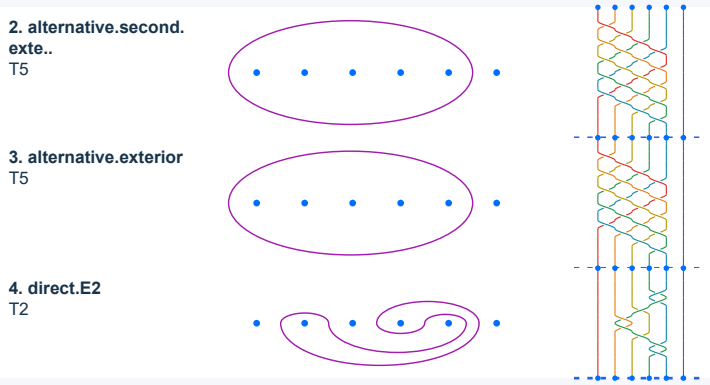
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



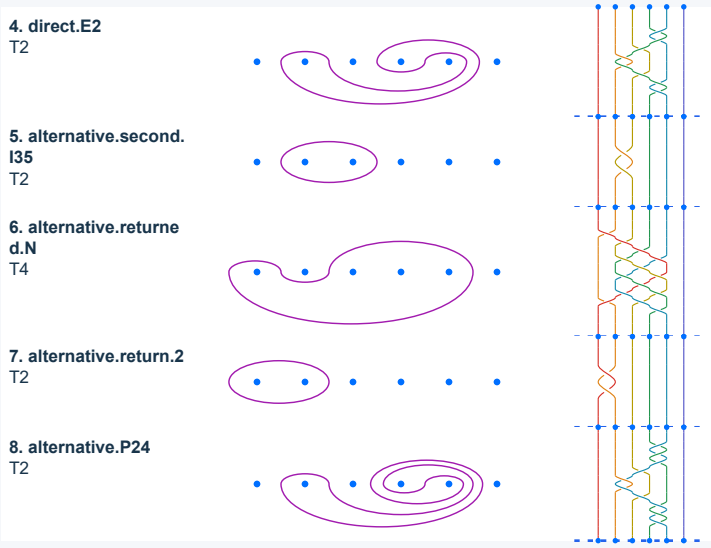
VII - step 077

SECOND actual PC daisy in its reached cyclic order

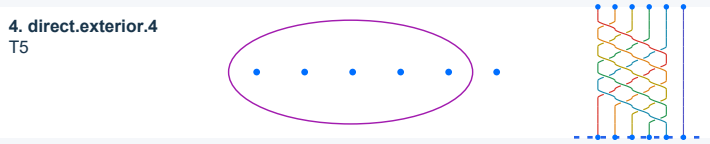
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 0 below. Only the changed interval is shown. Named PC daisy in its checked cyclic order; every input and the full internal conjugation are retained.

Before



After



VIII. Four exteriors and the sphere cap

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Four exteriors and the sphere cap

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. direct.exterior.3
T5

2. alternative.second.exte..
T5

3. alternative.exterior
T5

4. direct.exterior.4
T5



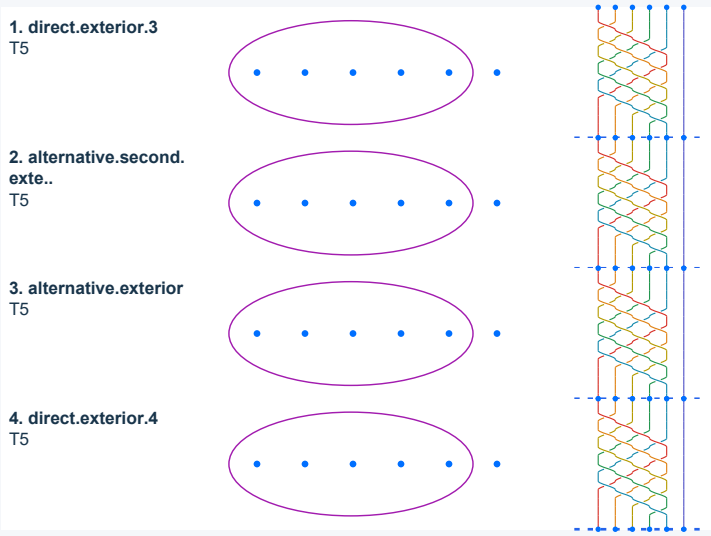
VIII - step 078

Combine the FOUR identical visible exterior twists

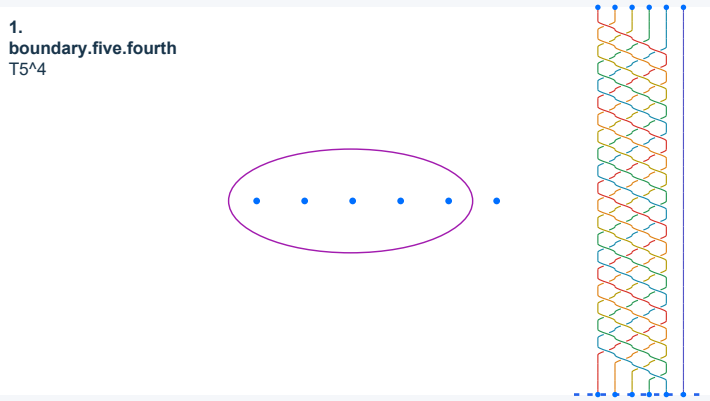
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



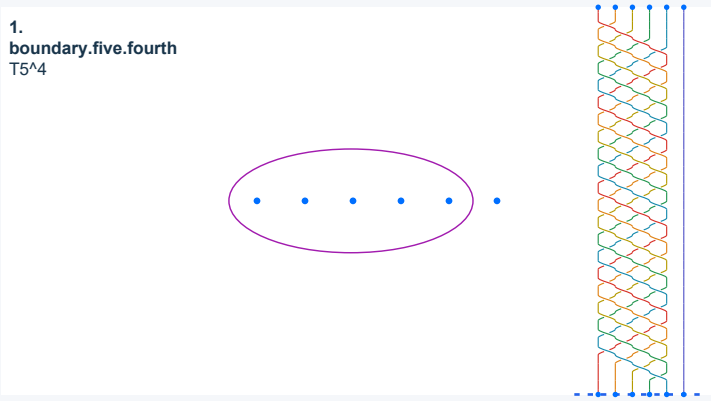
VIII - step 079

SPHERE MOVE: cap the FOURTH boundary power

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

IX. Sphere discrepancies and certificates

The following are ALL FOUR sphere operations in this named route. They are not disk braid equalities. Each discrepancy is written in signed generator indices; the exact sphere certificate and +I lift are retained online.

SPHERE MOVE at step 11

SPHERE LANTERN: keep F1 squared; absorb the two distinct actual half twists

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (70 signed braid letters):

5 -4 2 3 2 3 2 3 2 3 2 3 2 3 4 -5 -2 -2 -2 5 5 -4 3 4

-5 -5 2 2 2 -4 -2 3 2 4 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2

3 4 -5 -1 -1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 3 3 -5 -2 -3 -2

SPHERE MOVE at step 13

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (482 signed braid letters):

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 2 1 -4 2 3 2 3 2
3 2 3 2 3 2 3 4 2 3 2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2
-1 -2 -1 -2 -1 -2 -1 3 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 1
1 1 -4 -3 -2 -1 2 3 4 -1 -1 -1 4 4 4 1 1 2 -4 -3 -4 -3 -4 -3
-2 -1 -1 -4 -3 -4 -3 1 2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1 3 4
3 -4 -4 -3 -2 -1 -2 -1 -2 -1 3 4 4 1 -3 -2 -3 -2 -4 -3 -2 -3 -2 -3
-2 4 -3 -2 -1 4 -2 3 -4 -3 -3 2 -4 -3 -2 3 4 -2 3 3 4 -3 2 -4
-3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -1 -2 -4 -3 -3 -2 3 3 4 2 1
-2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -1 -1 -4 2 1 1 2 3 -4 -4 -3 -2
-1 -1 -2 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1

SPHERE MOVE at step 14

SPHERE MOVE: cap the SIX framing corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (120 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

SPHERE MOVE at step 79

SPHERE MOVE: cap the FOURTH boundary power

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (80 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4

Certificates and exact scope

Complete named graphical route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html?phase=namedfull

Full raw certificates:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/certificate.json

The independent six-point sphere cut comparison has six marked points and five consecutive-point arcs. It is separately jointly certified; the individual factor supports in this book are not joint cut complexes.

The early sphere lantern now accepts the original separating square and two distinct half twists directly. It bypasses the F1 root split, the half preconditioning move and the matched-half combine; its six-neighbor gather retains every crossing and rejoins the former step14 literally at new11. Both omitted crossings disappear from the actual route. The full-twist PC parent and its signed framing correction are retained; no half-twist convention for framed holes is asserted. The older23-operation closure remains available as a separate chapter. The original123/124 powers now enter a single powered cabled PC lantern, with the shared inner12 absorbed into its output square. Two explicit cyclic starts surround that substitution and rejoin the former six-operation prefix literally after only three operations; all original half twists stay unchanged. Both power decompositions and the inner/root commuting check remain in its certificate. An early partial PC daisy now replaces a power split followed later by a separate daisy: the actual three petals travel through the existing square drag, retaining five checks instead of three and rejoining literally. This saves one operation with two more adjacent checks. The first final daisy now accepts a doubled pair petal directly: A Q^m C = D (C^-1 Q^(m-1) C), where A Q C = D is the individually matched PC daisy. For m=2 this bypasses a displayed split and two separate petal carries. The spare positive twist remains visible. The generic expansion retains the split and both adjacent checks; this is a derived PC daisy rule, not a new sphere move. Every powered input and the literal later join are certified. The full disk tail remains within support maximum20. Four exterior twists remain visible until their separate fourth-power sphere cap. The under-twenty complete derivation remains open.

The archived 125-page/177-step PDF is preserved separately. This edition contains the new 79-step named proof, with no action-only boundary-recognition step.

Independent jointly certified six-point sphere cut comparison



